

Phenix Molecular Replacement

12 documentation pages

Generated from local Phenix documentation

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Molecular Replacement

xray-molecular-replacement.html

Why

One of the biggest challenges in crystallography is the phase problem, which can be addressed with experimental phasing methods. A popular alternative approach is molecular replacement (MR), which uses known structures to solve the phase problem. When a structurally similar model (e.g., a homologue) is available, this model is computationally placed in the unit cell containing the unknown structure. An initial electron density map is then calculated with the phases from the homologue and the observed structure factors.

The success of MR calculations depends on how much signal can be extracted from the data using the particular model, which in turn depends on a combination of the model quality and completeness, data resolution, and number of diffraction observations. For crystals of medium-sized proteins diffracting to moderate resolution, for example, the sequence identity between the molecule and homologue should be >25-30% and the r.m.s. deviation between C α atoms should be <2.0 Å.

How

In Phenix, the primary program for performing molecular replacement is Phaser. This requires experimental diffraction data, atomic model(s), the sequence of the molecule in the crystal, and typically some estimate of the number of molecules in the crystal. Phaser uses maximum-likelihood algorithms to determine the rotation of the model(s) with respect to the unit cell, and then the translation of the rotated model(s) within the unit cell. The output of this process, if successful, is a model file containing the placed model(s) and an MTZ file containing coefficients for the electron density made using the placed model(s) and the observed experimental amplitudes. For an introduction to molecular replacement in Phenix, [click here](#).

How to use the Phaser GUI in Phenix: [Click here](#)

Phenix reference manual for Phaser.

Common issues

- If more than one solution is found with Phaser-MR, this generally indicates that the solutions are all very low-confidence. The more solutions (more than 5 or ten is far too many), the lower the confidence in any of them. If you have more than one or two solutions, normally you will not want to go on with these solutions. Instead you will want to try a different search model or a different resolution or new data.
- If very low values of LLG (less than 30) or TFZ (less than 6) are obtained with Phaser-MR, this also generally indicates that the solutions are all very low confidence. Normally you will not want to go on with Phaser-MR solutions that have very low values of LLG. Instead you will want to try a different search model or a different resolution or new data.
- No solutions were found even with a good search model: There are multiple causes, but a common one is that a multiple domain protein has undergone a conformational change. In this case, you should split the structure and perform molecular replacement on the individual domains using separate search models. An alternative cause, thankfully less common, is that the crystallized molecule is not the intended one. For this reason, it is always prudent to use the "Search PDB Symmetry" tool in the Phenix GUI to check the cell dimensions and space group against the PDB. A third possibility is that the space group is incorrect (e.g., if twinning increases the apparent symmetry of the diffraction pattern), thus it is always good to run Xtriage in the "Data analysis" section of the Phenix GUI.

- Frequently asked questions about molecular replacement

Related programs

- phenix.MRage: This program uses Phaser to perform highly automated molecular replacement. In its simplest form, it uses experimental diffraction data and a sequence to solve a molecular replacement problem.
- phenix.sculptor: This program can improve models for molecular replacement by trimming off parts of the model that are unlikely to be preserved in the target structure.
- phenix.ensampler: This program prepares ensembles for molecular replacement. An ensemble is a set of structurally related models that have been superimposed with respect to each other, typically using a conserved structural core. Phenix.ensampler automates the process of superimposing models.
- phenix.morph_model: This program can improve an initial model after molecular replacement by locally moving the structure to better fit the electron density map. This is especially powerful when the molecular replacement solution is structurally too different to provide phases for map interpretation or automated model building.
- phenix.mr_rosetta: This program can improve an initial model after molecular replacement by using the Rosetta program to modify the model such that it moves closer to the true structure, while also improving the fit to electron density map. This is especially powerful when the molecular replacement solution is structurally too different to provide phases for map interpretation or automated model building.

phenix.ensemble_refinement

ensemble_refinement.html

Purpose

phenix.ensemble_refinement is designed to combine X-ray structure refinement with molecular dynamics in order to produce ensemble models fitted to diffraction data. These ensemble models can contain ~50-500 individual structures and simultaneously account for anisotropic and anharmonic distributions.

Development & implementation: Burnley BT, Afonine PV, Adams PD, Gros P. 2012. Modelling dynamics in protein crystal structures by ensemble refinement. *eLife* 1:e00311. doi: 10.7554/elife.00311.

Simple test case: Burnley BT, Gros P. 2013. phenix.ensemble_refinement: a test study of apo and holo BACE1. *Computational Crystallography Newsletter* Vol 4:2, 51-58.

Application: Forneris F, Burnley BT, Gros P. 2014. Ensemble refinement shows conformational flexibility in crystal structures of human complement factor D. *Acta D*, 70:3, 733-743.

General Procedure

The input structure should be a traditionally refined single structure model (i.e. post phenix.refine, publication quality). If TLS is used the TLS components must be present in the atom records (see phenix.tls). For optimal use explicit hydrogen should be added using either phenix.ready_set or phenix.pdbtools. The simulation is launched using the command line:

```
% phenix.ensemble_refinement model.pdb data.mtz [ligands.cif] [parameters]
```

For non-covalent ligands (excluding waters) it is recommended to use harmonic restraints e.g.:

```
% phenix.ensemble_refinement model.pdb data.mtz harmonic_restraints.selections='resname PO4 and  
element P'
```

Parameter Optimisation

There are three main empirical parameters should be optimised.

(1) pTLS:

```
% phenix.ensemble_refinement x.pdb x.mtz ptls=0.7
```

This defines the fraction of atoms included in the TLS fitting procedure (see Burnley et al. 2012). The optimum value for this parameter cannot be determined a priori. Setting up an array of simultaneous simulations with different values (e.g. 1.0, 0.9, 0.8, 0.6) is recommended. By default the TLS groups are applied to each separate chain. TLS groups can be user defined using:

```
% phenix.ensemble_refinement x.pdb x.mtz tls_group_selections=chain a tls_group_selections=chain b
```

Excessive numbers of TLS groups will result in intramolecular disorder being absorbed by the TLS B-factors rather than being modelled by atomic fluctuations in the simulation therefore it is recommended to use as few as possible, e.g. one per chain (see Burnley et al. 2012). The TLS fitting process requires that the number of non-hydrogen, non-solvent atoms, per TLS group * ptls to be greater than 63, otherwise ptls is automatically increased. If any TLS group has less than 63 non-solvent, non-hydrogen atoms then an isotropic model, based on the Wilson B-factor, is automatically applied.

(2) wxray_coupled_t bath_offset:

```
% phenix.ensemble_refinement x.pdb x.mtz wxray_coupled_t bath_offset=10.0
```

Unintuitively wxray_coupled_t bath_offset controls the X-ray weight! The X-ray weight is modulated in situ such that the simulation runs at the target temperature (cartesian_dynamics.temperature). The simulation uses velocity scaling to maintain the target temperature. This is coupled to a temperature thermostat which is set as:

```
thermostat = cartesian_dynamics.temperature - wxray_coupled_t bath_offset
```

The non-conservative X-ray force generates heat (see Burnley et al 2012), thus a larger offset increases the X-ray weight. The default value is 5 K. For example values of 2.5 and 10 K may also be tested.

(3) tx:

```
% phenix.ensemble_refinement x.pdb x.mtz tx=1.0
```

The relaxation time (or memory time) in picoseconds of the time-averaged restraints used in the simulation. As tx increases more structures contribute to the target function (see Burnley et al. 2012 for more details) and the optimum value is correlated with resolution (see Burnley et al. 2012). If tx is not defined (tx = None) a value will be automatically selected based on the dataset resolution. Values of 2 x automated Tx and 0.5 x automated Tx may also be tested. N.B. the simulation time is based on tx, larger values of tx will produce longer simulations.

The default parameters have been set to give best as possible results. However, it is strongly recommend that pTLS is optimised on a case by base basis and it may be necessary to optimise the temperature bath offset followed by tx.

The ens_pdb.py and ens_tools.py scripts referred to in "phenix.ensemble_refinement: a test study of apo and holo BACE1" are available within cctbx here: cctbx/mmtbx/refinement/ensemble_refinement/scripts/.

N.B. if you wish to display individual atom's b and q properties in PyMol the discrete=1 argument should be used when loading the ensemble (e.g. load ens.pdb, discrete=1).

List of all available keywords

extra_restraints_file = None File containing custom geometry restraints, using the same format as phenix.refine. On the command line this can be specified directly as a command-line argument, but this parameter is used by the Phenix GUI.
inputsequence = None scattering_table = wk1995 it1992 *n_gaussian
neutron electron wavelength = None energy = None use_experimental_phases = Auto twin_law = Auto Enter twin law if known.
xray_dataScope of X-ray data and free-R flags
file_name = None labels = None
high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
french_wilson_scale = True
sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True
force_anomalous_flag_to_be_equal_to = None
convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60
Maximum number of resolution bins
min_bin_size = 40 Minimum number of reflections per
binr_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.
label = None required = True Specify if free-r flags are must be present (or else generated)
test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!
ignore_r_free_flags = False Use all reflections in refinement (work and test)
disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
generate = False Generate R-free flags (if not

available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 experimental_phasesfile_name = None labels = None pdbfile_name = None Model file(s) name (PDB)monomersfile_name = None Monomer file(s) name (CIF)mapsmap_file_name = None A CCP4-formatted mapd_min = None Resolution of mapmap_coefficients_file_name = None MTZ file containing mapmap_coefficients_label = None Data label for complex map coefficients in MTZ fileensemble_refinementden_restraints = True Use DEN restraintsupdate_sigmaa_rfree = 0.001 test functionensemble_reduction = True Find minimum number of structures to reproduce simulation R-valuesensemble_reduction_rfree_tolerance = 0.0025 verbose = -1 output_file_prefix = None Prefix for all output filesgzip_final_model = True write_centroid_model = False write_mean_model = False random_seed = 2679941 Random seednproc = 1 tx = None Relaxation time (ps)equilibrium_n_tx = 10 Length of equilibration period, n times txacquisition_block_n_tx = 20 Length of acquisition block, n times txnumber_of_acquisition_periods = 1 Number of acquisition periodspdb_stored_per_block = 200 Number of model coordinates stored per acquisition blockwxray_coupled_t bath = True Use temperature control wxraywxray_coupled_t bath_offset = 5.0 Temperature offset, increasing offset increases wxraywxray = 1.0 Multiplier for xray weighting; used if wxray_coupled_t bath = False tls_group_selections = None TLS groups to use for TLS fitting (TLS details in PDB header not used)ptls = 0.80 The fraction of atoms to include in TLS fittingimport_tls_pdb = None PDB path to import TLS from external structuremax_ptls_cycles = 25 Maximum cycles to use in TLS fitting; TLS will stop prior to this if convergence is reachedisotropic_b_factor_model = False Use isotropic B-factor model instead of TLSpwilson = 0.8 Scale factor for isotropic b-factor model: all atoms = Bwilson * pwilsonset_occupancies = False Set all atoms aoccupancy to 1.0target_name = *ml mlhl ls_wunit_k1_fixed ls_wunit_k1 Choices for refinement targetremove_alt_conf_from_input_pdb = True Removes any alternative conformations if present in input PDB modelscale_wrt_n_calc_start = False Scale <Ncalc> to ...

- extra_restraints_file = None File containing custom geometry restraints, using the same format as phenix.refine. On the command line this can be specified directly as a command-line argument, but this parameter is used by the Phenix GUI.

- inputsequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = Auto twin_law = Auto Enter twin law if known.xray_dataScope of X-ray data and free-R flagsfile_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doingignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 experimental_phasesfile_name = None labels = None pdbfile_name = None Model file(s) name (PDB)monomersfile_name = None Monomer file(s) name (CIF)mapsmap_file_name = None A CCP4-formatted mapd_min = None Resolution of

mapmap_coefficients_file_name = None MTZ file containing mapmap_coefficients_label = None Data label for complex map coefficients in MTZ file

- sequence = None
- scattering_table = wk1995 it1992 *n_gaussian neutron electron
- wavelength = None
- energy = None
- use_experimental_phases = Auto
- twin_law = Auto Enter twin law if known.
- xray_dataScope of X-ray data and free-R flagsfile_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20
- file_name = None
- labels = None
- high_resolution = None
- low_resolution = None
- twin_law = None
- outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- french_wilson_scale = True
- sigma_fobs_rejection_criterion = None
- sigma_iobs_rejection_criterion = None
- ignore_all_zeros = True
- force_anomalous_flag_to_be_equal_to = None
- convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5
- french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per bin
- max_bins = 60 Maximum number of resolution bins
- min_bin_size = 40 Minimum number of reflections per bin

- `r_free_flagsfile_name = None` This is normally the same as the file containing Fobs and is usually selected automatically.
- `label = None` required = True Specify if free-r flags are must be present (or else generated)
- `test_flag_value = None` This value is usually selected automatically - do not change unless you really know what you're doing!
- `ignore_r_free_flags = False` Use all reflections in refinement (work and test)
- `disable_suitability_test = False` ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- `generate = False` Generate R-free flags (if not available in input files)
- `fraction = 0.1` max_free = 2000 lattice_symmetry_max_delta = 5
- `use_lattice_symmetry = True` use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- `n_shells = 20`
- `experimental_phasesfile_name = None` labels = None
- `file_name = None`
- `labels = None`
- `pdbfile_name = None` Model file(s) name (PDB)
- `file_name = None` Model file(s) name (PDB)
- `monomersfile_name = None` Monomer file(s) name (CIF)
- `file_name = None` Monomer file(s) name (CIF)
- `mapsmap_file_name = None` A CCP4-formatted map
- `mapmap_coefficients_file_name = None` MTZ file containing map
- `mapmap_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `map_file_name = None` A CCP4-formatted map
- `d_min = None` Resolution of map
- `map_coefficients_file_name = None` MTZ file containing map
- `map_coefficients_label = None` Data label for complex map coefficients in MTZ file

• ensemble_refinementden_restraints = True Use DEN restraintsupdate_sigmaa_rfree = 0.001 test functionensemble_reduction = True Find minimum number of structures to reproduce simulation R-valuesensemble_reduction_rfree_tolerance = 0.0025 verbose = -1 output_file_prefix = None Prefix for all output filesgzip_final_model = True write_centroid_model = False write_mean_model = False random_seed = 2679941 Random seednproc = 1 tx = None Relaxation time (ps)equilibrium_n_tx = 10 Length of equilibration period, n times txacquisition_block_n_tx = 20 Length of acquisition block, n times txnumber_of_acquisition_periods = 1 Number of acquisition periodspdb_stored_per_block = 200 Number of model coordinates stored per acquisition blockwxray_coupled_t bath = True Use temperature control wxraywxray_coupled_t bath_offset = 5.0 Temperature offset, increasing offset increases wxraywxray = 1.0 Multiplier for xray weighting; used if wxray_coupled_t bath = False tls_group_selections = None TLS groups to use for TLS fitting (TLS details in PDB header not used)ptls = 0.80 The fraction of atoms to include in TLS fittingimport_tls_pdb = None PDB path to import TLS from external structuremax_ptls_cycles = 25 Maximum cycles to use in TLS fitting; TLS will stop prior to this if convergence is reachedisotropic_b_factor_model = False Use isotropic B-factor model instead of TLSWilson = 0.8 Scale factor for isotropic b-factor model: all atoms = Bwilson * pwilsonset_occupancies = False Set all atoms aoccupancy to 1.0target_name = *ml mlhl ls_wunit_k1_fixed ls_wunit_k1 Choices for refinement targetremove_alt_conf_from_input_pdb = True Removes any alternative conformations if present in input PDB modelscale_wrt_n_calc_start = False Scale <Ncalc> to starting Ncalcoutput_running_kinetic_energy_in_occupancy_column = False Output PDB file contains running kinetic energy in place of occupancyordered_solvent_update = True Ordered water molecules automatically updated every nth macro cycleordered_solvent_update_cycle = 25 Number of macro-cycles per ordered solvent updatecartesian_dynamicsttemperature = 300 number_of_steps = 10 time_step = 0.0005 initial_velocities_zero_fraction = 0 n_print = 100 verbose = -1 random_seed = None n_collect = 10 stop_cm_motion = True protein_thermostat = True Use protein atoms thermostatdenreference_file = None gamma = 0.5 kappa = 0.1 weight = 30.0 sigma = 0.44 optimize = True If selected, will run a grid search to determine optimal values of the weight and gamma parameters for DEN refinement. This is very slow, but highly recommended, and can be parallelized across multiple CPU cores.opt_gamma_values = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 opt_weight_values = 3.0, 10.0, 30.0, 100.0, 300.0 num_cycles = 10 kappa_burn_in_cycles = 0 bulk_solvent_and_scale = True refine_adp = True final_refinement_cycle = False verbose = False annealing_type = *torsion cartesian select strategy to apply DEN restraintsminimize_c_alpha_only = False output_kinimage = False output kinimage representation of starting DEN restraintsrestraint_networklower_distance_cutoff = 3.0 upper_distance_cutoff = 15.0 sequence_separation_low = 0 sequence_separation_limit = 10 exclude_hydrogens = True ndistance_ratio = 1.0 export_den_pairs = False den_network_file = None harmonic_restraintsselections = None Atoms selections to apply harmonic restraintsweight = 0.001 Harmonic restraints weightslack = 1.0 Harmonic restraints slack distanceelectron_density_mapsapply_default_maps = True map_coefficientsmap_type = None format = *mtz phs mtz_label_amplitudes = None mtz_label_phases = None fill_missing_f_obs = False acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0: max(k-centrics_pre_scale,0)*Fo-max(n-centrics_pre_scale,0)*Fcsharpening = False Apply B-factor sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections = False Exclude free-R select...

- den_restraints = True Use DEN restraints
- update_sigmaa_rfree = 0.001 test function
- ensemble_reduction = True Find minimum number of structures to reproduce simulation R-values
- ensemble_reduction_rfree_tolerance = 0.0025

- verbose = -1
- output_file_prefix = None Prefix for all output files
- gzip_final_model = True
- write_centroid_model = False
- write_mean_model = False
- random_seed = 2679941 Random seed
- nproc = 1
- tx = None Relaxation time (ps)
- equilibrium_n_tx = 10 Length of equilibration period, n times tx
- acquisition_block_n_tx = 20 Length of acquisition block, n times tx
- number_of_acquisition_periods = 1 Number of acquisition periods
- pdb_stored_per_block = 200 Number of model coordinates stored per acquisition block
- wxray_coupled_t bath = True Use temperature control wxray
- wxray_coupled_t bath_offset = 5.0 Temperature offset, increasing offset increases wxray
- wxray = 1.0 Multiplier for xray weighting; used if wxray_coupled_t bath = False
- tls_group_selections = None TLS groups to use for TLS fitting (TLS details in PDB header not used)
- ptls = 0.80 The fraction of atoms to include in TLS fitting
- import_tls_pdb = None PDB path to import TLS from external structure
- max_ptls_cycles = 25 Maximum cycles to use in TLS fitting; TLS will stop prior to this if convergence is reached
- isotropic_b_factor_model = False Use isotropic B-factor model instead of TLS
- pwilson = 0.8 Scale factor for isotropic b-factor model: all atoms = Bwilson * pwilson
- set_occupancies = False Set all atoms aoccupancy to 1.0
- target_name = *ml mlhl ls_wunit_k1_fixed ls_wunit_k1 Choices for refinement target
- remove_alt_conf_from_input_pdb = True Removes any alternative conformations if present in input PDB model
- scale_wrt_n_calc_start = False Scale <Ncalc> to starting Ncalc
- output_running_kinetic_energy_in_occupancy_column = False Output PDB file contains running kinetic energy in place of occupancy
- ordered_solvent_update = True Ordered water molecules automatically updated every nth macro cycle
- ordered_solvent_update_cycle = 25 Number of macro-cycles per ordered solvent update
- cartesian_dynamicstemperture = 300 number_of_steps = 10 time_step = 0.0005
initial_velocities_zero_fraction = 0 n_print = 100 verbose = -1 random_seed = None n_collect = 10
stop_cm_motion = True protein_thermostat = True Use protein atoms thermostat
- temperature = 300
- number_of_steps = 10
- time_step = 0.0005
- initial_velocities_zero_fraction = 0
- n_print = 100

- verbose = -1
- random_seed = None
- n_collect = 10
- stop_cm_motion = True
- protein_thermostat = True Use protein atoms thermostat
- denreference_file = None gamma = 0.5 kappa = 0.1 weight = 30.0 sigma = 0.44 optimize = True If selected, will run a grid search to determine optimal values of the weight and gamma parameters for DEN refinement. This is very slow, but highly recommended, and can be parallelized across multiple CPU cores. opt_gamma_values = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 opt_weight_values = 3.0, 10.0, 30.0, 100.0, 300.0 num_cycles = 10 kappa_burn_in_cycles = 0 bulk_solvent_and_scale = True refine_adp = True final_refinement_cycle = False verbose = False annealing_type = *torsion cartesian select strategy to apply DEN restraints minimize_c_alpha_only = False output_kinimage = False output kinimage representation of starting DEN restraints restraint_network lower_distance_cutoff = 3.0 upper_distance_cutoff = 15.0 sequence_separation_low = 0 sequence_separation_limit = 10 exclude_hydrogens = True ndistance_ratio = 1.0 export_den_pairs = False den_network_file = None
- reference_file = None
- gamma = 0.5
- kappa = 0.1
- weight = 30.0
- sigma = 0.44
- optimize = True If selected, will run a grid search to determine optimal values of the weight and gamma parameters for DEN refinement. This is very slow, but highly recommended, and can be parallelized across multiple CPU cores.
- opt_gamma_values = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0
- opt_weight_values = 3.0, 10.0, 30.0, 100.0, 300.0
- num_cycles = 10
- kappa_burn_in_cycles = 0
- bulk_solvent_and_scale = True
- refine_adp = True
- final_refinement_cycle = False
- verbose = False
- annealing_type = *torsion cartesian select strategy to apply DEN restraints
- minimize_c_alpha_only = False
- output_kinimage = False output kinimage representation of starting DEN restraints
- restraint_network lower_distance_cutoff = 3.0 upper_distance_cutoff = 15.0 sequence_separation_low = 0 sequence_separation_limit = 10 exclude_hydrogens = True ndistance_ratio = 1.0 export_den_pairs = False den_network_file = None
- lower_distance_cutoff = 3.0
- upper_distance_cutoff = 15.0
- sequence_separation_low = 0

- sequence_separation_limit = 10
- exclude_hydrogens = True
- ndistance_ratio = 1.0
- export_den_pairs = False
- den_network_file = None
- harmonic_restraintsselections = None Atoms selections to apply harmonic restraintsweight = 0.001
Harmonic restraints weightslack = 1.0 Harmonic restraints slack distance
- selections = None Atoms selections to apply harmonic restraints
- weight = 0.001 Harmonic restraints weight
- slack = 1.0 Harmonic restraints slack distance
- electron_density_mapsapply_default_maps = True map_coefficientsmap_type = None format = *mtz
phs mtz_label_amplitudes = None mtz_label_phases = None fill_missing_f_obs = False acentrics_scale
= 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)centrics_pre_scale =
1.0 Centric reflections, k!=n and k*n != 0:
 $\max(k\text{-centrics_pre_scale}, 0) * F_o - \max(n\text{-centrics_pre_scale}, 0) * F_c$ sharpening = False Apply B-factor
sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections =
False Exclude free-R selected reflections from output map coefficientsisotropize = True
devcomplete_set_up_to_d_min = False apply_same_incompleteness_to_complete_set_at = randomly low
high
- apply_default_maps = True
- map_coefficientsmap_type = None format = *mtz phs mtz_label_amplitudes = None mtz_label_phases
= None fill_missing_f_obs = False acentrics_scale = 2.0 Scale terms corresponding to acentric reflections
(residual maps only: k==n)centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0:
 $\max(k\text{-centrics_pre_scale}, 0) * F_o - \max(n\text{-centrics_pre_scale}, 0) * F_c$ sharpening = False Apply B-factor
sharpeningsharpening_b_factor = None Optional sharpening B-factor valueexclude_free_r_reflections =
False Exclude free-R selected reflections from output map coefficientsisotropize = True
devcomplete_set_up_to_d_min = False apply_same_incompleteness_to_complete_set_at = randomly low
high
- map_type = None
- format = *mtz phs
- mtz_label_amplitudes = None
- mtz_label_phases = None
- fill_missing_f_obs = False
- acentrics_scale = 2.0 Scale terms corresponding to acentric reflections (residual maps only: k==n)
- centrics_pre_scale = 1.0 Centric reflections, k!=n and k*n != 0:
 $\max(k\text{-centrics_pre_scale}, 0) * F_o - \max(n\text{-centrics_pre_scale}, 0) * F_c$
- sharpening = False Apply B-factor sharpening
- sharpening_b_factor = None Optional sharpening B-factor value
- exclude_free_r_reflections = False Exclude free-R selected reflections from output map coefficients
- isotropize = True

- devcomplete_set_up_to_d_min = False apply_same_incompleteness_to_complete_set_at = randomly low high
- complete_set_up_to_d_min = False
- apply_same_incompleteness_to_complete_set_at = randomly low high
- maskuse_asu_masks = True solvent_radius = 1.1 shrink_truncation_radius = 0.9
use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factorstep = 0.6
Grid step (in Å) for solvent mask calculationn_real = None Gridding for mask calculationgrid_step_factor
= 4.0 Grid step is resolution divided by grid_step_factorverbose = 1 mean_shift_for_mask_update = 0.1
Value of overall model shift in refinement to updates the mask.ignore_zero_occupancy_atoms = True
Include atoms with zero occupancy into mask calculationignore_hydrogens = True Ignore H or D atoms in
mask calculationn_radial_shells = 1 Number of shells in a radial shell bulk solvent
modelradial_shell_width = 1.3 Radial shell widthFmask_res_high = 3.0 Assume Fmask=0 for reflections
outside the range $\inf > d \geq F_{\text{mask_res_high}}$
- use_asu_masks = True
- solvent_radius = 1.1
- shrink_truncation_radius = 0.9
- use_resolution_based_gridding = False If enabled, mask grid step = d_min / grid_step_factor
- step = 0.6 Grid step (in Å) for solvent mask calculation
- n_real = None Gridding for mask calculation
- grid_step_factor = 4.0 Grid step is resolution divided by grid_step_factor
- verbose = 1
- mean_shift_for_mask_update = 0.1 Value of overall model shift in refinement to updates the mask.
- ignore_zero_occupancy_atoms = True Include atoms with zero occupancy into mask calculation
- ignore_hydrogens = True Ignore H or D atoms in mask calculation
- n_radial_shells = 1 Number of shells in a radial shell bulk solvent model
- radial_shell_width = 1.3 Radial shell width
- Fmask_res_high = 3.0 Assume Fmask=0 for reflections outside the range $\inf > d \geq F_{\text{mask_res_high}}$
- ensemble_ordered_solventdiff_map_cutoff = 2.5 e_map_cutoff_keep = 0.5 e_map_cutoff_find = 0.5
tolerance = 0.9 ordered_solvent_map_to_model = True output_residue_name = HOH output_chain_id =
S output_atom_name = O scattering_type = O Defines scattering factors for newly added
watersprimary_map_type = mFo-DFmodel primary_map_cutoff = 3.0 secondary_map_type =
2mFo-DFmodel secondary_map_cutoff_keep = 1.0 secondary_map_cutoff_find = 1.0 dist_min = 1.8
dist_max = 3.2 dist_min_altloc = 0.5 new_solvent = *isotropic anisotropic Based on the choice, added
solvent will have isotropic or anisotropic b-factorsb_iso_min = 0.0 Minimum B-factor value, waters with
smaller value will be rejectedb_iso_max = 100.0 Maximum B-factor value, waters with bigger value will be
rejectedanisotropy_min = 0.1 For solvent refined as anisotropic: remove is less than this
valueoccupancy_min = 0.2 Minimum occupancy value, waters with smaller value will be
rejectedoccupancy_max = 1.0 Maximum occupancy value, waters with bigger value will be
rejectedoccupancy = 1.0 Initial occupancy value for newly added waterrefine_occupancies = False Refine
solvent occupancies.add_hydrogens = False Adds hydrogens to water molecules (except those on
special positions)refilter = True temperature = 300 Target temperature for random velocity
assignmentseed = 343534534 Fixes the random seed for velocity
assignmentpreserved_solvent_minimum_distance = 7.0 find_peakseuse_sigma_scaled_maps = True

Default is sigma scaled map, map in absolute scale is used otherwise.grid_step = 0.6 Grid step for map samplingmax_number_of_peaks = None map_next_to_modelmin_model_peak_dist = 1.8
 max_model_peak_dist = 3.0 min_peak_peak_dist = 1.8 use_hydrogens = False
 peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None
 general_positions_only = False min_cross_distance = 1.8 min_cubicle_edge = 5

- diff_map_cutoff = 2.5
- e_map_cutoff_keep = 0.5
- e_map_cutoff_find = 0.5
- tolerance = 0.9
- ordered_solvent_map_to_model = True
- output_residue_name = HOH
- output_chain_id = S
- output_atom_name = O
- scattering_type = O Defines scattering factors for newly added waters
- primary_map_type = mFo-DFmodel
- primary_map_cutoff = 3.0
- secondary_map_type = 2mFo-DFmodel
- secondary_map_cutoff_keep = 1.0
- secondary_map_cutoff_find = 1.0
- dist_min = 1.8
- dist_max = 3.2
- dist_min_altloc = 0.5
- new_solvent = *isotropic anisotropic Based on the choice, added solvent will have isotropic or anisotropic b-factors
- b_iso_min = 0.0 Minimum B-factor value, waters with smaller value will be rejected
- b_iso_max = 100.0 Maximum B-factor value, waters with bigger value will be rejected
- anisotropy_min = 0.1 For solvent refined as anisotropic: remove is less than this value
- occupancy_min = 0.2 Minimum occupancy value, waters with smaller value will be rejected
- occupancy_max = 1.0 Maximum occupancy value, waters with bigger value will be rejected
- occupancy = 1.0 Initial occupancy value for newly added water
- refine_occupancies = False Refine solvent occupancies.
- add_hydrogens = False Adds hydrogens to water molecules (except those on special positions)
- refilter = True
- temperature = 300 Target temperature for random velocity assignment
- seed = 343534534 Fixes the random seed for velocity assignment
- preserved_solvent_minimum_distance = 7.0
- find_peaksuse_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.grid_step = 0.6 Grid step for map samplingmax_number_of_peaks = None
 map_next_to_modelmin_model_peak_dist = 1.8 max_model_peak_dist = 3.0 min_peak_peak_dist = 1.8

use_hydrogens = False peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True
min_distance_sym_equiv = None general_positions_only = False min_cross_distance = 1.8
min_cubicle_edge = 5

- use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise.
- grid_step = 0.6 Grid step for map sampling
- max_number_of_peaks = None
- map_next_to_modelmin_model_peak_dist = 1.8 max_model_peak_dist = 3.0 min_peak_peak_dist = 1.8 use_hydrogens = False
- min_model_peak_dist = 1.8
- max_model_peak_dist = 3.0
- min_peak_peak_dist = 1.8
- use_hydrogens = False
- peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = False min_cross_distance = 1.8 min_cubicle_edge = 5
- peak_search_level = 1
- max_peaks = 0
- interpolate = True
- min_distance_sym_equiv = None
- general_positions_only = False
- min_cross_distance = 1.8
- min_cubicle_edge = 5
- amberParameters for using Amber in refinement.use_amber = False Use Amber for all the gradients in refinementtopology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.order_file_name = None A file that maps amber atom numbers to phenix atom numbers.wxc_factor = 1. restraint_wt = 0. restraintmask = " reference_file_name = " bellymask = " If given, turn on belly in sanderqmmask = " If given, turn on QM/MM with the given maskqmmcharge = 0 Charge of the QM/MM regionnetcdf_trajectory_file_name = " If given, turn on writing netcdf trajectoryprint_amber_energies = False Print details of Amber energies during refinement
- use_amber = False Use Amber for all the gradients in refinement
- topology_file_name = None A topology file needed by Amber. Can be generated using phenix.AmberPrep.
- coordinate_file_name = None A coordinate file needed by Amber. Can be generated using phenix.AmberPrep.
- order_file_name = None A file that maps amber atom numbers to phenix atom numbers.
- wxc_factor = 1.
- restraint_wt = 0.
- restraintmask = "
- reference_file_name = "
- bellymask = " If given, turn on belly in sander

- qmmask = " If given, turn on QM/MM with the given mask
- qmcharge = 0 Charge of the QM/MM region
- netcdf_trajectory_file_name = " If given, turn on writing netcdf trajectory
- print_amber_energies = False Print details of Amber energies during refinement
- qiQMworking_directory = None qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the environment of the corecalculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedcleanup = all *most None ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selections specific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- working_directory = None
- qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the environment of the corecalculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if not specifiedcleanup = all *most None ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False do_not_update_restraints = False For testing and maybe getting strain energy of standard restraintsbuffer_selection = None use this instead of distance from selections specific_atom_chargesatom_selection = None charge = None specific_atom_multiplicitiesatom_selection = None multiplicity = None freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1

read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation

- run_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the restraints generation is run

- buffer = 3.5 distance to include entire residues into the environment of the core

- calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to run

- write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to write

- protein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selection

- remove_water = False

- restraints_filename = Auto restraints filename is based on model name if not specified

- cleanup = all *most None

- ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisation

- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3

- capping_groups = True

- include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisation

- include_inter_residue_restraints = False

- do_not_update_restraints = False For testing and maybe getting strain energy of standard restraints

- buffer_selection = None use this instead of distance from selection

- specific_atom_chargesatom_selection = None charge = None

- atom_selection = None

- charge = None

- specific_atom_multiplicitiesatom_selection = None multiplicity = None

- atom_selection = None

- multiplicity = None

- freeze_specific_atomsatom_selection = None

- atom_selection = None

- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- program = *mopac test

- charge = Auto

- multiplicity = Auto

- method = Auto

- basis_set = Auto

- solvent_model = None
- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False
- correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None
- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None
- default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = True
- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
- restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False
- Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or
- aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond rest...

- `sort_atoms = True`
- `use_ncs_to_build_restraints = True`
- `show_restraints_histograms = True`
- `flip_symmetric_amino_acids = True`
- `superpose_ideal_ligand = *None all SF4 F3S DVT`
- `disable_uc_volume_vs_n_atoms_check = False`
- `allow_polymer_cross_special_position = False`
- `correct_hydrogens = True`
- `c_beta_restraints = True`
- `use_neutron_distances = False` Use neutron X-H distances (which are longer than X-ray ones)
- `disulfide_bond_exclusions_selection_string = None`
- `exclusion_distance_cutoff = 3` If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- `link_distance_cutoff = 3` Length of link between the linked residues
- `disulfide_distance_cutoff = 3`
- `add_angle_and_dihedral_restraints_for_disulfides = True`
- `dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic`
- `chir_volume_esd = 0.2`
- `max_reasonable_bond_distance = 50.0`
- `nonbonded_distance_cutoff = None`
- `default_vdw_distance = 1`
- `min_vdw_distance = 1`
- `nonbonded_buffer = 1` ****EXPERIMENTAL, developers only****
- `nonbonded_weight = None` Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- `const_shrink_donor_acceptor = 0` ****EXPERIMENTAL, developers only****
- `vdw_1_4_factor = 0.8`
- `min_distance_sym_equiv = 0.5`
- `custom_nonbonded_symmetry_exclusions = None`
- `translate_cns_dna_rna_residue_names = None`
- `proceed_with_excessive_length_bonds = True`
- `restraints_librarycdl = True` Use Conformation Dependent Library (CDL) for geometry
- `restraintsinclude_modified_amino_acid_in_cdl = False` `mcl = True` Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ...`cis_pro_eh99 = True` `omega_cdl = False` Use Omega Conformation Dependent Library (omega-CDL) for geometry `restraintscdl_svl = False` `rdl = False` `rdl_selection = *all TRP` `hpdI = False` `cdl_nucleotidesenable = False` Use RestraintsLib for DNA and RNA for geometry `restraintsesd = *phenix` `csd factor = 2.0` `user_suppliedpath = None` Root directory of user supplied restraints `saction = *pre` post Choice to look here before GeoStd, the default, or after

- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False
- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ...
- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuresss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for

h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown
 bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
 nucleic_acidenabled = True Turn on secondary structure restraints for nucleic
 acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be
 establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The
 smaller number is tighter restraints.base_pairenable = True Restrain this particular base-pairbase1 =
 None Selection string selecting at least one atom in the desired residuebase2 = None Selection string
 selecting at least one atom in the desired residuesaenger_class = 0 Type of
 base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176
 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True
 Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity
 restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenable = True
 Restrain this particular base-pairbase1 = None Selection string selecting at least one atom in the desired
 residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma =
 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Turn on secondary structure restraints for protein

- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None
- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value

- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restrict this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrict hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrict angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restrict this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenabled = True Restrict this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrict hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrict angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- enabled = True Restrict this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pair
- planarity_sigma = 0.176
- restrain_hbonds = True Restrict hydrogen bonds length for this base-pair
- restrain_hb_angles = True Restrict angles around hydrogen bonds for this base-pair
- restrain_parallelity = True Apply parallelity restraint to this base-pair

- parallelity_target = 0
- parallelity_sigma = 0.0335
- stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1.0 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- enabled = False
- exclude_outliers = True
- selection = all
- sigma = 0.2
- limit = 1.0
- top_out = False
- reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- automatic_restraintsside_chain_parallelity = False
- side_chain_parallelity = False
- automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3.0 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False
- link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.
- link_none = False
- link_metals = Auto
- link_residues = True
- link_amino_acid_rna_dna = False
- link_carbohydrates = True

- link_ligands = True
- link_small_molecules = False
- metal_coordination_cutoff = 3.0
- amino_acid_bond_cutoff = 1.9
- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!!
 Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.cis_threshold = 45
 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False
 omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield
 emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed =

True Apply restraints to residues in allowed region on Ramachandran

plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential. oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027

- ramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.

- cis_threshold = 45

- apply_all_trans = False

- discard_omega = False

- discard_psi_phi = True

- apply_peptide_plane = False

- omega_esd_override_value = None

- rama_weight = 1.0

- scale_allowed = 1.0

- rama_potential = *oldfield emsley

- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.

- restrain_rama_outliers = True Apply restraints to Ramachandran outliers

- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot

- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.

- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027

- esd = 10.0

- weight_scale = 1.0

- dist_weight_max = 10.0

- weight = None

- plot_cutoff = 0.027

- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set

up. inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$

weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region. distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region. plot_cutoff = 0.027

emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0
weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random
allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest
highest_probability random weighted_random

- enabled = False
- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$ weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region. distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region. plot_cutoff = 0.027
- weight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$
- weight_scale = 0.01
- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random

- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0 delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9 o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable = True
- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5 bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5 chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5 nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5 dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3 planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions = 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10

- `ncs_group` The definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.
`reference = None` Residue selection string for the complete master
`NCS.copyselction = None` Residue selection string for each NCS copy location in ASU
- `reference = None` Residue selection string for the complete master NCS copy
- `selection = None` Residue selection string for each NCS copy location in ASU
- `ncs_search` Set of parameters for NCS search procedure. Some of them also used for filtering user-supplied `ncs_group`.
`enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
`exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in
`NCS.chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
`chain_max_rmsd = 2.` limit of rms difference between chains to be considered as copies
`residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues
`try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
`minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
`validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 2.` limit of rms difference between chains to be considered as copies
- `residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `clash_guard`
`nonbonded_distance_threshold = -1.0`
`max_number_of_distances_below_threshold = 100000000`
`max_fraction_of_distances_below_threshold = 1.0`
- `nonbonded_distance_threshold = -1.0`
- `max_number_of_distances_below_threshold = 100000000`
- `max_fraction_of_distances_below_threshold = 1.0`
- `geometry_restraints`
`edit`
`excessive_bond_distance_limit = 10`
`bondaction = *add delete change`
`atom_selection_1 = None`
`atom_selection_2 = None`
`symmetry_operation = None`
The bond is between `atom_1` and `symmetry_operation * atom_2`, with `atom_1` and `atom_2` given in fractional coordinates.

Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None

- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0

- excessive_bond_distance_limit = 10

- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z

- distance_ideal = None

- sigma = None

- slack = None

- limit = -1.0

- top_out = False

- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None

- angle_ideal = None

- sigma = None

- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- guiPhenix GUI parameters, not used in command-line programjob_title = None Job title in PHENIX GUI, not used on command lineoutput_dir = None
- job_title = None Job title in PHENIX GUI, not used on command line
- output_dir = None

Ensemble creation with Ensembler

ensembler.html

Author(s)

- ensembler: Gabor Bunkoczi
- PHENIX GUI: Nathaniel Echols

Purpose

Ensembler can be used to superpose multiple chains to be used as an ensemble search model for molecular replacement.

See also

phenix.sculptor and phenix.ensembler have been superseded by phenix.sculpt_ensemble

Usage

Ensembler can be run from the PHENIX GUI and the command line, the only difference being the way commands are taken from the user.

The graphical user interface makes all settings accessible either as part of the main window (for frequently used options) or as a dialog box Ensemble generation settings.... Input files are specified either by the +/- button pair, or by drag-and-drop onto the window area. File types are automatically recognized, and added to the relevant input section.

Input files

- Structures (compulsory). The chains that are to be superposed. Protein chains are automatically identified, other chains are discarded. In case the structure contains multiple chains, these will all be used for superposition. Accepted formats: PDB. Recognized extensions: .pdb, .ent.
- Alignment (optional). In case a user-supplied residue mapping is desired, this can be achieved by selecting the appropriate mode and supplying alignment files. Accepted extensions (with the corresponding format) are .aln (CLUSTAL format), .pir (PIR-format) and .ali (relaxed PIR-like format).

Output files

The superposed chains can be written out either as a quasi multiple model PDB file that is readable by phaser directly (output style merged, output file name root_merged.pdb) or as a series of files containing each chain separately (output style separate, file name root_pdb_chain.pdb, where pdb is the name of the PDB file the chain was read from, and chain is the chain identifier). The file name root can be changed via the root parameter of the output (default: ensemble).

Description

The workflow consists of several stages that can be independently configured. These are listed in order of execution, and control parameters are accessible from the Ensemble generation settings... dialog box. For a

summary of all parameters with the corresponding defaults, see the Additional information section.

Residue mapping

Establishes the equivalence of residues among the input chains. There are several options available:

- `ssm` - Do structure-based SSM alignment (default).
- `muscle` - Generate a sequence alignment with MUSCLE and use that to align the residues.
- `multiple_alignment` - A user supplied multiple alignment is used. An error message will be raised if this is not present or does not cover all present chains.
- `alignments` - A series of alignments is used that all have the same first sequence (i.e. a series of pairwise alignments against the same target sequence). An error message will be raised if these do not cover all present chains.
- `resid` - Residue sequence number with insertion code will be used.

Most common residue mapping mode is `ssm`. If no SSM alignment can be done or this is imprecise (e.g. no secondary structure), `muscle` is a good second choice.

Atom mapping

Maps selected atoms within equivalent residues to each other. The mapping is done by name hence the order of atoms in the residue does not matter. If atoms are missing from certain residues (or if certain residues contain extra atoms), a gap will be filled where necessary. Atom selection is controlled by the `atoms` parameter of the configuration scope. Default atom selection: `CA`.

Superposition

Equivalent positions are superposed iteratively to find a globally optimal solution. There are two superposition algorithms implemented, which primarily differ in how they handle gaps in equivalent positions.

- `gapless`. This algorithm discards all positions where gaps are present, For further details, see Diamond (1992).
- `gapped`. This algorithm can use all positions as long as there are two sites presents (i.e. not gaps), and may give better results for chains with distant sequence identity. For the exact algorithm, see Wang & Snoeyink (2008).

Both algorithms use Diamond's formulation to solve the pairwise rotational superposition problem (Diamond, 1988).

An exception is raised if there are less than 3 sites present for superposition.

Multiple superposition is an iterative process and consists of a series of pairwise superpositions. The convergence criterion is controlled by the `convergence` parameter (in the superposition scope), which is the r.m.s. difference change between two consecutive iterations.

Weighting

Automatic weighting can be used to improve superposition, either to amplify highly homologous regions or to decrease the effect of incorrect site-equivalence (typically arises because of a wrong alignment). Implemented weighting schemes are as follows:

- `unit` - Unit weights (equivalent to no weighting).

unit - Unit weights (equivalent to no weighting).

- **robust_resistant** - Robust-resistant weighting scheme (default). This tends to converge fast and give reliable results. The exact formula for weighting is as follows: $w = 1 - (\delta^2 / \text{tolerance}^2)^2$ if $\delta < \text{tolerance}$ $w = 0$ (otherwise) where δ is the deviation from the average. Tolerance is an empirical value, and its optimal value is close to (unweighted r.m.s.d.)² and can be controlled by the critical parameter of the **robust_resistant** scope.

robust_resistant - Robust-resistant weighting scheme (default). This tends to converge fast and give reliable results. The exact formula for weighting is as follows:

```
w = 1 - ( delta2 / tolerance2 )2 if delta < tolerance
w = 0 (otherwise)
```

where δ is the deviation from the average. Tolerance is an empirical value, and its optimal value is close to (unweighted r.m.s.d.)² and can be controlled by the critical parameter of the **robust_resistant** scope.

Weighting is iterated with superposition until weights converge, which can be controlled by the convergence parameter of the weighting scope.

In case of highly dissimilar structures (or incorrect residue mapping), weight determination may temporarily need to be damped to avoid divergence. This is done automatically (in steps controlled by the **incremental_damping_factor** parameter of the weighting scope), until a preset value (controlled by the **max_damping_factor** parameter of the weighting scope) is reached, at which point an exception is raised.

Cluster analysis

Hierarchical cluster analysis is performed using the pairwise r.m.s. differences as a distance measure. The clustering parameter of the configuration score can be used to adjust cluster boundaries.

Chain trimming

This option trims residues from the final superposed model where the unweighted r.m.s.d. is above a certain threshold (threshold parameter of the trimming scope). Useful in removing flexible loops, etc. Default: no trimming.

Sorting

After superposition is complete, the chains can be sorted by sequence identity (identity), fraction of common sites wrt all aligned atom positions (overlap), weighted r.m.s.d. (wrmsd) or unweighted r.m.s.d. (unwrmsd). This is controlled by the sort parameter of the output scope. Default: input order (input).

Command line

```
phenix.enssembler \
[ command-line switches ] \
[ PHIL-format parameter files ] \
[ PHIL command-line assignments ] \
[ PDB-files ] \
[ alignment files ]
```

Command-line switches:

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
-v, --verbosity set verbosity level (DEBUG, INFO, WARNING, VERBOSE)
```

PHIL arguments:

Everything not starting with a dash('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters, command-line assignment or a file whose type is automatically recognized (based on file extension; structure files and alignment files are recognized automatically).

Specific limitations and possible problems

Processing features

- Ensembler considers each chain individually and therefore it is not possible to superpose assemblies. The rationale is that its primary purpose is ensemble generation for molecular replacement and since intermolecular interactions are weak, assemblies are unlikely to be preserved, and one would obtain better models by an assembly of ensembles of monomers than the other way round.
- Very short residue segments (shorter than 3 consecutive residues) cannot be reliably aligned to the sequence, and these will be discarded from the superposition.

Warning and error messages

- No alignments given: no alignments were specified and a user-supplied alignment based mapping method was requested.
- Unsuitable alignment set: different master sequences: this indicates that the series of input alignments cannot be assembled because of a different master sequence.
- Several alignment files specified; using first one only: this warning indicates that several alignment files were specified, but `multiple_alignment` was selected as mapping mode, which cannot assemble alignments.
- No matching alignment for chain...: no alignment sequence matches chain ... and a user-supplied alignment based mapping method was requested.
- SSM error:...: an error occurred during SSM superposition.
- Less than 2 chains for superposition: superposition cannot proceed because there are less than 2 chains found.
- Less than 3 sites for superposition: superposition cannot proceed because there are less than 3 sites available (this is checked after discarding gaps). If this error occurs, discarding the chain with the lowest overlap (printed just before the error is raised) may allow one to proceed.
- Excessive weight shift, damping weight change: this warning is emitted if weights need to be damped, because all weights are zero.
- Weight damping recovery exhausted: this error is raised if even after successive rounds of re-weighting, all weights are still zero. This either indicates incorrect alignment of the chains or too tight expected error.

References

Additional_information

List of all available keywords

inputInput filesmodel = None Input model filealignment = None Input alignment fileoutputOutput file(s)location = "" gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.root = ensemble Output file rootstyle = *merged separate Output options for ensemblesort = *input identity overlap wrmsd unwrmsd Sort ensemble componentsjob_title = None Job title in PHENIX GUI, not used on command lineconfigurationGeneral parameters for ensemble generationmapping = resid alignments multiple_alignment muscle *ssm Residue correspondence mappingatoms = CA Include atom in superpositionsclustering = 0.5 Cutoff distance for cluster analysistrim = False Trim differing loopssuperpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superpositionweightingParameters for weighting schemescheme = unit *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping factor is aboverobust_resistantSetting for robust-resistance schemecritical = 9 tolerancetrimmingTrimming parametersthreshold = 3.0 Max position RMSD for inclusion

- inputInput filesmodel = None Input model filealignment = None Input alignment file
- model = None Input model file
- alignment = None Input alignment file
- outputOutput file(s)location = "" gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.root = ensemble Output file rootstyle = *merged separate Output options for ensemblesort = *input identity overlap wrmsd unwrmsd Sort ensemble componentsjob_title = None Job title in PHENIX GUI, not used on command line
- location = ""
- gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.
- root = ensemble Output file root
- style = *merged separate Output options for ensemble
- sort = *input identity overlap wrmsd unwrmsd Sort ensemble components
- job_title = None Job title in PHENIX GUI, not used on command line
- configurationGeneral parameters for ensemble generationmapping = resid alignments multiple_alignment muscle *ssm Residue correspondence mappingatoms = CA Include atom in superpositionsclustering = 0.5 Cutoff distance for cluster analysistrim = False Trim differing loopssuperpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superpositionweightingParameters for weighting schemescheme = unit *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping factor is aboverobust_resistantSetting for robust-resistance schemecritical = 9 tolerancetrimmingTrimming parametersthreshold = 3.0 Max position RMSD for inclusion
- mapping = resid alignments multiple_alignment muscle *ssm Residue correspondence mapping
- atoms = CA Include atom in superpositions
- clustering = 0.5 Cutoff distance for cluster analysis
- trim = False Trim differing loops
- superpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superposition

- method = *gapless gapped Superposition algorithm
- convergence = 1.0E-4 Convergence criterion for superposition
- weightingParameters for weighting schemescheme = unit *robust_resistant Weighting
schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor =
1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping
factor is above
robust_resistantSetting for robust-resistance schemecritical = 9 tolerance
- scheme = unit *robust_resistant Weighting scheme
- convergence = 1.0E-3 Convergence criterion for weight iteration
- incremental_damping_factor = 1.5 Damping factor in recovery cycle
- max_damping_factor = 3.34 Quit recovery if cumulative damping factor is above
- robust_resistantSetting for robust-resistance schemecritical = 9 tolerance
- critical = 9 tolerance
- trimmingTrimming parametersthreshold = 3.0 Max position RMSD for inclusion
- threshold = 3.0 Max position RMSD for inclusion

Ensemble creation with Ensembler

ensembler_new.html

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Purpose

Ensembling is a superposition of multiple homologous structures, which often gives superior performance when used as search model for molecular replacement.

Usage

There is a standalone ensembler_new program available from the command line. The unified model preparation program sculpt_ensemble is available from the command line and also from the PHENIX GUI.

Input files

- Structures (compulsory). The chains that are to be superposed. Protein/DNA/RNA chains are automatically identified, other chains are not used in the superposition step. These non-recognized chains are either discarded, or if these are associated with a chain that is used for superposition (this is governed by whether these chains have the same chainID), these can also be transformed by the appropriate transformation matrix. In case the structure contains multiple chains, these are either superposed as independent entities (default), or as an assembly (e.g. a tetramer) - this is governed by the assembly keyword (defines a group of chains that should be superposed as a group - this can be used multiple times) or the read_single__assembly_from_file keyword (to treat the contents of the file as a single assembly). Accepted formats: PDB. Recognized extensions: .pdb, .ent.
- Alignment (optional). In case a user-supplied residue mapping is requested, this keywords allows the input for alignment files. In case assemblies are also defined, the user-supplied residue mapping option (and the corresponding alignment file) can be configured on an assembly-member basis, and this can be achieved by listing the mapping option/corresponding alignment in the same order as the assembly members. The program recognizes alignment files from their extensions [.aln (CLUSTAL format), .pir (PIR-format) and .ali (relaxed PIR-like format) are accepted], but in case multiple alignment input is required, it has to be used with a full assignment syntax, e.g. alignment="my1.aln my2.aln".

Output files

The superposed chains can be written out either as a quasi multiple model PDB file that is readable by phaser directly (output style merged, output file name root_merged.pdb) or as a series of files containing each chain separately (output style separate, file name root_pdb_chain.pdb, where pdb is the name of the PDB file the chain was read from, and chain is the chain identifier). The file name root can be changed via the root parameter of the output (default: ensemble).

Description

The workflow consists of several stages that can be independently configured. These are listed in order of execution. For a summary of all parameters with the corresponding defaults, see the Additional information section.

Model input

The syntax allows the definition of structural assemblies in a flexible manner, and is also backwards-compatible with the previous version, i.e. existing command lines do not have to be changed.

Single-chain structures

These only require the input of files that contain the structures to be superposed. In case multiple chains are present in a file, these are superposed independently, but it is also possible to restrict participating chains to a selection (using the selection keyword).

For example, to superpose all chains in a.pdb with all chains in b.pdb, the following PHIL is needed (these can also be specified as command-line arguments):

```
input.model.file_name=a.pdb
input.model.file_name = b.pdb
```

To superpose only chains A and B from a.pdb with chains A and C from b.pdb, the selection keyword can be used (please note that this can now longer be achieved from the command line):

```
input {
  model {
    file_name = a.pdb
    selection = chain A or chain B
  }

  model {
    file_name = b.pdb
    selection = chain A or chain C
  }
}
```

Multimeric assemblies

In this case the individual assemblies need to be specified using either the assembly keyword or the read_single_assembly_from_file keyword if the respective structure file contains a single assembly only.

In case the input PDB file contains multiple MODEL sections, these are automatically treated as separate assemblies and no further action is required. However, it is still possible to omit chains from the assembly by listing only the chainIDs on the assembly keyword. On the other hand, the read_single_assembly_from_file is not applicable to such a PDB file, and will result in an error message.

For example, the structure 4QTR contains two copies of a protein-DNA complex: protein chains A and B interact with DNA chains E and F, while protein chains C and D interact with DNA chains G and H.

To superpose these two assemblies, the following input is required:

```
input {
  model {
    file_name = 4qtr.pdb
    assembly = A B E F
    assembly = C D G H
  }
}
```

To superpose a subset of the assemblies, it is sufficient to specify the subset to be superposed. Additional chains are automatically discarded. For example, to superpose a partial assembly from the previous example, one can use the following command file:

```
input {
  model {
    file_name = 4qtr.pdb
    assembly = A E F
    assembly = C G H
  }
}
```

If the assemblies to be superposed are present in separate files, one can use a more convenient syntax:

```
input {
  model {
    file_name = 4qtr_1.pdb
    read_single_assembly_from_file = True
  }
  model {
    file_name = 4qtr_2.pdb
    read_single_assembly_from_file = True
  }
}
```


Input summary

The program lists the chains that were read from input files, and also displays the identified macromolecule type. It also does a consistency check to ensure that all assemblies have the same number of chains and these are of identical types.

Residue mapping

Establishes the equivalence of residues among the input chains. There are several options available:

- `ssm` - Do structure-based pairwise SSM alignment, using the first chain as reference (default).
- `ssm_multiple_alignment` - Do structure-based multiple SSM alignment.
- `muscle` - Generate a sequence alignment with MUSCLE and use that to align the residues.
- `multiple_alignment` - A user supplied multiple alignment is used. An error message will be raised if this is not present or does not cover all present chains.
- `resid` - Residue sequence number with insertion code will be used.

Residue mapping can be requested on an assembly-member basis, by specifying the requested options in the same order as the chains of the assembly. This is not compulsory, and if no options are specified, a default dependent on the macromolecule type will be used (for protein, the default is `ssm`, for nucleic acids it is `muscle`). It is also possible to request specific mapping algorithms for a subset of all chains of the assembly, but in this case these have to come first in the list.

For example, to request non-default mapping options for the 4QTR example, the following options are needed (input section repeated for convenience):

```
input {
  model {
    file_name = 4qtr.pdb
    assembly = A B E F
    assembly = C D G H
  }
}

configuration {
  mapping = ssm muscle muscle resid
}
```

This will use the `ssm` option for chains A and B, `muscle` for chains B and D, `muscle` for chains E and G and `resid` for chains F and H. Note that these coincide with the defaults for chains A and B (assembly position 1, protein chain) as well as for chains E and G (assembly position 3, DNA chains). To request non-default options only, one has to rearrange the assembly so that positions 2 and 4 in the current setup become positions 1 and 2:

```
input {
  model {
    file_name = 4qtr.pdb
    assembly = B F A E
    assembly = D H C G
  }
}

configuration {
  mapping = muscle resid
}
```

In this case, since no specific mapping is requested for assembly positions 3 and 4, the default options will be used.

Atom mapping

Maps selected atoms within equivalent residues to each other. The mapping is done by name hence the order of atoms in the residue does not matter. If atoms are missing from certain residues (or if certain residues contain extra atoms), a gap will be filled where necessary.

Atom selection is controlled by the atoms parameter of the configuration scope. If multiple atom names are requested, these need to be separated by a comma (.). If no selection is requested, a type dependent on the macromolecule will be used (CA for protein and P for nucleic acids). There are also type-dependent shorthands available (backbone and all).

For assembly structures, it is possible to request atom selection on an assembly-member basis, with the same limitation that was explained in the previous section. As an example, here is a selection for the 4QTR structure:

```
input {
  model {
    file_name = 4qtr.pdb
    assembly = A B E F
    assembly = C D G H
  }
}

configuration {
  atoms = C,CA,N,O backbone all backbone
}
```

This instructs the program to use atoms C, CA, N, O for assembly position 1 (chains A and C), backbone atoms for assembly position 2 (chains B and D; since these are protein chains, the shorthand backbone stands for atoms C, CA, N and O), all atoms for assembly position 3 (chains E and G) and backbone atoms for assembly position 4 (chains F and H; since these are DNA chains, backbone selection includes all atoms from the phosphate and the deoxy-ribose).

Superposition

Equivalent positions are superposed iteratively to find a globally optimal solution. There are two superposition algorithms implemented, which primarily differ in how they handle gaps in equivalent positions.

- gapless. This algorithm discards all positions where gaps are present, For further details, see Diamond (1992).
- gapped. This algorithm can use all positions as long as there are two sites presents (i.e. not gaps), and may give better results for chains with distant sequence identity. For the exact algorithm, see Wang & Snoeyink (2008).

Both algorithms use Diamond's formulation to solve the pairwise rotational superposition problem (Diamond, 1988).

An exception is raised if there are less than 3 sites present for superposition.

Multiple superposition is an iterative process and consists of a series of pairwise superpositions. The convergence criterion is controlled by the convergence parameter (in the superposition scope), which is the r.m.s. difference change between two consecutive iterations.

Weighting

Automatic weighting can be used to improve superposition, either to amplify highly homologous regions or to decrease the effect of incorrect site-equivalence (typically arises because of a wrong alignment). Implemented weighting schemes are as follows:

- unit - Unit weights (equivalent to no weighting).

unit - Unit weights (equivalent to no weighting).

- robust_resistant - Robust-resistant weighting scheme (default). This tends to converge fast and give reliable results. The exact formula for weighting is as follows: $w = 1 - (\text{delta}^2 / \text{tolerance}^2)^2$ if $\text{delta} < \text{tolerance}$ $w = 0$ (otherwise) where delta is the deviation from the average. Tolerance is an empirical value, and its optimal value is close to (unweighted r.m.s.d.)² and can be controlled by the critical parameter of the robust_resistant scope.

robust_resistant - Robust-resistant weighting scheme (default). This tends to converge fast and give reliable results. The exact formula for weighting is as follows:

```
w = 1 - ( delta2 / tolerance2 )2 if delta < tolerance
w = 0 (otherwise)
```

where delta is the deviation from the average. Tolerance is an empirical value, and its optimal value is close to (unweighted r.m.s.d.)² and can be controlled by the critical parameter of the robust_resistant scope.

Weighting is iterated with superposition until weights converge, which can be controlled by the convergence parameter of the weighting scope.

In case of highly dissimilar structures (or incorrect residue mapping), weight determination may temporarily need to be damped to avoid divergence. This is done automatically (in steps controlled by the incremental_damping_factor parameter of the weighting scope), until a preset value (controlled by the max_damping_factor parameter of the weighting scope) is reached, at which point an exception is raised.

Cluster analysis

Hierarchical cluster analysis is performed using the pairwise r.m.s. differences as a distance measure. The clustering parameter of the configuration score can be used to adjust cluster boundaries. Clustering information can be requested for multiple thresholds. Default: no clustering performed.

Chain trimming

This option trims residues from the final superposed model where the unweighted r.m.s.d. is above a certain threshold (threshold parameter of the trimming scope). Useful in removing flexible loops, etc. Default: no trimming.

Sorting

After superposition is complete, the chains can be sorted by sequence identity (identity), fraction of common sites wrt all aligned atom positions (overlap), weighted r.m.s.d. (wrmsd) or unweighted r.m.s.d. (unwrmsd). This is controlled by the sort parameter of the output scope. Default: input order (input).

Command line

```
phenix.enssembler \
[ command-line switches ] \
[ PHIL-format parameter files ] \
[ PHIL command-line assignments ] \
[ PDB-files ] \
```

[alignment files]

Command-line switches

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
-v, --verbosity set verbosity level (info,debug,verbose)
--version show program's version number and exit
--text-logfile FILE Verbatim copy of log stream
--html-logfile FILE Verbatim copy of log stream in HTML
```

PHIL arguments

Everything not starting with a dash('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters, command-line assignment or a file whose type is automatically recognized (based on file extension; structure files and alignment files are recognized automatically).

Specific limitations and possible problems

Processing features

- When an alignment-based residue mapping protocol is requested, very short residue segments (shorter than 3 consecutive residues) cannot be reliably aligned to the sequence, and these will be discarded from the superposition.

Warning and error messages

- No input pdb files: no PDB files were given.
- Cannot treat multiple MODELS as a single assembly: the `read_single_assembly_from_file` keyword was used in connection with a PDB file that contains multiple MODEL records.
- No chains read from input PDB files: no superposable chain was found in input files. Superposable chains are protein, DNA and RNA.
- No chain with allowed type found: no superposable chains found in structure file. Superposable chains are protein, DNA and RNA.
- No chain '..' found: chain '..' (requested as part of an assembly) is not present.
- Inconsistency between assembly 1 and assembly ..: different number of chains: assembly 1 and assembly .. do not have the same number of chains.
- Inconsistency between assembly 1 and assembly ..: different chain types: an assembly position in assembly 1 and assembly .. do not have the same macromolecule type. The information on chains types in the assembly is given prior to this error message.
- Mapping .. is not applicable to chain type ..: the mapping method .. is not applicable to a chain with macromolecule type .. (e.g. SSM superposition can only be used with protein chains).
- No alignment specified: no alignment file was specified and a user-supplied alignment based mapping method was requested.
- No matching alignment found: no alignment sequence matches from the user-supplied alignment matches the chain sequence.
- SSM error:....: an error occurred during SSM superposition.

- No sites selected: the requested atom selection results in no sites selected for superposition.
- Less than 2 chains for superposition: superposition cannot proceed because there are less than 2 chains found.
- Insufficient overlap among structures: superposition cannot proceed because there are less than 3 sites available (this is checked after discarding gaps). If this error occurs, discarding the chain with the lowest overlap (printed just before the error is raised) may allow one to proceed.

Insufficient overlap among structures: superposition cannot proceed

because there are less than 3 sites available (this is checked after discarding gaps). If this error occurs, discarding the chain with the lowest overlap (printed just before the error is raised) may allow one to proceed.

- Excessive weight shift, damping weight change: this warning is emitted if weights need to be damped, because all weights are zero.
- Weight damping recovery exhausted: this error is raised if even after successive rounds of re-weighting, all weights are still zero. This either indicates incorrect alignment of the chains or too tight expected error.

References

Additional information

List of all available keywords

inputInput filesmodelInput model filefile_name = None Input file nameselection = None Selection stringassembly = None Assembly informationread_single_assembly_from_file = False Consider contents of the file as a single assemblyoutputOutput file(s)location = "" gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.root = ensemble Output file rootstyle = *merged separate Output options for ensemblesort = *input wrmsd unwrmsd overlap cluster Sort ensemble componenterror_file_root = None Output file root for superposition error filesretain_associated_chains = False Keep (and transform) chains that are not assembly membersjob_title = None Job title in PHENIX GUI, not used on command lineconfigurationGeneral parameters for ensemble generationmapping = None Residue mapping methods. Valid choices: resid, ssm, ssm_multiple_alignment, muscle, multiple_alignmentalignment = None Alignment file (only necessary for 'alignments' modes)atoms = None Atom to include in superpositionclustering = None Cutoff distance for cluster analysissuperpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superpositionweightingParameters for weighting schemescheme = unit *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping factor is aboverobust_resistantSetting for robust-resistance schemecritical = 3 tolerancetrimmingTrimming parameterstrim = False Trim differing loopsthreshold = 3.0 Max position RMSD for inclusion

- inputInput filesmodelInput model filefile_name = None Input file nameselection = None Selection stringassembly = None Assembly informationread_single_assembly_from_file = False Consider contents of the file as a single assembly
- modelInput model filefile_name = None Input file nameselection = None Selection stringassembly = None Assembly informationread_single_assembly_from_file = False Consider contents of the file as a single assembly
- file_name = None Input file name

- selection = None Selection string
- assembly = None Assembly information
- read_single_assembly_from_file = False Consider contents of the file as a single assembly
- outputOutput file(s)location = "" gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.root = ensemble Output file rootstyle = *merged separate Output options for ensemblesort = *input wrmsd unwrmsd overlap cluster Sort ensemble componentserror_file_root = None Output file root for superposition error filesretain_associated_chains = False Keep (and transform) chains that are not assembly membersjob_title = None Job title in PHENIX GUI, not used on command line
- location = ""
- gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.
- root = ensemble Output file root
- style = *merged separate Output options for ensemble
- sort = *input wrmsd unwrmsd overlap cluster Sort ensemble components
- error_file_root = None Output file root for superposition error files
- retain_associated_chains = False Keep (and transform) chains that are not assembly members
- job_title = None Job title in PHENIX GUI, not used on command line
- configurationGeneral parameters for ensemble generationmapping = None Residue mapping methods. Valid choices: resid, ssm, ssm_multiple_alignment, muscle, multiple_alignmentalignment = None Alignment file (only necessary for 'alignments' modes)atoms = None Atom to include in superpositionclustering = None Cutoff distance for cluster analysissuperpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superpositionweightingParameters for weighting schemescheme = unit *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping factor is above robust_resistantSetting for robust-resistance schemecritical = 3 tolerancetrimmingTrimming parameterstrim = False Trim differing loopsthreshold = 3.0 Max position RMSD for inclusion
- mapping = None Residue mapping methods. Valid choices: resid, ssm, ssm_multiple_alignment, muscle, multiple_alignment
- alignment = None Alignment file (only necessary for 'alignments' modes)
- atoms = None Atom to include in superposition
- clustering = None Cutoff distance for cluster analysis
- superpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superposition
- method = *gapless gapped Superposition algorithm
- convergence = 1.0E-4 Convergence criterion for superposition
- weightingParameters for weighting schemescheme = unit *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping factor is above robust_resistantSetting for robust-resistance schemecritical = 3 tolerance

- scheme = unit *robust_resistant Weighting scheme
- convergence = 1.0E-3 Convergence criterion for weight iteration
- incremental_damping_factor = 1.5 Damping factor in recovery cycle
- max_damping_factor = 3.34 Quit recovery if cumulative damping factor is above
- robust_resistantSetting for robust-resistance schemecritical = 3 tolerance
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- trimmingTrimming parameterstrim = False Trim differing loopsthreshold = 3.0 Max position RMSD for inclusion
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Overview of molecular replacement in Phenix

mr_overview.html

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- Steps in structure solution with molecular replacement Model selection Molecular replacement Input files and mandatory parameters Outline of MR procedure Limitations
- Model selection
- Molecular replacement Input files and mandatory parameters Outline of MR procedure Limitations
- Input files and mandatory parameters
- Outline of MR procedure
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- Reference

Molecular replacement (MR) is a phasing method that uses prior information in the form of known structures that are related or homologous to components in the crystal. Because it requires no additional experimental procedures or data, and additionally simplifies model-building, MR is usually the method of choice for structure determination when a suitable search model is available.

For new users, we recommend reading this document first, followed by the Phaser-MR tutorial for a step-by-step introduction to running MR in the Phenix GUI.

Steps in structure solution with molecular replacement

Model selection

Structures of suitable models are normally found via homology searches (using e.g. NCBI Blast for reasonably close relatives or HHpred for more distant relatives), and can be characterized by the sequence identity with the target sequence. The main restriction on the use of molecular replacement is the requirement for a suitably similar search model. Although there is no exact rule for this, the relationship between sequence identity and MR success is roughly as follows: Better than 40% identity: usually easy, unless large conformational changes are involved. 30-40%: MR usually possible, but sometimes more difficult. 20-30%: MR usually difficult if at all possible, careful model search and preparation required. Less than 20%: MR unlikely to work, but MR-Rosetta can help in marginal cases. In terms of RMSD, above 2.5Å is very unlikely to work, while 1.5Å or less is preferable. Structures that undergo large conformational changes may need to be split into separate domains for searching, regardless of sequence identity. Where multiple similar search models are available, combining these into an ensemble will often improve the likelihood of success, particularly if the trimming option in Ensembler is used to trim the ensemble back to its conserved core. In Phenix, the Sculptor and Ensembler utilities are available for preparing search models. Sculptor can be used to improve the model by pruning residues and side chains according to an alignment between the model structure and the target sequence, and also to apply B-factor weights to the model to weight down unreliable parts. Ensembler can be used to superpose homologous models automatically. The recommended order is to run Sculptor and then Ensembler, for reasons explained in the MR FAQ.

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Molecular replacement

The procedure to place a search model is roughly divided into two steps, a rotation function (RF) to determine its orientation, and a translation function (TF) to determine its absolute position in the unit cell. Multiple components can be placed sequentially to solve the structure of a complex or the structure of a crystal containing multiple copies in the asymmetric unit. In Phenix, MR is performed by the program Phaser, written by Randy Read's group at the University of Cambridge. Although Phaser may be run on the command line with CCP4-style inputs, we recommend using the Phaser-MR GUI. This GUI requires manual preparation and specification of search models, but enables fine control over parameters and multi-step searches, which may be necessary for difficult structures. See the Phaser-MR GUI manual for additional details specific to that interface. (Phaser is also used for experimental phasing, but this functionality is exposed through the AutoSol wizard and Phaser-EP GUI.)

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Input files and mandatory parameters

All of the MR programs in Phenix require a single reflections file containing experimental data (with sigmas); the Phaser-MR GUI will accept any file format or data type, including intensities. The procedure traditionally uses all reflections, so R-free flags are not required. (Unlike refinement, this does not significantly bias the final R-free value, since there are so few degrees of freedom: six to rotate and translate each molecule, and one each for the overall B-factor and estimated RMS error.)

At least one search model is required; in the context of Phaser the search model is usually referred to as an "ensemble". In many cases this will be a single PDB file containing one structure. For more distant models, an actual ensemble model may be used instead - either a PDB file with multiple MODEL records, or multiple similar PDB files. When using a multi-model ensemble, all models must be superposed in the same orientation; the Ensembler utility is used for this. There are no limitations on the size, complexity, or number of search models. For complexes or multimers, if the relative positions of individual subunits do not change, the entire assembly (for instance, a ribosomal subunit) may be used as a search model instead of placing each component separately.

Another option is to search using electron density (or rather, an MTZ file containing pre-weighted map coefficients), often solved at low resolution or obtained from cryoEM image reconstruction. This requires additional information about the center and extent of the map section to search with. It is best to prepare this MTZ file using the `phenix.cut_out_density` tool, making sure that the density is placed in a unit cell that is at least 2.5 times the x, y and z extents of the density.

The maximum likelihood phasing methods used in Phaser require prior knowledge about the deviation (or variance) of the search model(s) from the real structure, and the expected ASU contents or scattering mass of the crystal. To specify the model variance, either an RMSD value or percent sequence identity may be used (these will be converted internally, using relationships determined from a database of test calculations). It is important to minimize the variance if possible (see Model selection for guidelines), which often requires eliminating atoms or modifying B-factors. The Sculptor utility will perform this step, given a PDB file and a sequence alignment. (This is usually unnecessary for search models with high sequence identity to the target molecule, however, especially at lower sequence identity, processing models with the Sculptor utility is highly recommended.) Additionally, the Ensembler utility can trim loops that deviate among members of the ensemble, leaving only the conserved core.

For ASU contents, you may supply a sequence file (protein or nucleic acid), or simply enter the molecular weight. The standalone versions of Phaser also accept the fractional ASU contents of each search ensemble, if known. Note that the ASU contents data does not necessarily have a 1:1 correspondence with the search ensembles (see MR FAQ for details); if you wish to simply specify the molecular weight of a complex while searching for multiple subunits, you may enter the mass for the complex as a single component. (You may also use multi-record FASTA-format sequence files.) Even if you are only searching for a single ensemble out of several (e.g. the protein in a protein-DNA complex), you must still supply the expected ASU contents of the entire crystal because Phaser needs to know what fraction of the asymmetric unit each search model comprises.

Outline of MR procedure

The automated molecular replacement method in Phaser involves several discrete steps:

Anisotropy correction: scales reflections as necessary to overcome anisotropy (weak data in a particular direction). Translational non-crystallographic symmetry (tNCS) correction: checks for the presence of tNCS. If present, parameters describing the translation and small orientation differences between copies are determined and used to compute correction factors. Rotation function: identifies possible orientations of the model. Translation function: given the orientation(s) from the RF, finds the absolute position(s) in the unit cell. Packing analysis: filters TF results based on number of clashes between atoms, given a certain cutoff. Refinement and phasing: performs simple rigid-body refinement of the placed molecules, and calculates phases from the final solution. Log-likelihood gain calculation: determines the final LLG, which can be used to evaluate the success of MR.

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- Refinement and phasing: performs simple rigid-body refinement of the placed molecules, and calculates phases from the final solution.
- Log-likelihood gain calculation: determines the final LLG, which can be used to evaluate the success of MR.

If multiple search models are used, these steps will be performed sequentially for each model. Although each step may be run individually in the Phaser-MR GUI, this is necessary only in exceptionally difficult cases. In fact, it is usually better to allow Phaser to work out an optimal search order and, if necessary, test different choices of search order.

Limitations

You should watch out for otherwise valid solutions being thrown out because of packing clashes due to model deviations and/or extra residues. Phaser will attempt to resolve clashes by pruning segments of chain that refine to low occupancy values. If this does not succeed, manually removing the offending residues or lowering the packing cutoff can circumvent the problem.

For cases where anomalous data from a SAD experiment are available, a poor (but genuine) MR solution may be used to identify heavy-atom sites and combined with SAD phases, a technique known as MR-SAD. This may provide a decent-quality map where neither technique is independently sufficient. At the end of MR in the Phaser-MR GUI, if SAD data are available a button will be presented allowing MR-SAD in the AutoSol wizard to be launched easily.

Automated Molecular replacement

The MRage framework integrates model processing steps and molecular replacement into one application. It provides a highly customizable interface with parallelization options and the ability to connect to the NCBI Blast and wwPDB websites to perform homology searches and fetch potential models. A separate MRage tutorial is available on the Phaser home page.

Frequently Asked Questions

These are now covered on a separate section.

Reference

Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. J Appl Crystallogr 40, 658-674 (2007).

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Molecular replacement and autobuilding using Phaser, Rosetta, and autobuild with mr_rosetta

mr_rosetta.html

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- Environment problems when running mr_rosetta
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- Specific limitations and problems
- Literature
- List of all available keywords

Author(s)

- mr_rosetta: Tom Terwilliger, Frank DiMaio, Randy Read, David Baker

Purpose

mr_rosetta is a procedure for extending the range of molecular replacement by combining tools from the structure-modeling field (Rosetta) with crystallographic molecular replacement, model-building, density modification and refinement. The approach is described in Dimaio et al. (2011). It can also be used to rebuild a model with a combination of Rosetta and Phenix tools.

A key requirement for using mr_rosetta is that you have to have a sequence alignment of the protein used as a template to model your target protein. You can try several different alignments, but a good alignment has to be in your set of alignments or the procedure will be unlikely to be successful. The reason is that Rosetta homology modeling makes strong use of the sequence, so if your alignment is incorrect you are essentially trying to build the wrong molecule.

The basic process is to find MR solutions with Phaser, rebuild them with Rosetta, then rebuild those models with phenix.autobuild. The combination of Rosetta rebuilding and phenix rebuilding is the key part of this method. In slightly more detail, this process is to select possible MR solutions (one of which must later be shown to be correct for the procedure to succeed) with Phaser, score with LLG following Rosetta relaxation, pick the best solutions, rebuild each of these with Rosetta including map information (density term), score the resulting models with Rosetta, select the highest and score with LLG, verify that the top solutions are all about the same (electron density maps are correlated), and rebuild the top models with autobuild.

During both the Rosetta and phenix.autobuild model-building the starting model may be modified by insertion and deletion of loops based on the sequence provided. The extent of loop fitting will normally depend on the quality of the map that is obtained by mr_rosetta.

mr_rosetta can handle a single copy of a single chain, or multiple copies of a single chain (NCS), or multiple copies of multiple chains (groups of NCS). If you supply one or more input search models, then the entire crystallographic asymmetric unit must contain some multiple of the search models you supply (Phaser will be used to find copies of the search model). NCS will be found automatically in your search model and in any models assembled by mr_rosetta.

NOTE: if your molecule has multiple chain types, then you cannot use the simple hhr file input (see below) and you cannot automatically run pre-refinement with Rosetta on your molecule. Instead you need to use mr_model_preparation and Phaser to place your model. Then you can supply the aligned, placed structure to mr_rosetta for rebuilding. Additionally in this case you will need to supply a different set of fragments files for each chain type.

Tools from Rosetta that are used in mr_rosetta

- Calculation of Rosetta energies for a model.

- Model relaxation (rebuilding) including just the Rosetta standard energy term. This optional rebuilding can be carried out before molecular replacement, improving the starting model.
- Model completion and relaxation (filling in missing sections and rebuilding) with Rosetta energies and fit to a map (Dimaio et al., 2009). This step is carried out after an MR solution has been found. The placed model is used to calculate a 2Fo-Fc map and fit to this map is used as a target in addition to the standard Rosetta energies. Note: Sequence-specific fragment libraries are used in this process (obtained from the Robetta webserver).

Steps where mr_rosetta uses structure-modeling algorithms

- Prior to MR. Edited templates are optionally rebuilt before carrying out molecular replacement.
- In scoring MR solutions. MR solutions are scored by Phaser LLG scores calculated after model relaxation with Rosetta including the term for fit to the current density map.
- In rebuilding MR solutions. Initial molecular replacement solutions are rebuilt using Rosetta model completion and relaxation including the term for fit to the current density map

Summary of the procedure used in mr_rosetta

The overall process in one cycle of mr_rosetta is: (a) edit the model and place it in the unit cell (e.g., MR, molecular replacement), (b) score all MR solutions and take the best ones by LLG for further steps, (c) rebuild each model 20-2000 times using Rosetta and density-modified 2Fo-Fc map to yield Rosetta models, (d) refine Rosetta models, average density from top 20%, continue rebuilding each Rosetta model using averaged density, and (e) take top models based on LLG score and rebuild with autobuild. An optional prerefinement step is to carry out Rosetta modeling in step (a) above, before carrying out molecular replacement.

Details of the procedure used in mr_rosetta

- Check installation and verify that Rosetta binary and libraries are available (can specify with keywords rosetta_path (overall path to rosetta directories), and the keywords (paths relative to overall rosetta_path) rosetta_binary_dir, rosetta_binary_name, rosetta_script_dir, rosetta_database_dir)
- Read reflections file (must be CCP4 mtz and have a freeR set; use phenix GUI or import_and_add_free to set up)
- An optional model editing step. If an hhpred .hhr file is supplied, read through this file, download the PDB files specified, apply alignments specified to generate pairs of alignment files and edited models (e.g., 2cng.ali, 2cng_mr.pdb based on 2cng.pdb). This step is carried out by sculptor using a default protocol.

NOTE 1: For tailoring of this step, use mr_model_preparation and then supply the aligned model to mr_rosetta.

NOTE 2: If your structure contains more than one chain or requires more than one homology model to represent the structure, then you need to use mr_model_preparation and Phaser to place your model. Then you can supply the aligned, placed structure to mr_rosetta for rebuilding.

- NOTE: The steps below are carried out for each model/alignment file pair supplied, (or for each pair generated by mr_rosetta if an hhpred .hhr file is supplied with alignment information).
- Check model, alignment file and sequence file to verify that they match (i.e., that the alignment file can be applied to the model to yield a model with the sequence in the sequence file). Copy overall B-factor and list of B-factors for atoms from the model to be substituted into subsequent models before scoring.

Rewrite the sequence file into standard (fasta) format.

- Optional Rosetta modelling prior to molecular replacement (prerefinement). The edited starting model (template) is rebuilt with Rosetta using the standard Rosetta energy functions and a fragments library specific to the target sequence to fill in gaps. (If no gaps are present, no fragments files are necessary).
- Run Phaser to find molecular replacement solutions (default number_of_output_models=5).
- Refine (if refine_after_mr=True) each MR solution with refine. Use resulting 2mFo-DFc map as starting point for density modification, yielding density-modified current density map for this refined solution, to be used in Rosetta rebuilding below. Ignore refined model. Note (but ignore) LLG of refined model.
- Determine the baseline LLG for model improvement. This LLG can either be the LLG obtained in the previous step, or an LLG obtained after rebuilding the MR model with Rosetta. Optionally perform rosetta relaxation (rebuilding rescore_mr.nstruct models, typically 5) models without filling in missing sections, including a density term from the map in the previous step. The sequence of all parts of the model at this point will match the target sequence. Score each solution by LLG. These models are not carried forward, but taking the best LLG for any rebuilt model as the score for the original model from MR. This score will be used later to prioritize MR solutions for further analysis.
- Sample relaxation script used to run relaxation in Rosetta:

```
#!/bin/sh
cd MR_ROSETTA_1/RESCORE_MR_1/RELAX_AND_SCORE_IN_SETS_1/RUN_1/WORK_1
/net/terwill/rosetta/rosetta_source/bin/mr_protocols.default.linuxgccrelease \
-database /net/terwill/rosetta/rosetta_database \
-MR:mode cm \
-in:file:extended_pose 1 \
-in:file:fasta MR_ROSETTA_1/WORK_1/EDITED_1crb_fasta.txt \
-in:file:alignment MR_ROSETTA_1/WORK_1/EDITED_1crb_2qo4.ali \
-in:file:template_pdb MR_ROSETTA_1/AutoMR_run_1/2Q04.1.pdb \
-relax:default_repeats 4 \
-relax:jump_move true \
-edensity:mapreso 3.00 \
-edensity:grid_spacing 1.5 \
-edensity:mapfile \
MR_ROSETTA_1/AutoMR_run_1/2Q04.1_refine_001_map_coeffs.map \
-edensity:sliding_window_wt 1.0 \
-edensity:sliding_window 5 \
-cm:aln_format grishin \
-MR:max_gaplength_to_model 0 \
-nstruct 1 \
-ignore_unrecognized_res \
-overwrite
```

- Take the top (max_solutions_to_rebuild=5) models from step 5 and rebuild them with Rosetta, this time filling in missing sections and including the density term with the current map (the same map as used above in relaxation) as part of the target function, generating total of (rosetta_rebuild.nstruct=20) rebuilt models. These are Rosetta models. The sequence of these models will be the same as that of the target (unless there are long gaps in the template that cannot be filled by Rosetta). Note: in many cases 20 models is sufficient...in others far more models will make the method work better (i.e., 1000 or 2000 models). This can take a lot of time unless you have a cluster to run on. This Rosetta rebuilding step uses a library of fragments specific to the target sequence to fill in any gaps. (If no gaps are present, no fragments files are necessary). Sample rebuild script used:

```
#!/bin/sh
cd MR_ROSETTA_1/WORK_1/REBUILD_IN_SETS_1/RUN_1/WORK_1
/net/terwill/rosetta/rosetta_source/bin/mr_protocols.default.linuxgccrelease \
-database /net/terwill/rosetta/rosetta_database \
-MR:mode cm \
-in:file:extended_pose 1 \
-in:file:fasta MR_ROSETTA_1/WORK_1/EDITED_1crb_fasta.txt \
```



```

-in:file:alignment MR_ROSETTA_1/WORK_1/EDITED_1crb_2qo4.ali \
-in:file:template_pdb MR_ROSETTA_1/AutoMR_run_1/2Q04.1.pdb \
-loops:frag_sizes 9 3 \
-loops:frag_files inputs/aa1crb_09_05.200_v1_3.gz \
  inputs/aa1crb_03_05.200_v1_3.gz none \
-loops:random_order \
-loops:random_grow_loops_by 5 \
-loops:extended \
-loops:remodel quick_ccd \
-loops:relax relax \
-relax:default_repeats 4 \
-relax:jump_move true \
-edensity:mapreso 3.00 \
-edensity:grid_spacing 1.5 \
-edensity:mapfile MR_ROSETTA_1/AutoMR_run_1/2Q04.1_refine_001_map_coeffs.map \
-edensity:sliding_window_wt 1.0 \
-edensity:sliding_window 5 \
-cm:aln_format grishin \
-MR:max_gaplength_to_model 8 \
-nstruct 1 \
-ignore_unrecognized_res \
-overwrite

```

- Choose top (percentage_to_rescore=10) rebuilt models based on Rosetta score (including density term) and rescore them based on LLG
- Determine whether the top (number_of_required_cc=5) best LLG score Rosetta models are all similar (map correlation between map for top model with each)
- Refine with phenix.refine the top Rosetta models based on LLG (if refine_top_models.run_refine_top_models=True, percent_to_refine=20). Save new 2mFo-DFc map, density modify the map, and then average the top density-modified maps (top based on Rosetta score) to yield an averaged density map used in the next relaxation step. Ignore the refined model.
- Relax (rebuild) the Rosetta models from into their corresponding density maps from with Rosetta, generating (relax_top_models.nstruct=5) models for each. Score each relaxed model with LLG, take best LLG as score. Save best relaxed model as new solution. Sample relax script used:

```

#!/bin/sh
cd MR_ROSETTA_1/GROUP_OF_RESCORE_MR_ROSETTA_2/RUN_1/RESCORE_MR_1/RELAX_AND_SCORE_IN_SETS_1/RUN_1/WORK_1
/net/terwill/rosetta/rosetta_source/bin/mr_protocols.default.linuxgccrelease \
-database /net/terwill/rosetta/rosetta_database \
-MR:mode relax \
-in::file::s \
  MR_ROSETTA_1/WORK_1/REBUILD_IN_SETS_1/RUN_8/WORK_1/S_2Q04B_0001_edited.pdb \
-relax:default_repeats 4 \
-relax:jump_move true \
-edensity:mapreso 3.00 \
-edensity:grid_spacing 1.5 \
-edensity:mapfile \
  MR_ROSETTA_1/WORK_1/REBUILD_IN_SETS_1/RUN_8/WORK_1/S_2Q04B_0001_edited_refine_001_map_coeffs.map \
-edensity:sliding_window_wt 1.0 \
-edensity:sliding_window 5 \
-nstruct 1 \
-overwrite

```

- Take the top (number_to_autobuild=5) relaxed refined rebuilt Rosetta models from the previous step (scored by LLG) and rebuild them with phenix.autobuild. Report the R/freeR of each model.

Notes on the procedure used in mr_rosetta

Viewing solutions and restarting with saved solutions

At each stage, existing solutions are saved as a python "pkl" file and can be read back in to mr_rosetta with "mr_rosetta_solutions=xxx.pkl". These solutions can be displayed with "display_solutions=True". Existing solutions are stored as "mr_rosetta_solution" objects which keep track of the model and its history, the map_coefficients and labels, etc. These can be read in to mr_rosetta with the keyword "mr_rosetta_solutions=results.pkl" and used as inputs for subsequent runs, starting at any step that can use those solutions.

NOTE: You can re-start mr_rosetta only at the beginning of major stages (like "place_model", "rosetta_rebuild" etc)...but not in between.

Normally at the end of a major stage a .pkl file is written out with text like "type this to see all the results". You can almost always give your original command, the command "start_point=xxx" and "mr_rosetta_solutions=my_pickle_file.pkl" and it should then continue on from there.

Running mr_rosetta on a cluster

Jobs can be run on a single machine or on a cluster. A run command for single jobs (single_run_command="sh") and a run command for batch jobs (group_run_command=qsub) can be specified as well as the number of processors to use (nproc=200).

The qsub command is used in Sun Grid Engine clusters. You can also use mr_rosetta on a Condor cluster, using group_run_command="condor_submit".

Single file system required for mr_rosetta

All files are stored on a single file system that must be accessible to all jobs.

Read/write delay allows for slow NFS disks

Read/write to files are (generally) accompanied by a wait for appearance of the new file of up to max_wait_time=100 sec.

Tracking your log files

mr_rosetta runs all cpu-intensive jobs as sub-processes. When it submits a sub process to do the work it lists the name of the corresponding log file. You can work your way down to the bottom level at any time by reading through these log files, copying the name of the next log file, and opening it until you get to the place where the actual work is done.

Re-running parts of your mr_rosetta jobs

Sub-processes are always run in sub-directories. Each sub-process has a file "RUN_FILE_1" that contains the information to run the sub-process, a parameter file PARAMS_1.eff and a log file "RUN_FILE_1.log" with the log file of running that sub-process. Note that you can use the parameters files to re-run any jobs that you want. You can say something like:

```
phenix.mr_rosetta PARAMS_1.eff
```

and that will rerun the job specified in that directory.

Failures in sub-processes

If some sub-processes fail, normally the failures will be ignored. This is useful as your overall job can often continue even if a few refinement or rosetta jobs fail. However if the failure is from the queueing system (rather than in the actual running of the jobs) then the overall job may still fail.

Stopping mr_rosetta

If you create a file "STOPWIZARD" in the top level directory (i.e., MR_ROSETTA_1/), then each job in the entire process will stop as soon as any Phenix part of the process takes over (i.e., as soon as Rosetta jobs finish).

Ignoring long-running place_model jobs and going on

There are two ways to avoid the problem of having one or more long-running (and very likely eventually unsuccessful) place_model sub-processes that prevent mr_rosetta from going on. One way is to set the parameter sufficient_number_finished=nn, where you are satisfied if any nn place_model jobs finish successfully. Then once nn jobs finish, all the rest are simply ignored and mr_rosetta goes on. The jobs that are ignored will continue on until they finish (and will still be ignored.)

A second way is to edit the value of sufficient_number_finished after mr_rosetta has started. This is convenient if you see that all the jobs except for one or two are done, and these seem to be going on forever. You can create a little file "GO_ON" in the directory where place_model is being run that contains the value you want for sufficient_number_finished (if you want it to stop right away and at least one job is finished, just put in 1). This directory is the directory where the log files for place_model are located. For example your overall log file might say,

```
Splitting work into 2 jobs and running with 2 processors using sh
background=True in /Users/terwill/unix/misc/junk/test_place_model/MR_ROSETTA_7/GROUP_OF_PLACE_MODEL_1
```

in which case that directory is where you would put GO_ON, where the file GO_ON just contains a number (the value for sufficient_number_finished). As in the use of sufficient_number_finished, the ignored jobs do not actually stop, so if you really want to stop them you will need to do that in another way (mr_rosetta does not capture the job numbers for sub-processes so it can't stop them, it can only monitor what they have done.)

NOTE: you can use this second method with the GO_ON file to tell mr_rosetta how many finished jobs to require for any set of sub-processes. Just put this file in the directory specified for that group of sub-processes. This also works for other phenix tools including autobuild, ligandfit and find_all_ligands that have the 'Splitting work into...' text in their log files.

Installing Rosetta for use with mr_rosetta

To run phenix.mr_rosetta, you must have installed Rosetta, software developed from the Baker laboratory at the University of Washington.

See the central installation notes for Rosetta

Setting up for a run of mr_rosetta A. Fragment files from the Robetta server

NOTE: For versions 3.6 and later of Rosetta, fragments files are optional.

To run mr_rosetta on your structure, you will need to use the Robetta fragment server at the Univ. of Washington to generate 9-mer and 3-mer fragments from the PDB that are compatible with your sequence file. This takes a few hours but is very easy to do.

To obtain the two optional fragments files:

- go to: <http://robetta.bakerlab.org/fragmentsubmit.jsp>
- register
- paste your sequence file into the form
- Receive an email from the server after a few hours that our files are ready
- Download the files (two files, with similar filenames, one containing a 9 and one a 3 like: aat000_09_05.200_v1_3.gz and aat000_03_05.200_v1_3.gz)
- These are your fragment files. They are optional in your mr_rosetta parameters file
- NOTE1: if your chain has more than 650 residues, then you will need to split it up into pieces of 650 residues or fewer before submitting the sequence to the Robetta server. Then you will get several 3-mer and 9-mer fragments files, one for each piece that you submit. You can then simply paste these together after editing all but the first to fix the residue numbers.
- NOTE2: if you have multiple chain types in your structure then you will want to have a separate set of fragments files for each chain type. You can specify these with: fragment_files_chain_list, fragment_files_3_mer_by_chain, and fragment_files_9_mer_by_chain instead of fragment_files. Use fragment_files_chain_list to define which chain ID each of your fragment_files_3_mer_by_chain and fragment_files_9_mer_by_chain go with. In this case you must make sure that there are no chains in your model other than the ones that you specify with fragment_files_chain_list (no waters, no ligands etc).
- NOTE3: You only need one set of fragments files for each UNIQUE chain. So if chains A and C are the same, you just need to specify fragments for chain A. If you have two different chains A and B and fragment files frag_A_3 frag_A_9 frag_B_3 frag_B_9 then you should use: fragment_files_chain_list=A fragment_files_chain_list=B fragment_files_3_mer_by_chain=frag_A_3 fragment_files_9_mer_by_chain=frag_A_9 fragment_files_3_mer_by_chain=frag_B_3 fragment_files_9_mer_by_chain=frag_B_9

Setting up for a run of mr_rosetta B. Alignment files from the hhpred server

You will need to tell mr_rosetta what to use as search models and the alignment between the search models and your target structure. The easiest way is to use the hhpred server (Söding J. (2005) Protein homology detection by HMM-HMM comparison. Bioinformatics 21, 951-960.) Here is what to do:

- go to: <http://toolkit.tuebingen.mpg.de/hhpred> and paste in your sequence and hit submit job (using all defaults).
- NOTE: choose only the PDB as your database. If you use other databases the hhr file you get may not work in mr_rosetta.
- In a few minutes there will be a new page with alignments in color. You want to click on the little Save button on the line above all the alignments. Save that .hhr file; this contains a list of all the PDB entries with similar sequences and the alignments.
- Optionally repeat the run of hhpred (hit "Rerun job"), this time selecting Alignment mode as global in the middle of the page. Save the resulting .hhr file as well. This creates an alternative, but also plausible, alignment file.
- The HHR analysis file from hhpred contains PDB entries similar in sequence to your target and sequence alignments. It is used to create a list of search models and alignment files. If you supply this file you do not need to specify alignment files or search models.

Search models and alignment files

If you supply an hhr analysis file from hhpred, you do not need to worry (usually) about the details of your search models and alignment files. However you can supply mr_rosetta with your own list of search models and a corresponding list of alignment files. This section describes what the alignment files need to look like (two ways you can format these files.)

Here are your options for supplying alignment information:

- If you have a pre-edited PDB file (i.e., you ran something beforehand to make the sequence be just what you want), then you just supply the PDB file and the sequence file (which may be identical or the PDB file may have deletions relative to the sequence file) as in the sample scripts.
- Otherwise if you have a .ali alignment file (see below) , you can supply that along with the PDB file and the sequence file, and then phenix will use sculpator to apply the alignment .
- Otherwise, and commonly, you supply a .hhr file, and mr_rosetta downloads the pdb and applies the alignment in the hhr file (and you don't have to supply either sequence or hhr file).
- If you want to apply an alignment yourself with an hhr file, you can use mr_model_preparation
- If you want to apply an alignment yourself with an alignment file that just contains 4 lines (1) > (greater-than-symbol), then title for target sequence (2) target sequence with - for gaps (3) >the sequence of the target and (4) the sequence of the protein in the template PDB you are supplying, then you can use mr_model_preparation

You can generate an alignment file with phenix.muscle if you do not have one from another source. Use a command like this:

```
phenix.muscle -in my_two_sequences.dat -out my_alignment.ali
```

where my_two_sequences.dat looks like:

```
> title text for sequence of target (your structure) to follow
LVLKWMSTKYVEAGELKEGSYVVIDGEPCRVVEIEKSKTGKHGSAKARIVAVGVFDGGKRTLSPVDAQVEVPIIEKFT
AQILSVSGDVIQLMDMRDYKTIEVPMKYVEEEAKGRLAPGAEEVWQILDRIYKIIRVKG
> title text for sequence of template (supplied PDB) to follow
qlmdmr AQILSVSGDVIQLMDMRDYKTIEVPMKYVEEEAKGRLAPGAEEVWQILDRIYKIIRVKG qlmdmr
```

and my_alignment.ali (your .ali file) looks like:

```
> title text for sequence of target (your structure) to follow
LVLKWMSTKYVEAGELKEGSYVVIDGEPCRVVEIEKSKTGKHGSAKARIVAVGVFDGGK
RTLSPVDAQVEVPIIEKFTAQILSVSGDVIQLMDMRDYKTIEVPMKYVEEEAKGRLAPG
AEVWQILDRIYKIIRVKG-----
> title text for sequence of template (supplied PDB) to follow
-----
-----QLMDMRDAQILSVSGDVIQLMDMRDYKTIEVPMKYVEEEAKGRLAPG
AEVWQILDRIYKIIRVKGQLMDMRD
```

You have two options for alignment files if you are going to use one.

- You can use an alignment file that sculpator can recognize. This file looks like this (there must be exactly the same number of characters for the target and the template sequences (including dashes for gaps):

```
> title text for sequence of target (your structure) to follow
VDFNGYWKMLSNENFEEYLRLADVNLKRIANLLKPKDKEIVQGDHMIIRTLSTFRNYIMDFQVGKEFEEDLTGIDD
> title text for sequence of template (supplied PDB) to follow
-AFSGTWQVYAQENYEEFLRAISLPPEEVIKLAQDVKPVTEIQNGSDFITISKTPGKVTNSFTIGKEAEIT--TMDG
```

- Alternatively, you can use a second format for the alignment file for mr_rosetta (This file is different than the alignment file for sculpator or mr_model_preparation; it is a MODELLER-style .ali file). Here is a sample:

```
## 1CRB_ 2qo4_A
# hhsearch
```

```
scores_from_program: 0 1.00
1 VDFNGYWKMLSNENFEEYLRALDVNVALRKIANLLKPDKEIVQDGDHMIIRTLSTFRNYIMDFQVGKEFEEDLTGIDD
0 -AFSGTWQVYAQENYEEFLRAISLPEEVIKLAQDKVPVTEIQNGSDFTITSKTPGKTVTNSFTIGKEAEIT - - TMDG
--
```

Here is what has to be on each line:

- Line 1: two ## signs, then target PDB ID then template PDB ID. NOTE: the template PDB ID must match the starting characters of your input search model file names (the file names themselves, not including the path to them)
- Line 2 just has a # sign and the word hhsearch
- Line 3 just has some text like scores_from_program: 0 1.00
- Line 4 has a number, then the entire sequence of the structure to be solved (the target), all on one line. The number is how many residues at the N-terminus of this sequence are to be ignored in generating a model. Usually this is 0, but if you supply a sequence that is not what is in your crystal, it could have some other number. If you are supplying a template PDB file that has residues to be removed, indicate these positions with a dash (-) in your sequence. Note: the sequence on this line cannot start with a dash.
- Line 5 has a number, then the matching sequence of the template PDB, using dashes (-) to indicate residues that are not present in the template PDB. There must be exactly the same number of characters in the sequence of your target and the sequence of your template. The number is how many residues at the N-terminus of your template PDB are to be ignored. If you have fully edited your template PDB to match the target sequence, the number will be 0.
- line 6 has two dashes: --
- NOTE: if your model has more than one chain, then you can create two of the second kind of alignment file (the ones starting with ##) and just paste them one after the other into an alignment file containing both.

Output files from mr_rosetta

The output files from mr_rosetta are the same as those from .autobuild: a model and map coefficients. These will be in a subdirectory listed at the end of your log file. The files will be something like:

```
MR_ROSETTA_1/.../AutoBuild_run_1/overall_best.pdb and
MR_ROSETTA_1/.../AutoBuild_run_1/overall_best_denmod_map_coeffs.mtz.
```

Graphical interface

A GUI for MR-Rosetta is now available in the "Molecular replacement" category. Its function is essentially identical to that of the command-line version, but many of the details are not shown by default. In addition to the methods described above for configuring your system to use Rosetta from PHENIX, the GUI also includes a preferences setting in the "Wizards" section for defining the path to the Rosetta installation. If the GUI does not detect that you have the environment set up correctly, it will issue a warning when started.

The configuration tab in the GUI includes a list into which any combination of input files may be added; the file types should be recognized (and any relevant data they contain extracted, such as space group and MTZ label information) automatically. For more complex inputs involving fragment files, click the button labeled "Other inputs" below the list of files.

The number of processors to use will be set to one fewer than the total the number of CPU cores PHENIX thinks are available, but if you are using a queueing system this number can be increased. You can change how MR-Rosetta runs child processes by clicking the "Job control" button in the lower left-hand corner of the

configuration tab.

Because it usually takes hours to run, MR-Rosetta will always be launched by the GUI as a "detached" job, meaning that you can close the GUI without killing the process, and resume it later. While MR-Rosetta is running, the current set of solutions will be continuously updated in a tab labeled "Current results", with the relevant score (LLG from Phaser, Rosetta score, or R-factor). You may view any of these solutions by clicking the buttons next to them.

Once the job is complete, a simply summary tab will be displayed, listing the output files and basic statistics such as R-factors. If the program was successful, the R-free will usually be below 50%, although this may vary depending on resolution and data quality. Buttons are provided to start additional programs or view the model and maps.

Adding a specific Rosetta command (disulfides to fix) to mr_rosetta

When you run mr_rosetta you can specify a command or commands to be added to the Rosetta scripts. For example, if your model has disulfide bonds between residues 12 and 15 and between 22 and 39 you can say:

```
rosetta_command="-MR::disulf 12:15 22:39"
```

In this command, each disulfide is colon-separated, and the numbering corresponds to the input fasta file. If you have multiple commands you can just give multiple rosetta_command statements. Note that any commands will be applied to all rosetta scripts in this mr_rosetta run. That means that you can't have different commands for different steps that use Rosetta. It also means that you cannot specify chain names or use different commands for different chains. At the moment this feature is most useful if you are supplying a single chain (or multiple chains with identical sequences).

Parameters files in mr_rosetta

When you run mr_rosetta it will write out a mr_rosetta_params.eff parameter file that can be used to re-run mr_rosetta (just as for essentially all PHENIX methods).

Examples

Standard run of mr_rosetta

Before you run mr_rosetta, you can optionally get fragment files from the Robetta server (see Setting up for a run of mr_rosetta, part A, above). Then you need an hhr alignment information file from the hhpred server (see Setting up for a run of mr_rosetta, part B, above), or else a search model and an alignment file to go with it.

Once you have these files, running mr_rosetta is easy. If you have a search model (coords1.pdb) and an alignment file for it (coords1.ali), and a data file fobs.mtz with FP SIGFP and FreeR_flag, you can type:

```
phenix.mr_rosetta \
  seq_file=seq.dat \
  data=coords1.mtz \
  alignment_files=coords1.ali \
  search_models=coords1.pdb \
  already_placed=False \
  rescore_mr.relax=False \
  rosetta_models=20 \
  ncs_copies=2 \
  space_group=p212121 \
  use_all_plausible_sg=False \
  nproc=200 \
```

```
group_run_command=qsub
```

and mr_rosetta will run automatically, generating 20 rosetta models during structure determination.

If you have an hhr alignment information file, you can specify that instead of search_models and alignment_files, with the command hhr_files=myhhpred.hhr. Then you can tell mr_rosetta how many of the PDB files to use with read_hhpred.number_of_models=1 (to use just the best one, for example).

Running mr_rosetta with a model that is already place in the unit cell

You can run mr_rosetta as a purely model-building tool as well. This is convenient if you have found a MR solution but cannot rebuild it successfully. Here is an example. The keyword to use is already_placed=True:

```
phenix.mr_rosetta \  
  seq_file=seq.dat \  
  data=coords1.mtz \  
  search_models=coords1.pdb \  
  already_placed=True \  
  rescore_mr.relax=False \  
  rosetta_models=20 \  
  ncs_copies=2 \  
  space_group=p212121 \  
  use_all_plausible_sg=False \  
  nproc=200 \  
  group_run_command=qsub
```

Rebuilding your model with Rosetta before MR

If your search model is too distant to find a molecular replacement solution, you can prerefine your model with Rosetta before carrying out molecular replacement. Here is an example. The keyword to use is: run_prerefine=True. NOTE 1: It is best to specify the number of ncs_copies if you use run_prerefine. If you do not, then you may end up running several parallel jobs, each of which is independently carrying out prerefinement on the same input model (to be used later with different numbers of ncs copies). Once you have run your job with one value of ncs_copies, you can just use the best prerefined model from that job as a search model in your other runs.

```
phenix.mr_rosetta \  
  seq_file=seq.dat \  
  data=coords1.mtz \  
  search_models=coords1.pdb \  
  run_prerefine=True \  
  number_of_prerefine_models=1000 \  
  rescore_mr.relax=False \  
  rosetta_models=20 \  
  ncs_copies=2 \  
  space_group=p212121 \  
  use_all_plausible_sg=False \  
  nproc=200 \  
  group_run_command=qsub
```

NOTE 2: if you have a model and just want to run pre-refinement and not anything else...then you can do so without any data:

```
phenix.mr_rosetta \  
  seq_file=seq.dat \  
  search_models=coords1.pdb \  
  run_prerefine=True \  
  number_of_prerefine_models=1000
```

Your pre-refined model(s) will be listed in

```
MR_ROSETTA_1/GROUP_OF_PLACE_MODEL_1/RUN_FILE_1.log
```


and you can pick the best of these (most negative score, listed first).

Running mr_rosetta from a homology search (with an hhr file)

If you have run hhpred and obtained a .hhr file with a list of alignments of proteins in the PDB with your sequence, you can run starting from your sequence file and this .hhr file. Here is an example. The keyword to use is: hhr_files=my_hhr_file.hhr.

```
phenix.mr_rosetta \
  seq_file=bfr258e.fasta \
  data=bfr258e_data.mtz \
  hhr_files=bfr258e.hhr \
  read_hhpred.number_of_models=1 \
  read_hhpred.number_of_models_to_skip=0 \
  rescore_mr.relax=False \
  rosetta_models=20 \
  ncs_copies=1 \
  nproc=200 \
  group_run_command=qsub
```

NOTE: it is generally a good idea to run several separate mr_rosetta jobs, one for each homology model you want to extract from the PDB, and possibly also separately for each possible number of NCS copies. You can do this by adjusting the "read_hhpred.number_of_models_to_skip" from 0 to N and the value of "ncs_copies" in the script above. In this way, you can just pick the first job that gives you a good solution. If you run them all at once, then all jobs will wait for the slowest job to finish at each step. If there are multiple NCS copies and some search models are poor, this can sometimes take a very long time.

Getting a default parameters file for mr_rosetta

Usually you will want to edit a parameters file so that you can specify more details of the run. You can get a default parameters file with:

```
phenix.mr_rosetta
```

and then just edit that file.

Testing mr_rosetta

You can do a test of mr_rosetta to make sure everything is ok with:

```
phenix_regression.wizards.test_command_line_rosetta_quick
```

Possible Problems

Environment problems when running mr_rosetta

If you get an error message something like ...

/opt/rosetta3.4/rosetta_source/bin/mr_protocols.default.linuxgccrelease: error while loading shared libraries: libprotocols.7.so: cannot open shared object file: No such file or directory" this can mean that the LD_LIBRARY_PATH is not correctly interpreted by mr_rosetta. Try running this command just before running phenix, or put in your bash.bashrc file (if using sh/bash) or .cshrc file (if using csh):
PHENIX_TRUST_OTHER_ENV="yes"

Debugging problems with running mr_rosetta

If `mr_rosetta` fails, the first thing (after just checking the commands you used) is to run the `mr_rosetta` regression tests to make sure that the installations of `phenix` and `rosetta` are both ok:

```
phenix_regression.wizards.test_command_line_rosetta_quick
```

That should take 10-20 minutes to run and say "OK" for all the tests. If one or more of these say instead "FAILED" ...you can go into the failed run (for example, `test_autobuild/`) and run the script there (e.g., `./test_autobuild.com`) which should fail.. and you can track down what is not working.

NOTE: On some systems there may be some really minor (numerical) differences between the standard results and those on your system. These can cause the "FAILED" to be printed out but can safely be ignored. You can tell by looking at the file "diff.dat" that should be in the failed run directory and you'll see that the differences are very minor.

If the tests all are OK, then there is something specific to your data or script. The best way to debug this is to go to the last sub-process that has failed or hung and look at the log file, and possibly re-run that step from the terminal. Here is how to get there:

- In your main log file the last lines will be something like...

```
Starting job 1...Log will be: /net/omega/raid1/scratch1/terwillMR_ROSETTA_2/GROUP_OF_PLACE_MODEL_1/RUN_FILE_1.log
```

- This log file in turn may say that further jobs were submitted...if so, go to the end of that log file...find the name of the next log file...etc... until you are at the very last thing done.
- Your last run is in the directory where `RUN_FILE_1.log` is located. There will be the following files (more if there are lots of runs in this directory of course):

```
terwill@sigma> cd MR_ROSETTA_2/GROUP_OF_PLACE_MODEL_1/
terwill@sigma> ls -tlr
total 60
-rwx----- 1 terwill lanl 1495 Feb 5 14:54 RUN_FILE_1.sh*
-rwx----- 1 terwill lanl 282 Feb 5 14:54 RUN_FILE_1*
-rw-r--r-- 1 terwill lanl 6431 Feb 5 14:54 PARAMS_1.eff
-rw-r--r-- 1 terwill lanl 6564 Feb 5 14:54 mr_rosetta_params.eff
-rw-r--r-- 1 terwill lanl 130 Feb 5 14:54 INFO_FILE_1
drwxr-xr-x 6 terwill lanl 4096 Feb 5 16:44 RUN_1/
-rw-r--r-- 1 terwill lanl 21575 Feb 5 16:45 RUN_FILE_1.log
-rw-r--r-- 1 terwill lanl 51 Feb 5 16:46 JOBS_RUNNING
```

Here:

- `PARAMS_1.eff` are the parameters used in the run
- `RUN_FILE_1.sh` actually runs the job (e.g., `phenix.mr_rosetta PARAMS_1.eff`) NOTE: usually this is `mr_rosetta` but it could also be another routine, so you do have to look at it or the first line of `PARAMS_1.eff` which will name the routine used.
- `RUN_FILE_1.log` is the log file for this run. Look at the end of this file.
- The job is run in `RUN_1/`

The key here is that you can type

```
phenix.mr_rosetta PARAMS_1.eff
```

and the exact same job that failed or ran will be run again. You can use this to debug what is going on.

- Look at the log file `RUN_FILE_1.log` and the files in `RUN_1/`. Notice what the last file written in `RUN_1/` is...this may give a clue as to when and where the problem occurred. Usually there will be an error message in `RUN_FILE_1.log` that may be informative.

- If the run in question is a Rosetta job, then the actual Rosetta job is run in a subdirectory of RUN_1/ This will be in a directory like:

MR_ROSETTA_2/GROUP_OF_ROSETTA_REBUILD_1/RUN_1/REBUILD_IN_SETS_1/RUN_5/WORK_1

Here this is in RUN_1 of a group of rosetta models, set 1, run 5, working directory. In this directory you will find something like:

```
terwill@sigma> cd WORK_1/
terwill@sigma> ls -tlr
total 684
-rw-r--r-- 1 terwill lanl 1475 Feb 5 16:48 rebuild.flags
-rwxr-xr-x 1 terwill lanl 304 Feb 5 16:48 run_rebuild.sh*
-rw-r--r-- 1 terwill lanl 422921 Feb 5 17:26 S_3DZB__0001.pdb
-rw-r--r-- 1 terwill lanl 665 Feb 5 17:26 score.sc
-rw-r--r-- 1 terwill lanl 97437 Feb 5 17:26 rebuild.log
-rw-r--r-- 1 terwill lanl 158717 Feb 5 18:17 S_3DZB__0001_ed.pdb
```

Here:

- rebuild.flags are the commands to Rosetta
- run_rebuild.sh is a command file to run Rosetta with rebuild.flags
- rebuild.log is the log file

You can look at the log file and see if there are any messages. Then you can rerun the Rosetta job in a scratch directory with:

```
mkdir junk
cd junk
../run_rebuild.sh
```

With luck, you will get the same errors and you can debug from there by changing the parameters or input files in rebuild.flags to see what was causing the problems.

Specific limitations and problems

mr_rosetta does not have the full flexibility of autobuild, so you may want to get a nearly-complete model with mr_rosetta and then use autobuild to increase the completeness and quality. You may also want to take the output of mr_rosetta and then put it back in as input to mr_rosetta and re-run it to improve your model.

File names of PDB files for mr_rosetta need to have at least 4 characters before the .pdb. So test.pdb is fine, but my.pdb is not.

Literature

Improved molecular replacement by density- and energy-guided protein structure optimization. F. DiMaio, T.C. Terwilliger, R.J. Read, A. Wlodawer, G. Oberdorfer, U. Wagner, E. Valkov, A. Alon, D. Fass, H.L. Axelrod, D. Das, S.M. Vorobiev, H. Iwaï, P.R. Pokkuluri, and D. Baker. Nature 473, 540-3 (2011).

- Improved molecular replacement by density- and energy-guided protein structure optimization. F. DiMaio, T.C. Terwilliger, R.J. Read, A. Wlodawer, G. Oberdorfer, U. Wagner, E. Valkov, A. Alon, D. Fass, H.L. Axelrod, D. Das, S.M. Vorobiev, H. Iwaï, P.R. Pokkuluri, and D. Baker. Nature 473, 540-3 (2011).

phenix.mr_rosetta: molecular replacement and model rebuilding with Phenix and Rosetta. T.C. Terwilliger, F. Dimaio, R.J. Read, D. Baker, G. Bunkóczi, P.D. Adams, R.W. Grosse-Kunstleve, P.V. Afonine, and N. Echols. J Struct Funct Genomics 13, 81-90 (2012).

- phenix.mr_rosetta: molecular replacement and model rebuilding with Phenix and Rosetta. T.C. Terwilliger, F. Dimaio, R.J. Read, D. Baker, G. Bunkóczi, P.D. Adams, R.W. Grosse-Kunstleve, P.V.

Afonine, and N. Echols. J Struct Funct Genomics 13, 81-90 (2012).

Refinement of protein structures into low-resolution density maps using rosetta. F. DiMaio, M.D. Tyka, M.L. Baker, W. Chiu, and D. Baker. J Mol Biol 392, 181-90 (2009).

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List of all available keywords

mr_rosettainput_filesdata = None Data file with experimental data (FP SIGFP or I SIGI). NOTE: May optionally instead contain F SIGF FreeR_flag for backward compatibilitydata_labels = None Optional labels for experimental data Normally these would be something like I,SIGI or F,SIGFfree_r_data = None Optional data file with free_r flags. By default this is the same file as used for experimental data.free_r_labels = None Optional labels for free_r flags. Normally these would be something like FreeR_flags or R_free_flagsremove_free = True Remove free data from mapsabin = None Optional Labin line for file with data. This is present for backward compatibility only. Normally use data_labels instead. Only allowed if the data file contains myFP mySIGF and myFreeR_flag: LABIN FP=myFP SIGFP=mySIGFP FreeR_flag=myFreeR_flagseq_file = None File with 1-letter code sequence of molecule. Chains separated by blank line or greater-than signsearch_models = None Search model PDB file Not needed if you supply an hhr alignment analysis file You can supply several with search_models=model_1.pdb search_models=model_2.pdb (and matching list of alignment files)... NOTE: If you supply a search model that contains more than one chain, then the entire search model (all chains) will be used. If your search model has multiple chains and you specify model_already_placed=False (i.e., run MR) then MR will be run with your entire search model as a single search model. If you need to run MR on parts of your search model, or if you need to combine several search models, then you will need to run automr first, then take the resulting model and use it as a search_model for mr_rosetta, specifying model_already_placed=True.copies_in_search_models = None You can specify how many ncs copies are in each search model (Not usually necessary, used to skip combinations of ncs_copies and search models that are implausible)hhr_files = None Optional HHR analysis file from hhpred. This file contains PDB entries similar in sequence to your target and sequence alignments. It is used to create a list of search models and alignment files. If you supply this file you do not need to specify alignment files or search models. To obtain this file go to: <http://toolkit.tuebingen.mpg.de/tools/hhpred> and paste in your sequence. Run it twice, once with default parameters and once with alignment mode=global. Download the '.hhr' output file each time and enter them as your hhr_files. NOTE: If your model has more than one chain type then you cannot use an hhr file to start the analysis. Instead you will need to use phenix.mr_model_preparation and phenix.automr to create your model; then you can start phenix.mr_rosetta with already_placed=Truealignment_files = None Alignment file. Supply a list if you have a list of search models, with alignment_files=model_1.ali alignment_files=model_2.ali etc. NOTE 1: Not needed if you supply an hhr alignment analysis file. NOTE 2: Not needed if your search model has the same sequence as your sequence file. NOTE 3: If your model has more than one unique chain, your alignment file should just be the contents of several single alignment files, one after the other in a single file. NOTE 4: Alignment file format: .ali Looks like: Line 1: target PDB ID then template PDB ID. NOTE: the template PDB ID must match the first 5 characters of your input search model file names (the file names themselves, not including the path to them) Lines 2, 3 keep as is Line 4: OFFSET, then entire sequence of target PDB line5: OFFSET, then matching sequence of template PDB line 6: as is --"" NOTE OFFSET is residue position, starting with ZERO, where the alignment starts. For line 4 (target PDB sequence) take number from hhpred output and subtract 1 For line 5 (template PDB sequence) should always be 0 ## 1CRB_2qp4_A # hhsearch scores_from_program: 0 1.00 1 VDFNGYWKMLSNENFEEYLRALDVNV ALRKIANLLKPDKEIVQDGDHMIIRTLSTFRNYIMDFQVGKEFEEDLTGIDDRKCMTTVSWDGDKLQCVQKGE KEGRGWTQWIEGDELHLEMRAE...

- `mr_rosettainput_filesdata` = None Data file with experimental data (FP SIGFP or I SIGI). NOTE: May optionally instead contain F SIGF `FreeR_flag` for backward compatibility `data_labels` = None Optional labels for experimental data Normally these would be something like I,SIGI or F,SIGF `free_r_data` = None Optional data file with `free_r` flags. By default this is the same file as used for experimental data. `free_r_labels` = None Optional labels for `free_r` flags. Normally these would be something like `FreeR_flags` or `R_free_flags` `remove_free` = True Remove free data from mapslabin = None Optional Labin line for file with data. This is present for backward compatibility only. Normally use `data_labels` instead. Only allowed if the data file contains myFP mySIGF and myFreeR_flag: LABIN FP=myFP SIGFP=mySIGFP `FreeR_flag`=myFreeR_flag `seq_file` = None File with 1-letter code sequence of molecule. Chains separated by blank line or greater-than sign `search_models` = None Search model PDB file Not needed if you supply an hhr alignment analysis file You can supply several with `search_models=model_1.pdb search_models=model_2.pdb` (and matching list of alignment files)... NOTE: If you supply a search model that contains more than one chain, then the entire search model (all chains) will be used. If your search model has multiple chains and you specify `model_already_placed=False` (i.e., run MR) then MR will be run with your entire search model as a single search model. If you need to run MR on parts of your search model, or if you need to combine several search models, then you will need to run automr first, then take the resulting model and use it as a `search_model` for mr_rosetta, specifying `model_already_placed=True`. `copies_in_search_models` = None You can specify how many ncs copies are in each search model (Not usually necessary, used to skip combinations of ncs_copies and search models that are implausible) `hhr_files` = None Optional HHR analysis file from hhpred. This file contains PDB entries similar in sequence to your target and sequence alignments. It is used to create a list of search models and alignment files. If you supply this file you do not need to specify alignment files or search models. To obtain this file go to: <http://toolkit.tuebingen.mpg.de/tools/hhpred> and paste in your sequence. Run it twice, once with default parameters and once with alignment mode=global. Download the '.hhr' output file each time and enter them as your `hhr_files`. NOTE: If your model has more than one chain type then you cannot use an hhr file to start the analysis. Instead you will need to use phenix.mr_model_preparation and phenix.automr to create your model; then you can start phenix.mr_rosetta with `already_placed=True` `alignment_files` = None Alignment file. Supply a list if you have a list of search models, with `alignment_files=model_1.ali alignment_files=model_2.ali` etc. NOTE 1: Not needed if you supply an hhr alignment analysis file. NOTE 2: Not needed if your search model has the same sequence as your sequence file. NOTE 3: If your model has more than one unique chain, your alignment file should just be the contents of several single alignment files, one after the other in a single file. NOTE 4: Alignment file format: .ali Looks like: Line 1: target PDB ID then template PDB ID. NOTE: the template PDB ID must match the first 5 characters of your input search model file names (the file names themselves, not including the path to them) Lines 2, 3 keep as is Line 4: OFFSET, then entire sequence of target PDB line5: OFFSET, then matching sequence of template PDB line 6: as is --"" NOTE OFFSET is residue position, starting with ZERO, where the alignment starts. For line 4 (target PDB sequence) take number from hhpred output and subtract 1 For line 5 (template PDB sequence) should always be 0 ## 1CRB_2qp4_A # hhsearch
scores_from_program: 0 1.00 1 VDFNGYWKMLSNENFEEYLRALDVNVALRKIANLLKPDKEIVQDGDHMI
IIRTLSTFRNYIMDFQVGKEFEEDLTGIDDRKCMTTVSWDGDQLQCVQKGEKEGRGWTQWIEGDELHLE
MRAE...
- `input_filesdata` = None Data file with experimental data (FP SIGFP or I SIGI). NOTE: May optionally instead contain F SIGF `FreeR_flag` for backward compatibility `data_labels` = None Optional labels for experimental data Normally these would be something like I,SIGI or F,SIGF `free_r_data` = None Optional data file with `free_r` flags. By default this is the same file as used for experimental data. `free_r_labels` = None Optional labels for `free_r` flags. Normally these would be something like `FreeR_flags` or `R_free_flags` `remove_free` = True Remove free data from mapslabin = None Optional Labin line for file

with data. This is present for backward compatibility only. Normally use data_labels instead. Only allowed if the data file contains myFP mySIGF and myFreeR_flag: LABIN FP=myFP SIGFP=mySIGFP FreeR_flag=myFreeR_flagseq_file = None File with 1-letter code sequence of molecule. Chains separated by blank line or greater-than signsearch_models = None Search model PDB file Not needed if you supply an hhr alignment analysis file You can supply several with search_models=model_1.pdb search_models=model_2.pdb (and matching list of alignment files)... NOTE: If you supply a search model that contains more than one chain, then the entire search model (all chains) will be used. If your search model has multiple chains and you specify model_already_placed=False (i.e., run MR) then MR will be run with your entire search model as a single search model. If you need to run MR on parts of your search model, or if you need to combine several search models, then you will need to run automr first, then take the resulting model and use it as a search_model for mr_rosetta, specifying model_already_placed=True.copies_in_search_models = None You can specify how many ncs copies are in each search model (Not usually necessary, used to skip combinations of ncs_copies and search models that are implausible)hhr_files = None Optional HHR analysis file from hhpred. This file contains PDB entries similar in sequence to your target and sequence alignments. It is used to create a list of search models and alignment files. If you supply this file you do not need to specify alignment files or search models. To obtain this file go to: <http://toolkit.tuebingen.mpg.de/tools/hhpred> and paste in your sequence. Run it twice, once with default parameters and once with alignment mode=global. Download the '.hhr' output file each time and enter them as your hhr_files. NOTE: If your model has more than one chain type then you cannot use an hhr file to start the analysis. Instead you will need to use phenix.mr_model_preparation and phenix.automr to create your model; then you can start phenix.mr_rosetta with already_placed=Truealignment_files = None Alignment file. Supply a list if you have a list of search models, with alignment_files=model_1.ali alignment_files=model_2.ali etc. NOTE 1: Not needed if you supply an hhr alignment analysis file. NOTE 2: Not needed if your search model has the same sequence as your sequence file. NOTE 3: If your model has more than one unique chain, your alignment file should just be the contents of several single alignment files, one after the other in a single file. NOTE 4: Alignment file format: .ali Looks like: Line 1: target PDB ID then template PDB ID. NOTE: the template PDB ID must match the first 5 characters of your input search model file names (the file names themselves, not including the path to them) Lines 2, 3 keep as is Line 4: OFFSET, then entire sequence of target PDB line5: OFFSET, then matching sequence of template PDB line 6: as is --"" NOTE OFFSET is residue position, starting with ZERO, where the alignment starts. For line 4 (target PDB sequence) take number from hhpred output and subtract 1 For line 5 (template PDB sequence) should always be 0 ## 1CRB_2qo4_A # hhsearch scores_from_program: 0 1.00 1 VDFNGYWKMLSNENFEEY LRALDVNVALRKIANLLKPDKEIVQDGDHMIIRTLSTFRNYIMDFQVGKEFEEDLTGIDDRKCMTTVSWDG DKLQCVQKGEKEGRGWTQWIEGDELHLEMRAEGVTCKQVFKK...

- data = None Data file with experimental data (FP SIGFP or I SIGI). NOTE: May optionally instead contain F SIGF FreeR_flag for backward compatibility
- data_labels = None Optional labels for experimental data Normally these would be something like I,SIGI or F,SIGF
- free_r_data = None Optional data file with free_r flags. By default this is the same file as used for experimental data.
- free_r_labels = None Optional labels for free_r flags. Normally these would be something like FreeR_flags or R_free_flags
- remove_free = True Remove free data from maps
- labin = None Optional Labin line for file with data. This is present for backward compatibility only. Normally use data_labels instead. Only allowed if the data file contains myFP mySIGF and

myFreeR_flag: LABIN FP=myFP SIGFP=mySIGFP FreeR_flag=myFreeR_flag

- seq_file = None File with 1-letter code sequence of molecule. Chains separated by blank line or greater-than sign
- search_models = None Search model PDB file Not needed if you supply an hhr alignment analysis file You can supply several with search_models=model_1.pdb search_models=model_2.pdb (and matching list of alignment files)... NOTE: If you supply a search model that contains more than one chain, then the entire search model (all chains) will be used. If your search model has multiple chains and you specify model_already_placed=False (i.e., run MR) then MR will be run with your entire search model as a single search model. If you need to run MR on parts of your search model, or if you need to combine several search models, then you will need to run automr first, then take the resulting model and use it as a search_model for mr_rosetta, specifying model_already_placed=True.
- copies_in_search_models = None You can specify how many ncs copies are in each search model (Not usually necessary, used to skip combinations of ncs_copies and search models that are implausible)
- hhr_files = None Optional HHR analysis file from hhpred. This file contains PDB entries similar in sequence to your target and sequence alignments. It is used to create a list of search models and alignment files. If you supply this file you do not need to specify alignment files or search models. To obtain this file go to: <http://toolkit.tuebingen.mpg.de/tools/hhpred> and paste in your sequence. Run it twice, once with default parameters and once with alignment mode=global. Download the '.hhr' output file each time and enter them as your hhr_files. NOTE: If your model has more than one chain type then you cannot use an hhr file to start the analysis. Instead you will need to use phenix.mr_model_preparation and phenix.automr to create your model; then you can start phenix.mr_rosetta with already_placed=True
- alignment_files = None Alignment file. Supply a list if you have a list of search models, with alignment_files=model_1.ali alignment_files=model_2.ali etc. NOTE 1: Not needed if you supply an hhr alignment analysis file. NOTE 2: Not needed if your search model has the same sequence as your sequence file. NOTE 3: If your model has more than one unique chain, your alignment file should just be the contents of several single alignment files, one after the other in a single file. NOTE 4: Alignment file format: .ali Looks like: Line 1: target PDB ID then template PDB ID. NOTE: the template PDB ID must match the first 5 characters of your input search model file names (the file names themselves, not including the path to them) Lines 2, 3 keep as is Line 4: OFFSET, then entire sequence of target PDB line5: OFFSET, then matching sequence of template PDB line 6: as is --"" NOTE OFFSET is residue position, starting with ZERO, where the alignment starts. For line 4 (target PDB sequence) take number from hhpred output and subtract 1 For line 5 (template PDB sequence) should always be 0 ## 1CRB_2qo4_A # hhsearch scores_from_program: 0 1.00 1 VDFNGYWKM LSNENFEEYLRALDVNVALRKIANLL KPDKEIVQDGDHMIIRTLSTFRNYIMDFQVGKEFEEDLTGIDDRKCM TTVSWDGDQLQCVQKGEKEGRG WTQWIEGDELHLEMRAEGVTCKQVFKKV 0 -AFSGTWQVYAQENYEEFLRAISLP EEVIKLAKDVKPVTEI QQNGSDF TITSKTPGKTVTNSFTIGKEAEIT--TMDGKKLKCIVKLDGGKLCV CRT----DRFSHIQEIKAGEMV ETLTVGGTTMIRKSKKI --
- model_info_file = None Pickled file containing information about starting model
- mr_rosetta_solutions = None You can read in prior solutions (.pkl files) Then you can skip steps and it will pick up where it left off. You can load just some solutions with the keyword ids_to_load
- ids_to_load = None You can restrict the load of mr_rosetta solutions to just one or more id's
- map_coeffs = None Data file (mtz format) with map coeffs (FP PHIB FOM) for rosetta electron density Normally use None; this is used in iterations of mr_rosetta to pass map coeffs from autobuild to the next cycle where they are used instead of refinement or phaser map coeffs. If used, also should set run_refine_top_models=False Always used with labin_map_coeffs and map

- `labin_map_coeffs` = None Labin line for map coeffs file Something like FP=FWT PHIB=PHWT Normally use None; used by `mr_rosetta` in iteration
- `map` = None Map file (ccp4 format) with map coeffs for rosetta electron density. This is expected if `map_coeffs` is specified. Normally this map should not contain freeR reflection information. Normally use None; this is used in iterations of `mr_rosetta` to pass map from autobuild to the next cycle where it is used instead of refinement or phaser map
- `refinement_params` = None You can specify a parameters file for refinement
- `display_solutions` = False You can display any solutions in `mr_rosetta_solutions` and select some of them with `ids_to_load`
- `fragment_files` = None Note: Starting with Rosetta version 2013wk35 you can specify `generate_fragment_files=True` and leave `fragment_files=None` and Rosetta will generate the files for you. Fragment files (optionally used for Rosetta rebuilding if your template has any gaps to fill) Enter one at a time with `fragment_files=myfragments3.gz` `fragment_files=myfragments9.gz` To obtain the two required files...go to: <http://rosetta.bakerlab.org/fragmentsubmit.jsp> and register, then submit your sequence file, and when the server finishes, download the two files '`...9...gz`' and '`...3...gz`' These are your fragment files of length 9 and 3 NOTE: if your molecule has multiple chains, use instead `fragment_files_chain_list`, `fragment_files_3_mer_by_chain`, and `fragment_files_9_mer_by_chain`.
- `fragment_files_chain_list` = None If your molecule has multiple chains, use `fragment_files_chain_list`, `fragment_files_3_mer_by_chain`, and `fragment_files_9_mer_by_chain` instead of `fragment_files`. Use `fragment_files_chain_list` to define which chain ID each of your `fragment_files_3_mer_by_chain` and `fragment_files_9_mer_by_chain` go with. NOTE: You only need one set of fragments files for each UNIQUE chain. So if chains A and C are the same, you just need to specify fragments for chain A. If you have two different chains A and B and fragment files `frag_A_3` `frag_A_9` `frag_B_3` `frag_B_9` then you should use: `fragment_files_chain_list=A` `fragment_files_chain_list=B`
`fragment_files_3_mer_by_chain=frag_A_3` `fragment_files_9_mer_by_chain=frag_A_9`
`fragment_files_3_mer_by_chain=frag_B_3` `fragment_files_9_mer_by_chain=frag_B_9`
- `fragment_files_9_mer_by_chain` = None See `fragment_files_chain_list`.
- `fragment_files_3_mer_by_chain` = None See `fragment_files_chain_list`.
- `use_dummy_fragment_files` = False You can use dummy fragment files (this is ok if your template matches your sequence file already). If True, then you do not need to supply fragment files
- `sort_fragment_files` = True Sort the fragment files by name so that 9 is first then 3
- `output_fileslog` = `mr_rosetta.log` Output log file
`sort_score_type` = None You can specify the scoring method for choosing top models to list at the end
`params_out` = `mr_rosetta_params.eff` Parameters file to rerun `mr_rosetta`
- `log` = `mr_rosetta.log` Output log file
- `sort_score_type` = None You can specify the scoring method for choosing top models to list at the end
- `params_out` = `mr_rosetta_params.eff` Parameters file to rerun `mr_rosetta`
- `directoriestemp_dir` = "" Optional temporary work directory
`workdir` = "" Optional work directory. Base path for all work
`output_dir` = "" Output directory where files are to be written
`gui_output_dir` = None Output directory for PHENIX GUI. Not used when run from the command line.
`top_output_dir` = None Output directory for entire set of runs
`rosetta_path` = "" Location of rosetta directories All rosetta files are located relative to this path You can set the environment variable 'PHENIX_ROSETTA_PATH' to indicate where rosetta is to be found. In csh/tcsh use something like: `setenv PHENIX_ROSETTA_PATH /Users/Shared/unix/rosetta` In bash/sh use: `export`

PHENIX_ROSETTA_PATH=/Users/Shared/unix/rosettarosetta_binary_dir = "rosetta_source/bin"
 Directory with rosetta scripts for mr_rosetta Path is relative to rosetta_pathrosetta_binary_name =
 "mr_protocols.default" Name of rosetta binary Path is relative to rosetta_path+rosetta_binary_dir NOTE:
 any suffixes such as '.default', '.macosgccrelease', '.linuxrelease' are ignoredrosetta_script_dir =
 "rosetta_source/src/apps/public/electron_density" Directory with rosetta scripts for mr_rosetta Path is
 relative to rosetta_pathrosetta_database_dir = "rosetta_database" Location of rosetta database Path is
 relative to rosetta_path

- temp_dir = "" Optional temporary work directory
- workdir = "" Optional work directory. Base path for all work
- output_dir = "" Output directory where files are to be written
- gui_output_dir = None Output directory for PHENIX GUI. Not used when run from the command line.
- top_output_dir = None Output directory for entire set of runs
- rosetta_path = "" Location of rosetta directories All rosetta files are located relative to this path You can set the environment variable 'PHENIX_ROSETTA_PATH' to indicate where rosetta is to be found. In csh/tcsh use something like: setenv PHENIX_ROSETTA_PATH /Users/Shared/unix/rosetta In bash/sh use: export PHENIX_ROSETTA_PATH=/Users/Shared/unix/rosetta
- rosetta_binary_dir = "rosetta_source/bin" Directory with rosetta scripts for mr_rosetta Path is relative to rosetta_path
- rosetta_binary_name = "mr_protocols.default" Name of rosetta binary Path is relative to rosetta_path+rosetta_binary_dir NOTE: any suffixes such as '.default', '.macosgccrelease', '.linuxrelease' are ignored
- rosetta_script_dir = "rosetta_source/src/apps/public/electron_density" Directory with rosetta scripts for mr_rosetta Path is relative to rosetta_path
- rosetta_database_dir = "rosetta_database" Location of rosetta database Path is relative to rosetta_path
- read_hhprednumber_of_models = 1 Take the first number_of_models models from the hhpred similarity analysis that are specified with hhr_files. (this will give you number_of_models models for each hhr file)number_of_models_to_skip = 0 Skip the first number_of_models_to_skip models (most similar) in hhpred file (Useful along with number_of_models to pick any one or group of templates from your hhpred file)copies_to_extract = None Number of copies of the unique chain defined in your hhr_file to extract from the template PDB file (if possible). You can specify more than one value: copies_to_extract='1 2 4' will try to run MR with a monomer, dimer, and tetramer from each template PDB file (if available). If None, then the values used will be: 1, ncs_copies, and all other divisors of ncs_copies, so if ncs_copies=6, the values will be 1, 2, 3, and 6. Note: if ncs_copies is also None and the number of copies that can fit in the cell is large, then this can lead to a lot of different combinations being tried.only_extract_proper_symmetry = False Only extract groups of copies from template that form proper symmetry (i.e., do not extract 2 molecules from a trimer). Note: not implemented. All are currently extracted.
- number_of_models = 1 Take the first number_of_models models from the hhpred similarity analysis that are specified with hhr_files. (this will give you number_of_models models for each hhr file)
- number_of_models_to_skip = 0 Skip the first number_of_models_to_skip models (most similar) in hhpred file (Useful along with number_of_models to pick any one or group of templates from your hhpred file)
- copies_to_extract = None Number of copies of the unique chain defined in your hhr_file to extract from the template PDB file (if possible). You can specify more than one value: copies_to_extract='1 2 4' will try

to run MR with a monomer, dimer, and tetramer from each template PDB file (if available). If None, then the values used will be: 1, ncs_copies, and all other divisors of ncs_copies, so if ncs_copies=6, the values will be 1, 2, 3, and 6. Note: if ncs_copies is also None and the number of copies that can fit in the cell is large, then this can lead to a lot of different combinations being tried.

- only_extract_proper_symmetry = False Only extract groups of copies from template that form proper symmetry (i.e., do not extract 2 molecules from a trimer). Note: not implemented. All are currently extracted.

- place_modelrun_place_model = True Run place_model: use AutoMR or place existing model Each model will be used to generate number_of_prerefine_models. Note: this can take a lot of CPU time. You might want to only do this on a single input modelmodel_already_placed = False Use model_already_placed to indicate that your model is already placed in the correct locationmodel_already_aligned = False Use model_already_aligned to indicate that your model is already edited to match your sequence. Note: if model_already_aligned=True then any alignment files supplied are ignored.force_alignment = False If set and alignment file is set, use alignment as supplied.number_of_output_models = 5 Number of Phaser molecular replacement models to consideralign_with_sculptor = True Use phenix.sculptor and phenix.mr_model_preparation to apply alignments and edit templates (alternative is to use Rosetta scripts).identity = None Percent identity between search model and target Normally set automatically based on your alignmentfileidentity_for_scoring_only = 25 Percent identity between search model and target to be used for LLG scoring. This is normally a fixed value so that scores from different templates can be compared.use_all_plausible_sg = True Often you will want to search all space groups with the same point group as you may not know which is correct from your data. overlap_allowed = 10 Solutions will be accepted by default if fewer than 10 percent of residues are involved in clashes. You can choose to increase the percent clashes if the packing is tight and your search molecule is not exactly the same as the molecule in the cell. selection_criteria_rot_value = 75 Choose a value for your criterion for keeping rotation solutions at each stage. Percent of Best Score: AutoMR looks down the list of LLG scores and only keeps the ones that differ from the mean by more than the chosen percentage, compared to the top solution. fast_search_mode = True Run phaser with selection_criteria_rot_value and then if no obvious solution, repeat with cutoff lowered by peak_rota_downpeak_rota_down = 25 Used if fast_search_mode=True. Run phaser with selection_criteria_rot_value and then if no obvious solution, repeat with cutoff lowered by peak_rota_downmr_resolution = None Resolution for molecular replacement. Normally leave at None to allow phaser to determine this automatically.refine_after_mr = True Refine placed model for map calculation only before rescoring and rebuilding. Required for denmod_after_refinedenmod_after_refine = True After refinement, density-modify map before rosetta scoring and rebuilding Note: denmod_after_refine appears separately in the scopes place_model and refine_top_models so you need to set it separately in each placeps_in_rebuild = False You can choose to use a prime-and-switch resolve map in map calculation/density modification in the place_model step.find_ncs_after_mr = True Find NCS in model after placing model if ncs_copies is greater than 1fixed_model = None A fixed model can be added. This will be added as a fixed model in molecular replacement. The sequence for this model should be included in your sequence file.fixed_model_identity = None Percent identity between fixed model and target This is a required parametersufficient_number_finished = None If specified and sufficient_number_finished place_model jobs have finished, then all other jobs are ignored (they become zombie jobs and finish on their own) and mr_rosetta goes on to the next step. This is a way to avoid the problem of a really long-running and ultimately unsuccessful MR job slowing everything down. Note: you can set sufficient_number_finished while mr_rosetta is running by putting the number you want in a file called GO_ON in the directory where the log file RUN_FILE_1.log is located (something like

MR_ROSETTA_3/GROUP_OF_PLACE_MODEL_1)/copies_of_search_model_to_place = None
(Optional) number of copies of search model to place with MR. This is how many new copies of s...

- run_place_model = True Run place_model: use AutoMR or place existing model Each model will be used to generate number_of_prerefine_models. Note: this can take a lot of CPU time. You might want to only do this on a single input model
- model_already_placed = False Use model_already_placed to indicate that your model is already placed in the correct location
- model_already_aligned = False Use model_already_aligned to indicate that your model is already edited to match your sequence. Note: if model_already_aligned=True then any alignment files supplied are ignored.
- force_alignment = False If set and alignment file is set, use alignment as supplied.
- number_of_output_models = 5 Number of Phaser molecular replacement models to consider
- align_with_sculptor = True Use phenix.sculptor and phenix.mr_model_preparation to apply alignments and edit templates (alternative is to use Rosetta scripts).
- identity = None Percent identity between search model and target Normally set automatically based on your alignment file
- identity_for_scoring_only = 25 Percent identity between search model and target to be used for LLG scoring. This is normally a fixed value so that scores from different templates can be compared.
- use_all_plausible_sg = True Often you will want to search all space groups with the same point group as you may not know which is correct from your data.
- overlap_allowed = 10 Solutions will be accepted by default if fewer than 10 percent of residues are involved in clashes. You can choose to increase the percent clashes if the packing is tight and your search molecule is not exactly the same as the molecule in the cell.
- selection_criteria_rot_value = 75 Choose a value for your criterion for keeping rotation solutions at each stage. Percent of Best Score: AutoMR looks down the list of LLG scores and only keeps the ones that differ from the mean by more than the chosen percentage, compared to the top solution.
- fast_search_mode = True Run phaser with selection_criteria_rot_value and then if no obvious solution, repeat with cutoff lowered by peak_rota_down
- peak_rota_down = 25 Used if fast_search_mode=True. Run phaser with selection_criteria_rot_value and then if no obvious solution, repeat with cutoff lowered by peak_rota_down
- mr_resolution = None Resolution for molecular replacement. Normally leave at None to allow phaser to determine this automatically.
- refine_after_mr = True Refine placed model for map calculation only before rescoring and rebuilding. Required for denmod_after_refine
- denmod_after_refine = True After refinement, density-modify map before rosetta scoring and rebuilding Note: denmod_after_refine appears separately in the scopes place_model and refine_top_models so you need to set it separately in each place
- ps_in_rebuild = False You can choose to use a prime-and-switch resolve map in map calculation/density modification in the place_model step.
- find_ncs_after_mr = True Find NCS in model after placing model if ncs_copies is greater than 1
- fixed_model = None A fixed model can be added. This will be added as a fixed model in molecular replacement. The sequence for this model should be included in your sequence file.

- `fixed_model_identity = None` Percent identity between fixed model and target This is a required parameter
- `sufficient_number_finished = None` If specified and `sufficient_number_finished` place_model jobs have finished, then all other jobs are ignored (they become zombie jobs and finish on their own) and `mr_rosetta` goes on to the next step. This is a way to avoid the problem of a really long-running and ultimately unsuccessful MR job slowing everything down. Note: you can set `sufficient_number_finished` while `mr_rosetta` is running by putting the number you want in a file called `GO_ON` in the directory where the log file `RUN_FILE_1.log` is located (something like `MR_ROSETTA_3/GROUP_OF_PLACE_MODEL_1/`)
- `copies_of_search_model_to_place = None` (Optional) number of copies of search model to place with MR. This is how many new copies of search model to add to anything already present. By default, calculated from `ncs_copies`, copies in the search model, and number of already-placed copies. Note difference from `ncs_copies` which is the total copies in the asymmetric unit
- `preremoteUsed` if you need to improve your models before `MRrun_preremote = False` Pre-refine models before `MRnumber_of_preremote_models = 1000` Number of models to generate in `preremotementnumber_of_models_in_ensemble = 1` Number of top-scoring models to use as an ensemble for MR NOTE: Not implemented: only one model used at this point
- `run_preremote = False` Pre-refine models before MR
- `number_of_preremote_models = 1000` Number of models to generate in preremotement
- `number_of_models_in_ensemble = 1` Number of top-scoring models to use as an ensemble for MR NOTE: Not implemented: only one model used at this point
- `fixed_ensembles` If you already know the placement of one or more molecules you can specify them as fixed ensembles. NOTE 1: you are specifying location and orientation of one or more copies of the search model NOTE 2: you cannot specify `use_all_plausible_sg` if you have fixed ensembles `fixed_ensembleID_list = None` Enter the word 'ensemble_1' to indicate that you want to specify a copy of your search model that is to be fixed. To specify more than one placement just say 'ensemble_1' more than once. For example if you specify `fixed_ensembleID_list=ensemble_1` and do not specify `fixed_euler_list` or `fixed_frac_list`, it will be assumed that you have one copy of your search model already placed (with the input coordinates), and you are looking to place an additional `copies_of_search_model_to_place` copies of the search model. `fixed_euler_list = 0.0 0.0 0.0` Enter Euler angles (from AutoMR or Phaser) for fixed component. NOTE 2: you can enter more than one fixed component if you want. If you do, then enter `fixed_euler_list` in multiples of 3 numbers and also `fixed_frac_list` in multiples of 3 numbers. `fixed_frac_list = 0.0 0.0 0.0` Enter fractional offset (location) for fixed component (from AutoMR or Phaser) for fixed component. NOTE 2: you can enter more than one fixed component if you want. If you do, then enter `fixed_euler_list` in multiples of 3 numbers and also `fixed_frac_list` in multiples of 3 numbers. `fixed_frac_list_is_fractional = True` Normally `fixed_frac_list` is fractional coordinates. You can say `fixed_frac_list_is_fractional=False` to instead use orthogonal angstroms to specify the locations of your ensembles.
- `fixed_ensembleID_list = None` Enter the word 'ensemble_1' to indicate that you want to specify a copy of your search model that is to be fixed. To specify more than one placement just say 'ensemble_1' more than once. For example if you specify `fixed_ensembleID_list=ensemble_1` and do not specify `fixed_euler_list` or `fixed_frac_list`, it will be assumed that you have one copy of your search model already placed (with the input coordinates), and you are looking to place an additional `copies_of_search_model_to_place` copies of the search model.

- `fixed_euler_list = 0.0 0.0 0.0` Enter Euler angles (from AutoMR or Phaser) for fixed component. NOTE 2: you can enter more than one fixed component if you want. If you do, then enter `fixed_euler_list` in multiples of 3 numbers and also `fixed_frac_list` in multiples of 3 numbers.
- `fixed_frac_list = 0.0 0.0 0.0` Enter fractional offset (location) for fixed component (from AutoMR or Phaser) for fixed component. NOTE 2: you can enter more than one fixed component if you want. If you do, then enter `fixed_euler_list` in multiples of 3 numbers and also `fixed_frac_list` in multiples of 3 numbers.
- `fixed_frac_list_is_fractional = True` Normally `fixed_frac_list` is fractional coordinates. You can say `fixed_frac_list_is_fractional=False` to instead use orthogonal angstroms to specify the locations of your ensembles.
- `rescore_mrrun_rescore_mr = True` Rescore MR solutions, optionally by rosetta modeling `nstruct = 5` Number of models to build with rosetta in rescoring if `relax=True` `relax = False` Relax solution with rosetta modeling before rescoring NOTE 1: if you only have one solution to rescore (as in the case where you supplied a placed model) you might want to say `relax=False` to not bother to relax the model. NOTE 2: if your model has multiple chain types then you have to use `relax=False`. `include_unrelaxed_in_scoring = False` Include unrelaxed (original) model in scoring `align = True` Use alignment file in relax procedure `edit_model = False` Edit model before rescoring using `model_info_filestage_to_rescore = mr_solution` You can specify the stage of solutions to consider for rescoring (i.e., `mr_solution`, `rosetta_solution`) Default is `mr_solution`; during scoring of rosetta solutions it is set automatically to `rosetta_solution`
- `run_rescore_mr = True` Rescore MR solutions, optionally by rosetta modeling
- `nstruct = 5` Number of models to build with rosetta in rescoring if `relax=True`
- `relax = False` Relax solution with rosetta modeling before rescoring NOTE 1: if you only have one solution to rescore (as in the case where you supplied a placed model) you might want to say `relax=False` to not bother to relax the model. NOTE 2: if your model has multiple chain types then you have to use `relax=False`.
- `include_unrelaxed_in_scoring = False` Include unrelaxed (original) model in scoring
- `align = True` Use alignment file in relax procedure
- `edit_model = False` Edit model before rescoring using `model_info_file`
- `stage_to_rescore = mr_solution` You can specify the stage of solutions to consider for rescoring (i.e., `mr_solution`, `rosetta_solution`) Default is `mr_solution`; during scoring of rosetta solutions it is set automatically to `rosetta_solution`
- `rosetta_rebuildrun_rosetta_rebuild = True` Run rosetta modeling on best rescored MR solutions `stage_to_rebuild = rescored_mr_solution` Normally set automatically. You can specify the stage of solutions to consider for rebuilding (i.e., `mr_solution`, `rosetta_solution`) Default is `rescored_mr_solution` `max_solutions_to_rebuild = 5` Keep all solutions with at least `llg_percent_of_max_to_keep`, up to `max_solutions_to_rebuild`, and at least `min_solutions_to_rebuild` `min_solutions_to_rebuild = 1` Keep all solutions with at least `llg_percent_of_max_to_keep`, up to `max_solutions_to_rebuild`, and at least `min_solutions_to_rebuild` `llg_percent_of_max_to_keep = 50` Keep all solutions with at least `llg_percent_of_max_to_keep`, up to `max_solutions_to_rebuild`, and at least `min_solutions_to_rebuild` `rosetta_models = 100` Number of models to build with rosetta in rebuilding `chunk_size = 1` If `background=False`, divide the `nstruct` models into chunks of `chunk_size` or smaller to keep the length of individual jobs shorter `edit_model = True` Edit model before using `superpose_model = False` Superpose the rebuilt model on the original model. (Restore original location and orientation as much as possible.)

- `run_rosetta_rebuild = True` Run rosetta modeling on best rescored MR solutions
- `stage_to_rebuild = rescored_mr_solution` Normally set automatically. You can specify the stage of solutions to consider for rebuilding (i.e., `mr_solution`, `rosetta_solution`) Default is `rescored_mr_solution`
- `max_solutions_to_rebuild = 5` Keep all solutions with at least `llg_percent_of_max_to_keep`, up to `max_solutions_to_rebuild`, and at least `min_solutions_to_rebuild`
- `min_solutions_to_rebuild = 1` Keep all solutions with at least `llg_percent_of_max_to_keep`, up to `max_solutions_to_rebuild`, and at least `min_solutions_to_rebuild`
- `llg_percent_of_max_to_keep = 50` Keep all solutions with at least `llg_percent_of_max_to_keep`, up to `max_solutions_to_rebuild`, and at least `min_solutions_to_rebuild`
- `rosetta_models = 100` Number of models to build with rosetta in rebuilding
- `chunk_size = 1` If `background=False`, divide the nstruct models into chunks of `chunk_size` or smaller to keep the length of individual jobs shorter
- `edit_model = True` Edit model before use
- `superpose_model = False` Superpose the rebuilt model on the original model. (Restore original location and orientation as much as possible.)
- `rosetta_rescorerun_rosetta_rescore = True` Run Phaser rescoring on top rosetta rebuilt models
`percentage_to_rescore = 20` Rescore percentage_to_rescore of top rosetta solutions with Phaser RNP LLG scoring
`min_solutions_to_rescore = 2` Rescore at least percentage_to_rescore of the rosetta, models, and at least `min_solutions_to_rescore` Usually choose at least 2 so that they can be compared
- `run_rosetta_rescore = True` Run Phaser rescoring on top rosetta rebuilt models
- `percentage_to_rescore = 20` Rescore percentage_to_rescore of top rosetta solutions with Phaser RNP LLG scoring
- `min_solutions_to_rescore = 2` Rescore at least percentage_to_rescore of the rosetta, models, and at least `min_solutions_to_rescore` Usually choose at least 2 so that they can be compared
- `similarityrun_similarity = False` Identify similarity of top solutions
`required_cc = 0.20` Value of required_cc for number_of_required_cc top solutions to carry on
`number_of_required_cc = 5` Number of top solutions with CC of required_cc or better to top solution required to carry on
- `run_similarity = False` Identify similarity of top solutions
- `required_cc = 0.20` Value of required_cc for number_of_required_cc top solutions to carry on
- `number_of_required_cc = 5` Number of top solutions with CC of required_cc or better to top solution required to carry on
- `refine_top_modelsrun_refine_top_models = True` Refine models to get map before relaxing them to get new map coeffs. Use unrefined model in relaxation however
`stage_to_refine = None` You can specify the stage of solutions to consider for rescoring (i.e., `mr_solution`, `rosetta_solution`) Default is `rosetta_solution`; during scoring of rosetta solutions
`sort_score_type = None` You can specify the scoring method for choosing top models to refine
`percent_to_refine = 20` Percentage of top models to refined
`enmod_after_refine = True` After refinement, density-modify map before rosetta scoring and rebuilding. Note: this appears separately in the scopes `place_model` and `refine_top_models` so you need to set it separately in each place
`remove_clashing_residues = None` Remove clashing residues before refinement and autobuilding
`clash_cutoff = 1.5` Remove clashing residues before refinement and autobuilding if `remove_clashing_residues=True` and clash is worse than `clash_cutoff`.
- `run_refine_top_models = True` Refine models to get map before relaxing them to get new map coeffs. Use unrefined model in relaxation however

- `stage_to_refine = None` You can specify the stage of solutions to consider for rescoring (i.e., `mr_solution`, `rosetta_solution`) Default is `rosetta_solution`; during scoring of rosetta solutions
- `sort_score_type = None` You can specify the scoring method for choosing top models to refine
- `percent_to_refine = 20` Percentage of top models to refine
- `denmod_after_refine = True` After refinement, density-modify map before rosetta scoring and rebuilding. Note: this appears separately in the scopes `place_model` and `refine_top_models` so you need to set it separately in each place
- `remove_clashing_residues = None` Remove clashing residues before refinement and autobuilding
- `clash_cutoff = 1.5` Remove clashing residues before refinement and autobuilding if `remove_clashing_residues=True` and `clash` is worse than `clash_cutoff`.
- `average_density_top_modelsrun_average_density_top_models = True` Average density from top models
`percent_to_average = 100` percentage of refined models to use in averaging
- `run_average_density_top_models = True` Average density from top models
- `percent_to_average = 100` percentage of refined models to use in averaging
- `relax_top_modelsrun_relax_top_models = True` Relax rosetta rebuilt solutions, scoring with LLG
`stage_to_relax = None` You can specify the stage of solutions to consider for relaxing (i.e., `mr_solution`, `rosetta_solution`) Default is `rescored_rosetta_solution`; during
`relax_top_modelsnumber_to_relax = 2` Number of top models to relax
`nstruct = 5` Number of rosetta relaxed models to build for each starting model in relaxation (best will be chosen)
`sort_score_type = None` You can specify the scoring method for choosing top models to relax
- `run_relax_top_models = True` Relax rosetta rebuilt solutions, scoring with LLG
- `stage_to_relax = None` You can specify the stage of solutions to consider for relaxing (i.e., `mr_solution`, `rosetta_solution`) Default is `rescored_rosetta_solution`; during `relax_top_models`
- `number_to_relax = 2` Number of top models to relax
- `nstruct = 5` Number of rosetta relaxed models to build for each starting model in relaxation (best will be chosen)
- `sort_score_type = None` You can specify the scoring method for choosing top models to relax
- `autobuild_top_modelsrun_autobuild_top_models = True` Autobuild top relaxed_rosetta_solutions
`number_to_autobuild = 2` Number of top models to autobuild
`quick = False` Use fewer cycles
`phase_and_build = False` Use `phase_and_build` to rebuild models instead of autobuild (much faster, but not quite as good)
`macro_cycles = None` Number of overall cycles for `phase_and_build` (`macro_cycles`) or autobuild (`n_cycle_rebuild_max`) Set by default if None (recommended)
`remove_residues_on_special_positions = True` It is a good idea to set `remove_residues_on_special_positions=True` so that if a poor starting model has some atoms on special positions then they will be removed so that refinement can proceed.
`morph = False` You can choose whether to distort your model in order to match the current working map. This may be useful for MR models that are quite distant from the correct structure. [See also `repeats_with_morph` which will instead morph your model at the beginning of each repeat cycle up to `repeats_with_morph` times.]
`edit_model = True` Edit model before rescoring using `model_info_file`
`use_map_coeffs = True` Use current best map as starting map coeffs in autobuild (Only applies if `phase_and_build=False`)
- `run_autobuild_top_models = True` Autobuild top relaxed_rosetta_solutions
- `number_to_autobuild = 2` Number of top models to autobuild
- `quick = False` Use fewer cycles

- `phase_and_build = False` Use `phase_and_build` to rebuild models instead of autobuild (much faster, but not quite as good)
- `macro_cycles = None` Number of overall cycles for `phase_and_build` (`macro_cycles`) or autobuild (`n_cycle_rebuild_max`) Set by default if None (recommended)
- `remove_residues_on_special_positions = True` It is a good idea to set `remove_residues_on_special_positions=True` so that if a poor starting model has some atoms on special positions then they will be removed so that refinement can proceed.
- `morph = False` You can choose whether to distort your model in order to match the current working map. This may be useful for MR models that are quite distant from the correct structure. [See also `repeats_with_morph` which will instead morph your model at the beginning of each repeat cycle up to `repeats_with_morph` times.]
- `edit_model = True` Edit model before rescoring using `model_info_file`
- `use_map_coeffs = True` Use current best map as starting map coeffs in autobuild (Only applies if `phase_and_build=False`)
- `setup_repeat_mr_rosettarun_setup_repeat_mr_rosetta = True` Set up for running `mr_rosetta` again using results from current run. Must be run before `repeat_mr_rosetta`
- `repeats = 1` Maximum repeats of running `mr_rosetta`. (Runs one cycle if `repeats=0`)
- `template_repeats = 0` Number of repeat cycles in which to restart from the template (the MR solution corresponding to best current model) instead of continuing with best current model. This may be useful for very poor starting models. This is normally combined with `morph_repeats` (i.e., morph the template at the end of one cycle and use that in the next cycle)
- `morph_repeats = 0` Number of repeat cycles in which to morph the starting model corresponding to best current model instead of continuing as is with best current model. This may be useful for very poor starting models. This is normally combined with `template_repeats` (i.e., morph the template at the end of one cycle and use that in the next cycle. [See also `morph` which instead applies morphing during autobuilding.]
- `number_to_repeat = 1` Number of top models to re-run in MR rosetta Usually this should be 1. If you are using condor, or set `one_subprocess_level=True`, it must be 1
- `acceptable_r = 0.25` Used to decide whether the model is acceptable enough to quit if it is not improving much. A good value is 0.25
- `minimum_delta_r = None` Used to decide whether the model is improving. Skip additional cycles if improvement in R since last is less than `minimum_delta_r`

- `minimum_delta_r = None` Used to decide whether the model is improving. Skip additional cycles if improvement in R since last is less than `minimum_delta_r`
- `repeat_mr_rosettarun_repeat_mr_rosetta = True` Run mr_rosetta again using results from current run
- `copies_in_new_search_group = 1` Number of copies of model to used to create a search model that is to be placed with MR on repeat cycles (if not all ncs copies are found on the first cycle). If you are searching with a dimer on the first cycle, you might want to consider `copies_in_new_search_group=2`.
- `update_map_coeffs_with_autobuild = True` Update map coeffs only during autobuild (not with refinement) on cycles of iteration
- `run_repeat_mr_rosetta = True` Run mr_rosetta again using results from current run
- `copies_in_new_search_group = 1` Number of copies of model to used to create a search model that is to be placed with MR on repeat cycles (if not all ncs copies are found on the first cycle). If you are searching with a dimer on the first cycle, you might want to consider `copies_in_new_search_group=2`.
- `update_map_coeffs_with_autobuild = True` Update map coeffs only during autobuild (not with refinement) on cycles of iteration
- `rosetta_modelingmap_resolution = 3` Map resolution in rosetta modeling
- `map_grid_spacing = 1.5` Grid spacing in map in Rosetta rebuilding
- `map_weight = 1` Weighting on map in Rosetta rebuilding. Ignored if `fast=True`
- `map_window = 5` Smoothing distance (residues) for map scoring. Ignored if `fast=True`
- `include_solvation_energy = True` Include solvation energy term in Rosetta modeling If you are modeling a membrane protein you may want to turn this off. Note: if False, then a `weights_file` is created with the specification of `fa_sol 0.0`
- `weights_file = None` Optional weights file for Rosetta. If specified, this will be used instead of the file `$PHENIX_ROSETTA_PATH/rosetta_database/scoring/weights/score12_full.wts`
- `map_resolution = 3` Map resolution in rosetta modeling
- `map_grid_spacing = 1.5` Grid spacing in map in Rosetta rebuilding
- `map_weight = 1` Weighting on map in Rosetta rebuilding. Ignored if `fast=True`
- `map_window = 5` Smoothing distance (residues) for map scoring. Ignored if `fast=True`
- `include_solvation_energy = True` Include solvation energy term in Rosetta modeling If you are modeling a membrane protein you may want to turn this off. Note: if False, then a `weights_file` is created with the specification of `fa_sol 0.0`
- `weights_file = None` Optional weights file for Rosetta. If specified, this will be used instead of the file `$PHENIX_ROSETTA_PATH/rosetta_database/scoring/weights/score12_full.wts`
- `crystal_inforesolution = 0` high-resolution limit for map calculations
- `space_group = None` You can specify the space group. If None then the space group in your input data file or its inverse will be used unless you specify `use_all_plausible_sg=True`
- `chain_type = *PROTEIN DNA RNA` Chain type (for identifying main-chain and side-chain atoms)
- `ncs_copies = Auto` Number of copies of unique sequence defined in sequence file expected in the a.u. This is how many molecules there are in the asymmetric unit. Default is Auto, in which case the value of ncs copies leading to solvent content closest to 50% (based on unique chains in sequence file and volume of asymmetric unit) will be chosen. NOTE: you can specify more than one value with `ncs_copies='1 2 7'`, in which case each will be tried. You can also specify None in which case all likely values (those leading to solvent content from 0.35 to 0.65) will be tried.
- `resolution = 0` high-resolution limit for map calculation
- `space_group = None` You can specify the space group. If None then the space group in your input data file or its inverse will be used unless you specify `use_all_plausible_sg=True`
- `chain_type = *PROTEIN DNA RNA` Chain type (for identifying main-chain and side-chain atoms)

- `ncs_copies` = Auto Number of copies of unique sequence defined in sequence file expected in the a.u. This is how many molecules there are in the asymmetric unit. Default is Auto, in which case the value of `ncs_copies` leading to solvent content closest to 50% (based on unique chains in sequence file and volume of asymmetric unit) will be chosen. NOTE: you can specify more than one value with `ncs_copies='1 2 7'`, in which case each will be tried. You can also specify None in which case all likely values (those leading to solvent content from 0.35 to 0.65) will be tried.

- `controljob_title` = None Job title in PHENIX GUI, not used on command line
- `verbose` = False Verbose output
- `debug` = False Debugging output
- `raise_sorry` = False Raise sorry if problems
- `dry_run` = False Just read in and check parameter names
- `nproc` = 1 Number of processors to use
- `group_run_command` = "sh " Command to use to run multiple jobs This may be sh if you are using a single machine (where you might set `background=True`) or something like 'qsub' or 'qsub -q all.q@theta' or 'qsub -q all.q@theta -o \$HOME/logs -e \$HOME/logs' on a cluster (where you should leave `background=False`)
- `queue_commands` = None You can add any commands that need to be run for your queueing system. These are written before any other commands in the file that is submitted to your queueing system. For example on a PBS system you might say: `queue_commands='#PBS -N mr_rosetta'`
- `queue_commands='#PBS -j oe'` `queue_commands='#PBS -l walltime=03:00:00'`
- `queue_commands='#PBS -l nodes=1:ppn=1'` NOTE: you can put in the characters '<path>' in any `queue_commands` line and this will be replaced by a string of characters based on the path to the run directory. The first character and last two characters of each part of the path will be included, separated by '_', up to 15 characters. For example 'test_autobuild/WORK_5/AutoBuild_run_1/TEMP0/RUN_1' would be represented by: 'tld_W_5_A1__TP0_1'
- `condor_universe` = vanilla The universe for condor is usually vanilla. However you might need to set it to local for your cluster
- `add_double_quotes_in_condor` = True You might need to turn on or off double quotes in condor job submission scripts. These are already default elsewhere but may interfere with condor paths.
- `condor` = None Specifies if the `group_run_command` is submitting a job to a condor cluster. Set by default to True if `group_run_command=condor_submit`, otherwise False. For condor job submission `mr_rosetta` uses a customized script with condor commands. Also uses `one_subprocess_level=True`
- `one_subprocess_level` = None Specifies that a subprocess cannot submit a job
- `single_run_command` = "sh " Command to use to run single jobs Normally this is `sh`
- `last_process_is_local` = True If true, run the last process in a group in background with `sh` as part of the job that is submitting jobs. This prevents having the job that is submitting jobs sit and wait for all the others while doing nothing
- `background` = None Run in background. If None, automatically set to True if `nproc` is greater than one and `group_run_command` is `sh`
- `ignore_errors_in_subprocess` = True Generally use `ignore_errors_in_subprocess=True` to ignore errors in sub-processes. This allows you to continue even if a few jobs crash. If all jobs in a group crash, the process will stop. NOTE: if a job hangs or never runs...this will not be detected and you will have to either put a file with the name FINISHED in the directory where the job was to run (e.g., `MR_ROSETTA_3/GROUP_OF_PLACE_MODEL_1/RUN_1/FINISHED`) or stop the whole job by putting a file with the name STOPWIZARD in the main run directory (e.g., `MR_ROSETTA_3/STOPWIZARD`)
- `check_run_command` = False Try out run command to make sure it works Use False if your queue may not be available at the beginning of your run. Use True if you want to check things out
- `max_wait_time` = 100 Maximum time (sec) to wait for a file to be written (Useful for queues or nfs-mounted systems)
- `check_wait_time` = 10 Time between checks for finishing jobs (sec)
- `wait_between_submit_time` = 1.0 You can specify the length of time (seconds) to wait between each job that is submitted when running sub-processes. This can be helpful on NFS-mounted systems when running with multiple processors to avoid file conflicts. The symptom of too short a `wait_between_submit_time` is File exists:....
- `wizard_directory_number` = None Directory number for `MR_ROSETTA_xx`. Normally None except if called from GUI
- `dir_max` = 100000 Maximum number of directories to create (must be as big as `nproc` or `nstruct/chunk`)
- `number_to_print` = 5 Number of entries to

print in l...

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- `background` = None Run in background. If None, automatically set to True if `nproc` is greater than one and `group_run_command` is sh
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- `wizard_directory_number` = None Directory number for MR_ROSETTA_xx. Normally None except if called from GUI
- `n_dir_max` = 100000 Maximum number of directories to create (must be as big as `nproc` or `nstruct/chunk`)
- `number_to_print` = 5 Number of entries to print in long lists
- `write_run_directory_to_file` = None The working directory name is written to this file
- `rosetta_command` = None Command for Rosetta (like `-MR::disulf 12:13 22:39`)
- `rosetta_3_6_or_later` = None Rosetta inputs change in version 3.6
- `fast` = None NOTE: Not tested. For Rosetta versions 2013wk42 and later you can speed up Rosetta calculations (this sets `-MR::fast` flag in Rosetta)
- `generate_fragment_files` = True For Rosetta versions 2013wk35 and later you can generate fragment files automatically using the Rosetta database
- `resolve_command_list` = None You can supply any resolve command here for autobuild NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`;
- `start_point` = `*place_model rescore_mr rosetta_rebuild rosetta_rescore similarity refine_top_models average_density_top_models relax_top_models autobuild_top_models setup_repeat_mr_rosetta repeat_mr_rosetta` You can specify what point to start at by supplying a `rosetta_solutions .pkl` file and specifying a place to start
- `stop_point` = `place_model rescore_mr rosetta_rebuild rosetta_rescore similarity refine_top_models average_density_top_models relax_top_models autobuild_top_models setup_repeat_mr_rosetta repeat_mr_rosetta` You can specify a step to stop at (after completing this step) For example, to carry out just `place_model`, you can say `start_point=place_model stop_point=place_model`.
- `clean_up` = True At the end of the autobuild runs the TEMP directories will be removed if `clean_up` is True.
- `add_id` = True Add solution id to names of models
- `test_flag_value` = None This parameter sets the value of the test set that is to be free. Normally phenix sets up test sets with values of 0 and 1 with 1 as the free set. The CCP4 convention is values of 0 through 19 with 0 as the free set. Either of these is recognized by default in Phenix. If you have any other convention (for example values of 0 to 19 and test set is 1) then you can specify this with `test_flag_value`.
- `real_space_optimize` = None Flag for keeping track of whether `real_space_optimize` is used.
- `set_real_space_optimize` = None If set, `set_real_space_optimize` sets the following parameter values (overrides the values that you specify and those that are default): `real_space_optimize = True clean_up=True is_sub_process=True model_already_placed=True refine_after_mr=False place_model.denmod_after_refine=False find_ncs_after_mr=False run_rescore_mr=False`

```

stage_to_rebuild="mr_solution" max_solutions_to_rebuild=1 run_rosetta_rescore=False
run_refine_top_models=False refine_top_models.denmod_after_refine=False
run_average_density_top_models=False stage_to_relax="rescored_rosetta_solution"
run_autobuild_top_models=False run_setup_repeat_mr_rosetta=False run_repeat_mr_rosetta=False
ncs_copies=1 dummy_refinement=True dummy_rosetta=False
relax_top_models.sort_score_type='ROSETTA SCORE' output_files.sort_score_type='ROSETTA
SCORE' . Normal run will look like: phenix.mr_rosetta real_space_optimize=true remove_free=False
nproc=20 group_run_command=qsub data=data.mtz map_coeffs=map_coeffs.mtz
labin_map_coeffs=FWT,PHWT seq_file=seq.dat search_models=search_model.pdb
rosetta_rebuild.rosetta_models=20

```

- `non_user_paramsfile_base` = None String defining intermediate file names Normally set automatically. If given, must match the 3rd word on the first line of the alignment file
- `print_citations` = True Print citation information at end of run
- `highest_id` = 0 Start ID numbers with `highest_id+1`
- `is_sub_process` = False identifies if this is a sub-process or top-level job
- `dummy_autobuild` = False Allows you to skip actual run of autobuild
- `dummy_refinement` = False Allows you to run refinements but not to change coordinates
- `dummy_rosetta` = False Allows you to skip models from rosetta steps
- `preremote_only` = False Set internally to allow pre-refinement without data
- `skip_clash_guard` = True Skip clash guard check in refinement
- `correct_special_position_tolerance` = None Adjust tolerance for special position check. If 0., then check for clashes near special positions is not carried out. This sometimes allows phenix.refine to continue even if an atom is near a special position. If 1., then checks within 1 Å of special positions. If None, then uses phenix.refine default. (1)
- `ncs_in_refinement` = *torsion cartesian None Use torsion_angle refinement of NCS. Alternative is cartesian or None
- `comparison_mtz` = None Allows you to compare results with an existing map file
- `labin_comparison_mtz` = None labin line for comparison mtz

- `write_local_files = False` Used to create test pickle files only with `phenix.mr_rosetta`
`mr_rosetta_solutions=results.pkl display_solutions=True write_local_files=true`
- `rosetta_fixed_seed = None` Fixed seed for rosetta (so that the same answer is always obtained. Use for regression tests only).

Automated Molecular Replacement with Phaser

phaser.html

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- Composition by Sequence
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- "sol" Files
- "rlist" Files
- Fixed partial structure
- How to Select Peaks
- How to Control Output
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- Default tNCS detection and correction No tNCS Pairs of molecules
- No tNCS

- Pairs of molecules
- Non-default tNCS correction Higher order tNCS Complex tNCS
- Higher order tNCS
- Complex tNCS

Quicklink to example scripts -> MR using keyword input

Quicklink to phaser.famos (find_alt_orig_sym_mate) documentation -> Famos

Phaser should be able to solve most structures with the Automated Molecular Replacement mode, and this is the first mode that you should try. Give Phaser your data (How to Define Data) and your models (How to Define Models), tell Phaser what to search for, and a list of possible spacegroups (in the same point group).

If this doesn't work (see Has Phaser Solved It?), you can try selecting peaks of lower significance in the rotation function in case the real orientation was not within the selection criteria. By default peaks above 75% of the top peak are selected (see How to Select Peaks). See What to do in Difficult Cases for more hints and tips. If the automated molecular replacement mode doesn't work even with non-default input you need to run the modes of Phaser separately. The possibilities are endless - you can even try exhaustive searches (translations of all orientations) if you want - but experience has shown that most structures that can be solved by Phaser can be solved by relatively simple strategies.

Automated Molecular Replacement

Automated Molecular Replacement combines the anisotropy correction, likelihood enhanced fast rotation function, likelihood enhanced fast translation function, packing and refinement modes for multiple search models and a set of possible spacegroups to automatically solve a structure by molecular replacement. Top solutions are output to the files `FILEROOT.sol`, `FILEROOT.#.mtz` and `FILEROOT.#.pdb` (where "#" refers to the sorted solution number, 1 being the best, and only 1 is output by default). Many structures can be solved by running an automated molecular replacement search with defaults, giving the ensembles that you expect to be easiest to find first.

At the completion of Molecular Replacement you may wish to place your solutions on a common origin with a previous solution, for which Famos can be used.

Flow Diagram for Automated MR

Should Phaser Solve It?

The difficulty of a molecular replacement problem depends primarily on two major factors: how well the model will be able to explain the diffraction data (which depends both on the accuracy of the model and on its completeness), and how many reflections can be explained, at least in part. Each reflection provides a piece of information that helps to identify correct MR solutions.

It is possible to make a reasonable prediction of whether or not a solution will be found. If the quality of the model (its accuracy and completeness) can be estimated, then the expected contribution of each reflection to the total LLG can also be estimated. From a large battery of tests, we know that an LLG of 60 or greater usually indicates a correct solution (at least in the absence of complicating factors such as translational non-crystallographic symmetry, tNCS). Building on this understanding, if it is estimated that the LLG will be 60 or less, then Phaser will assume that the problem is a difficult one, and will implement search procedures optimised for difficult problems.

What Resolution of Data Should be Used?

The signal for a molecular replacement solution should be very clear if the expected value of the LLG is much higher than the minimum required to be fairly certain of a solution. Currently Phaser aims for a minimum LLG of 120 and, if it is possible to achieve an even higher value, given the quality of the model and the quantity of diffraction data, then the resolution for the initial search is limited to the value required to achieve an expected LLG of 120. Data to the full resolution are still used for a final rigid-body refinement, or in a second pass if a clear solution is not found in the first attempt.

However, if the model is expected to have a large RMS error (based usually on the correlation between sequence identity and RMS error), then data to high resolution will not contribute any significant signal. Regardless of the expected LLG at the highest resolution limit, the resolution used is limited to 1.8 times the estimated RMS error of the model, because this resolution limit gives about 99% of the LLG that could be achieved.

Because Phaser implements strategies designed to solve structures with as much confidence as possible, as efficiently as possible, it is best to leave the choice of resolution to Phaser, at least in the first instance.

Has Phaser Solved It?

Ideally, a unique solution with a strong signal will be found at the end of the search. If you are searching for multiple components, then ideally the search for each component will also give a strong signal. However if the signal-to-noise of your search is low, there will be noise peaks and multiple ambiguous solutions.

Signal-to-noise is judged using the Z-score, which is computed by comparing the LLG values from the rotation or translation search with LLG values for a set of random rotations or translations. The mean and the RMS deviation from the mean are computed from the random set, then the Z-score for a search peak is defined as its LLG minus the mean, all divided by the RMS deviation, "i.e. " "the number of standard deviations above (or below) the mean. "

For a rotation function, the correct orientation may be well down the list with a Z-score (number of standard deviations above the mean value, or RFZ) under 4, and it is often not possible to identify the correct orientation until a translation function is performed and yields a clear solution. Note that the signal-to-noise of the rotation function drops with increasing number of primitive symmetry operations (the number of different orientations for symmetry-related molecules), because there is more uncertainty about how the structure factor contributions from symmetry-related copies will add up.

For a translation function the correct solution will generally have a Z-score (TFZ) over 5 and be well separated from the rest of the solutions. Of course, there will always be exceptions! The table gives a very rough guide to interpreting TFZ scores. This table will be updated, as we learn more from systematic molecular replacement trials.

When you are searching for multiple components, the signal may be low for the first few components but, as the model becomes more complete, the signal should become stronger. Finding a clear solution for a new component is a good sign that the partial solution to which that component was added was indeed correct.

You should always at least glance through the summary of the logfile. One thing to look for, in particular, is whether any translation solutions with a high Z-score have been rejected by the packing step. By default up to 5 percent of marker atoms (C-alpha atoms for protein) are allowed to be involved in clashes. A solution with more clashes may still be correct, and the clashes may arise only because of differences in small surface loops. If this happens, repeat the run allowing a suitable number of clashes. Note that, unless there is specific evidence in the logfile that a high TFZ-score solution is being rejected with a few clashes, it is much better to edit the model to remove the loops than to increase the number of allowed clashes. Packing criteria are a very powerful constraint on the translation function, and increasing the number of allowed clashes beyond the default will increase the search time enormously without the possibility of generating any correct solutions that

would not have otherwise been found.

Note that, by default, Phaser will produce a single PDB file corresponding to the top solution found (if any), so finding a single PDB file in your output directory is not an indication that the search succeeded! You have to look, at least, at the summary of the logfile, or at the list of possible solutions in the .sol file that is produced if you run Phaser from ccp4i or command-line scripts.

Annotation

A highly compact summary of the history of the statistics of a solution is given in the SOLUTION SET in the .sol file. This is a good place to start your analysis of the output. The annotation gives the Z-score of the solution at each rotation and translation function, the number of clashes in the packing, and the refined LLG.

Two versions of TFZ (the translation function Z-score) now appear for each component. The first ("TFZ=") is the Z-score from the actual translation search, which depends on the accuracy of the orientation used for that search. The second ("TFZ==") is the TFZ-equivalent, which indicates what the TFZ score would have been with the correct (refined) orientation. You should see the TFZ-equivalent is high at least for the final components of the solution, and that the LLG (log-likelihood gain) increases as each component of the solution is added. For example, in the case of beta-blip the annotation for the single solution output in the .sol file shows these features

Note that the Euler angles in Phaser follow the same convention as those defined for the Crowther fast rotation function, i.e. z-y-z (rotate around the z-axis, followed by the new y-axis, followed by the new z-axis).

History

A highly compact summary of the history of the peak positions of a solution is given in the SOLUTION HISTORY in the .sol file. Together with the SOLUTION SET annotation, this is useful in your analysis of the output.

For example, in the case of beta-blip the annotation for the single solution output in the .sol file shows these features

A more complicated structure solution may have

```
SOLU HISTORY RF/TF(7/1:10)PAK(10:10)RNP(10,12,13,11,17,16,18,25,3,8,22,21,20,7,969,6,5,201,9,4,390,2,1,19:1)RNP(1:1)
```

What to do in Difficult Cases

Not every structure can be solved by molecular replacement, but the right strategy can push the limits. What to do when the default jobs fail depends on why your structure is difficult.

- **Flexible Structure** The relative orientations of the domains may be different in your crystal than in the model. If that may be the case, break the model into separate PDB files containing rigid-body units, enter these as separate ensembles, and search for them separately. If you find a convincing solution for one domain, but fail to find a solution for the next domain, you can take advantage of the knowledge that its orientation is likely to be similar to that of the first domain. The ROTate AROund option of the brute rotation search can be used to restrict the search to orientations within, say, 30 degrees of that of the known domain. Allow for close approach of the domains by increasing the allowed clashes with the PACK keyword by, say, 1 for each domain break that you introduce. Note that it is possible to use the brute rotation search as part of the automated molecular replacement pipeline, by changing the choice of the type of rotation search. Alternatively, you could try generating a series of models perturbed by normal

modes, with the NMAPdb keyword. One of these may duplicate the hinge motion and provide a good single model.

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- Poor or Incomplete Model Signal-to-noise is reduced by coordinate errors or incompleteness of the model. Since the rotation search has lower signal to begin with than the translation search, it is usually more severely affected. For this reason, it can be very useful to use the subsequent translation search as a way to choose among many (say 1000) orientations. The MR_AUTO FAST search mode automatically reduces the cutoff for accepting peaks from the fast rotation function if the default pass does not find a solution with a high z-score, but you can manually reduce this further with the PEAKS and PURGE keywords. You can also try turning off the clustering of fast rotation function peaks because the correct orientation may sit on the shoulder of a peak in the rotation function. As shown convincingly by Schwarzenbacher et al. (Schwarzenbacher, Godzik, Grzechnik and Jaroszewski, Acta Cryst. D60, 1229-1236, 2004), judicious editing can make a significant difference in the quality of a distant model. In a number of tests with their data on models below 30% sequence identity, we have found that Phaser works best with a "mixed model" (non-identical sidechains longer than Ser replaced by Ser). In agreement with their results, the best models are generally derived using more sophisticated alignment protocols, such as their FFAS protocol. Use phenix.sculptor to edit your model.

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rotation function peaks because the correct orientation may sit on the shoulder of a peak in the rotation function. As shown convincingly by Schwarzenbacher et al. (Schwarzenbacher, Godzik, Grzechnik and Jaroszewski, Acta Cryst. D60, 1229-1236, 2004), judicious editing can make a significant difference in the quality of a distant model. In a number of tests with their data on models below 30% sequence identity, we have found that Phaser works best with a "mixed model" (non-identical sidechains longer than Ser replaced by Ser). In agreement with their results, the best models are generally derived using more sophisticated alignment protocols, such as their FFAS protocol. Use phenix.sculptor to edit your model.

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- High Degree of Non-crystallographic Symmetry If there are clear peaks in the self-rotation function, you can expect orientations to be related by this known NCS. Methods to automatically use such information will be implemented in a future version of Phaser. In the meantime, you can work out for yourself the orientations that would be consistent with NCS and use the ROTate AROund option to sample similar orientations. Alternatively, you may have an oligomeric model and expect similar NCS in the crystal. First search with the oligomeric model; if this fails, search with a monomer. If that succeeds, you can again use the ROTate AROund option to force a subsequent monomer to adopt an orientation similar to the one you expect.

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- What not to do The automated mode of Phaser is fast when Phaser finds a high Z-score solution to your problem. When Phaser cannot find a solution with a significant Z-score, it "thrashes", meaning it maintains a list of 100-1000's of low Z-score potential solutions and tries to improve them. This can lead

to exceptionally long Phaser runs (over a week of CPU time). Such runs are possible because the highly automated script allows many consecutive MR jobs to be run without you having to manually set 100-1000's of jobs running and keep track of the results. "Thrashing" generally does not produce a solution: solutions generally appear relatively quickly or not at all. It is more useful to go back and analyse your models and your data to see where improvements can be made. Your system manager will appreciate you terminating these jobs. It is also not a good idea to effectively remove the packing test. Unless there is specific evidence in the logfile that a high TF-function Z-score solution is being rejected with a few clashes, it is much better to edit the model to remove the loops than to increase the number of allowed clashes. Packing criteria are a very powerful constraint on the translation function, and increasing the number of allowed clashes beyond a few (e.g. 1-5) will increase the search time enormously without the possibility of generating any correct solutions that would not have otherwise been found.

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- Other suggestions Phaser has powerful input, output and scripting facilities that allow a large number of possibilities for altering default behaviour and forcing Phaser to do what you think it should. However, you will need to read the information in the manual below to take advantage of these facilities!

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How to Define Data

You need to tell Phaser the name of the mtz file containing your data and the columns in the mtz file to be used using the HKLIN and LABIN keywords. Additional keywords (BINS CELL OUTLIER RESOLUTION SPACegroup) define how the data are used.

How to Define Models

Phaser must be given the models that it will use for molecular replacement. A model in Phaser is referred to as an "ensemble", even when it is described by a single file. This is because it is possible to provide a set of aligned structures as an ensemble, from which a statistically-weighted averaged model is calculated. A molecular replacement model is provided either as one or more aligned pdb files, or as an electron density map, entered as structure factors in an mtz file. Each ensemble is treated as a separate type of rigid body to be placed in the molecular replacement solution. An ensemble should only be defined once, even if there are several copies of the molecule in the asymmetric unit.

Fundamental to the way in which Phaser uses MR models (either from coordinates or maps) is to estimate how the accuracy of the model falls off as a function of resolution, represented by the Sigma(A) curve. To generate the Sigma(A) curve, Phaser needs to know the RMS coordinate error expected for the model and the fraction of the scattering power in the asymmetric unit that this model contributes.

A Babinet-style correction is used to account for the effects of disordered solvent on the completeness of the model at low resolution.

Molecular replacement models are defined with the ENSEMBLE keyword and the COMPOSITION keyword. The ENSEMBLE keyword gives (amongst other things) the RMS deviation for the Sigma(A) curve. The COMPOSITION keyword is used to deduce the fraction of the scattering power in the asymmetric unit that each ensemble contributes. The composition of the asymmetric unit is defined either by entering the molecular weights or sequences of the components in the asymmetric unit, and giving the number of copies of each. Expert users can also enter the fraction of the scattering of each component directly, although the composition must still be entered for the absolute scale calculation. Please note that the composition supplied to Phaser has to include everything in the asymmetric unit, not just what is being looked for in the current search!

Building an Ensemble from Coordinates

The RMS deviation is determined directly from RMS or indirectly from IDENTITY in the ENSEMBLE keyword using a formula that depends on the sequence identity and the number of residues in the model.

The RMS deviation estimated from ID may be an underestimate of the true value if there is a slight conformational change between the model and target structures. To find a solution in these cases it may be necessary to increase the RMS from the default value generated from the ID, by say 0.5 Angstroms. On the other hand, when Phaser succeeds in solving a structure from a model with sequence identity much below 30%, it is often found that the fold is preserved better than the average for that level of sequence identity. So it may be worth submitting a run in which the RMS error is set at, say, 1.5, even if the sequence identity is low. The table below can be used as a guide as to the default RMS value corresponding to ID.

If you construct a model by homology modelling, remember that the RMS error you expect is essentially the error you expect from the template structure (if not worse!). So specify the sequence identity of the template, not of the homology model.

Only the model with the highest sequence identity is reported in the output pdb file. Also, HETATM cards in the input pdb file are ignored in the calculation of the structure factors for the ensemble, but are carried through to the output pdb file. Thus, the phases on the output mtz file (which come from the structure factors of the ensemble) do not correspond to those that would be calculated from the output pdb file, when there is more than one pdb file in an ensemble and/or the pdbfile(s) have HETATM records.

Coordinate Editing

HETATM/LIGANDS

Phaser ignores the scattering from HETATM records. The HETATM records are carried through to output with occupancy set to zero. Ligands will therefore not contribute to the scattering used for molecular replacement. The exceptions to this rule are the HETATM records for MSE (seleno-methionine) MSO (seleno-methionine selenoxide) CSE (seleno-cysteine) CSO (seleno-cysteine selenoxide) ALY (acetyllysine) MLY (n-dimethyl-lysine) and MLZ (n-methyl-lysine) which are used in the scattering and carried through to output with their original occupancy. If you wish to include any HETATM records in the scattering the record name use the keyword ENSE modlid HETATOM ON

WATER

Water molecules (identified by the residue name OW WAT HOH H2O OH2 MOH WTR or TIP) are deleted from the pdb file on input, are not used in the scattering and are not carried through to file output. If you want to retain water molecules you will need to change the residue name to something other than this (e.g. WWW) so that the atoms are not identified as water. To include the water molecules in the scattering, the HETATM records will also have to be changed to ATOM records as described above.

Building an Ensemble from Electron Density

When using density as a model, it is necessary to specify both the extent (x,y,z limits) of the cut-out region of density, and the centre of this region. With coordinates, Phaser can work this out by itself. This information is needed, for instance, to decide how large rotational steps can be in the rotation search and to carry out the molecular transform interpolation correctly. In the case of electron density, the RMS value does not have the same physical meaning that it has when the model is specified by atomic coordinates, but it is used to judge how the accuracy of the calculated structure factors drops off with resolution. A suitable value for RMS can be obtained, in the case of density from an experimentally-phased map, by choosing a value that makes the SigmaA curve fall off with resolution similarly to the mean figures-of-merit. In the case of density from an EM image reconstruction, the RMS value should make the SigmaA curve fall off similarly to a Fourier correlation curve used to judge the resolution of the EM image.

For detailed information, including a tutorial with example scripts, see Using density as a model

How to Define Composition

The composition defines the total amount of protein and nucleic acid that you have in the asymmetric unit not the fraction of the asymmetric unit that you are searching for.

Default Composition

For convenience, the composition defaults to 50% protein scattering by volume (the average for protein crystals). It is better to enter it explicitly, even if only to check that you have correctly deduced the probable content of your crystal. If your crystal has higher or lower solvent content than this, or contains nucleic acid, then the composition should be entered explicitly.

Composition by Solvent Content

Scattering is determined from the solvent content of the crystal, assuming that the crystal contains protein only, and the average distribution of amino acids in protein. If your crystal contains nucleic acid or your protein has an unusual amino acid distribution then the composition should be entered explicitly using the MW or sequence options.

Composition by Number of Residues in ASU

Scattering is determined from the number of residues in the asymmetric unit, assuming that the crystal contains protein only or nucleic acid only, and assuming an average distribution of residues for either. If your crystal contains a mixture then the composition should be entered explicitly using the MW or sequence options. If your crystal has an unusual residue distribution then the composition should be entered explicitly using the sequence options.

Composition by Molecular Weight

The composition is calculated from the molecular weight of the protein and nucleic acid assuming the protein and nucleic acid have the average distribution of amino acids and bases. If your protein or nucleic acid has an unusual amino acid or base distribution the composition should be entered by sequence. You can mix compositions entered by molecular weight with those entered by sequence.

Composition by Sequence

The composition is calculated from the amino acid sequence of the protein and the base sequence of the nucleic acid in fasta format. You can mix compositions entered by molecular weight with those entered by sequence. Individual atoms can be added to the composition with the COMPOSITION ATOM keyword. This allows the explicit addition of heavy atoms in the structure e.g. Fe atoms.

Composition by Percentage Scattering

The fraction scattering of each ensemble can be entered directly. The fraction scattering of each ensemble is normally automatically worked out from the average scattering from each ensemble (calculated from the pdb files if entered as coordinates, or from the protein and nucleic acid molecular weights if entered as a map) divided by the total scattering given by the composition, but entering the fraction scattering directly overrides this calculation. This option is for use when the pdb files of the models in the ensemble are unusual e.g. consist only of C-alpha atoms, or only of hydrogen atoms (as in the CLOUDS method for NMR).

How to Define Solutions

Phaser writes out files ending in ".sol" and ".rlist" that contain the solution information from the job. The root of the files is given by the ROOT keyword. By default, the root filename is PHASER. These files can be read back into subsequent runs of Phaser to build up solutions containing more than one molecule in the

asymmetric unit.

"PHASER.sol" files are generated by all modes (rotation function modes with VERBOSE output), and contain the current idea of potential molecular replacement solutions.

"PHASER.rlist" files are generated by the rotation function modes, and are used as input for performing translation functions.

For simple MR cases you don't really need to know how to define molecular replacement solutions. However, for difficult cases you might need to edit the files "PHASER.sol" and "PHASER.rlist" files manually

"sol" Files

SOLUtion 6DIM keywords describe Ensembles that have been oriented by a rotation search and positioned by a translation search. Each Ensemble in the asymmetric unit has its own SOLUtion keyword. When more than one (potential) molecular replacement solution is present, the solutions are separated with the SOLUTION SET keywords.

"rlist" Files

These files define a rotation function list. The peak list is given with a series of SOLUtion TRIAl keywords.

If a partial solution is already known, then the information for the currently "known" parts of the asymmetric unit is given in the form used for the PHASER.sol file, followed by the list of trial orientations for which a translation function is to be performed.

Fixed partial structure

If you have the coordinates of a partial solution with the pdb coordinates of the known structure in the correct orientation and position, then you can force Phaser to use these coordinates. Use the SOLUTION keyword to fix a rotation of 0 0 0 and a position of 0 0 0 for these coordinates.

How to Select Peaks

The selection of peaks saved for output in the rotation and translation functions can be done in four different ways.

- Select by Percentage Percentage of the top peak, where the value of the top peak is defined as 100% and the value of the mean is defined as 0%. Default, cutoff=75%. This criteria has the advantage that at least one peak (the top peak) always survives the selection. If the top solution is clear, then only the one solution will be output, but if the distribution of peaks is rather flat, then many peaks will be output for testing in the next part of the MR procedure (e.g. many peaks selected from the rotation function for testing with a translation function).

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- Select by Z-score Number of standard deviations (sigmas) over the mean (the Z-score). Absolute significance test. Not all searches will produce output if the cutoff value is too high (e.g. 5 sigma).

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- Select by Number Number of top peaks to select. If the distribution is very flat then it might be better to select a fixed large number (e.g. 1000) of top rotation peaks for testing in the translation function.

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- No selection All peaks are selected. Enables full 6 dimensional searches, where all the solutions from the rotation function are output for testing in the translation function. This should never be necessary; it would be much faster and probably just as likely to work if the top 1000 peaks were used in this way.

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Selection criteria

Peaks can also be clustered or not clustered prior to selection in steps 1 and 2.

- Clustering Off All high peaks on the search grid are selected

Clustering Off

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- Clustering On Points on the search grid with higher neighbouring points are removed from the selection

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Clustering

How to Control Output

The output of Phaser can be controlled with optional keywords.

The ROOT keyword is not compulsory (the default root filename is "PHASER"), but should always be given, so that your jobs have separate and meaningful output filenames.

The TOPFiles keyword controls the number of potential MR solutions for which PDB and (in the appropriate modes) MTZ files are produced.

For the MR_AUTO, MR_RNP and MR_LLG modes, unless HKLOut OFF is given as an optional keyword, Phaser produces an MTZ file with "SigmaA" type weighted Fourier map coefficients for producing electron density maps for rebuilding.

Translational Non-crystallographic Symmetry

Warning Solution by MR in the presence of translational non-crystallographic symmetry is not fully automated.

Phaser calculates correction factors for the expected intensities in the presence of translational non-crystallographic symmetry (tNCS), and is able to solve structures with complex patterns of tNCS. However, the use of Phaser in the presence of tNCS requires the nature of the tNCS to be understood by the user. In simple cases, solution is no more difficult than solution without tNCS, but in complex cases, separate Phaser runs with tNCS turned on and off, and/or the use of different tNCS vectors, may be necessary.

The output of Phaser will help the user in detecting and understanding the tNCS, but the tNCS is not completely characterised by Phaser. The default behaviour may or may not be correct for the particular crystal under study.

Characterization of the tNCS involves understanding the number of copies of the molecule in the asymmetric unit and the translation vectors between them. Molecules related by a tNCS vector will have an associated peak in the native Patterson. Phaser calculates the native Patterson (MODE TNCS) and lists the peaks that are more than 20% of the origin peak. Any given crystal with tNCS may have one or more peaks meeting this criteria.

Default tNCS detection and correction

Documentation for Phaser-2.7.16 and above

No tNCS

No tNCS correction is applied by default if there is

- no peak in the native Patterson
- more than one peak in the native Patterson over 20% of the origin and these peaks are not all the result of a commensurate modulation

Pairs of molecules

By default, if Phaser detects a peak in the native Patterson then Phaser will search for molecules in pairs related by the tNCS vector given by the peak in the native Patterson.

This will be the correct behaviour if and only if there are an even number of copies of the molecule in the asymmetric unit, clustered into two groups related by a single tNCS vector. There will only be one significant

peak in the native Patterson. Fortunately, this is a reasonably common scenario.

Phaser refines the relative orientation of the molecules in the two groups (rotations of up to 10 degrees will still give rise to a significant native Patterson peak) and uses this information to generate expected intensity factors for the reflections. Solution should be straightforward, with the usual caveat for MR that there is a sufficiently good model.

Where there is a single peak in the native Patterson, it is often located at a position half way along a unit cell axis or diagonal, representing a pseudo-halving of the unit cell dimensions. However, Phaser is by no means restricted to these sorts of pseudo-cells in its handling of two-fold tNCS, and the tNCS vector can be in a general position.

Non-default tNCS correction

Higher order tNCS

Frequently, tNCS does not associate 2 clusters of molecules in the asymmetric unit, but rather there are 3 or more (n) clusters of molecules associated by a series of vectors that are multiples of 1, 2, 3 ... (n-1) times a basic translation vector. Where n times the basic translation vector equates to (very close to) integer multiples of unit cell axes, the tNCS represents a pseudo-cell, and this case is known as commensurate modulation.

Phaser attempts to automatically detect commensurate modulation. The peaks of the native Patterson are analyzed to find the n-fold relationship. The series will not generally have all peaks the same height. Lower peaks in the series represent relationships where the relative rotations between related molecules are larger. Missing peaks in the series may be below the default 20% of origin cut-off. This can be lowered with TNCS PATT PERCENT

Phaser then sets TNCS NMOL and the vector for the tNCS, and searches for ensembles in multiples of NMOL.

When there are more than two molecules related by tNCS, Phaser does not refine the orientations between the molecules related by the tNCS.

However, as for two-fold tNCS, Phaser is not restricted to these sorts of pseudo-cells and the basic tNCS vector can be in a general position, as can the number of copies.

The automatic detection may not give the true tNCS relationship. For example, the true commensurate modulation may be a factor of the NMOL automatically detected by Phaser, or there may not be commensurate modulation at all, or commensurate modulation may not be found with the default Patterson peak height cutoff. In difficult cases, please inspect the Patterson for peaks.

Complex tNCS

If there are many molecules in the asymmetric unit but they are not all related by tNCS, or there are sub-groups of molecules related by different tNCS vectors, then the modulations of the expected intensities due to the tNCS will be much less significant than the cases described above. In these cases it is possible that structure solution will be achieved without any tNCS correction factors being applied. Indeed, searching for all the copies as tNCS-related multiples when some molecules are not related by tNCS will cause structure solution to fail. To turn off the automatic detection and use of tNCS use the keyword TNCS USE OFF.

If turning off the TNCS correction factors fails to give a solution, then a good approach is to proceed step-wise. Consider the highest native Patterson peak first and determine that nature of the tNCS associated with it. Use the appropriate correction factors to locate all the molecules with this tNCS. Then take the second

independent native Patterson peak and apply the correction factors associated with it to find the second set of molecules, fixing the first, etc. Finally, turn TNCS off to find any orphan molecules.

SAD and MR-SAD phasing with the Phaser GUI

phaser_ep.html

Contents

- Introduction
- Input files and mandatory parameters
- Output
- Frequently Asked Questions
- References

Introduction

Since the two Phaser GUIs use a common base, much of the experimental phasing interface is similar to the Phaser-MR GUI. This section only covers features unique to SAD phasing, including instructions for MR-SAD.

Phaser is used internally in the AutoSol wizard, which automates a complete experimental phasing pipeline including heavy atom search, phasing, density modification, and preliminary model-building. (It is also capable of running MR-SAD experiments.) For that reason, we recommend trying AutoSol first, and falling back to standalone Phaser for more difficult problems where finer control over parameters is required.

This document only covers configuration of the Phaser GUI; for a detailed explanation of the various terms and search procedures referenced here as well as interpretation of results, consult the documentation for AutoSol or Phaser, or the Phaser WIKI.

Input files and mandatory parameters

Phaser requires a reflections file (any format) containing anomalous data, and (for normal SAD mode) a file containing heavy atom sites, in either PDB or SHELX format. Either the wavelength or the energy used is also required. If you are using normal SAD mode, you may choose whether to use the given enantiomer ("hand") of the heavy-atom sites or the inverse or both.

To run an MR-SAD experiment, change the mode to "MR-SAD phasing", and enter a partial model (PDB file) and the expected variance from the actual structure (RMSD or percent identity). In MR-SAD, the chirality of the partial model restricts the sites to the given enantiomer. You could optionally provide a file containing heavy atom sites, but these are normally found by log-likelihood-gradient completion from the partial model; an advantage is that these sites are guaranteed to correspond to the same hand and origin as the partial model. Instead of a partial model specified with atoms, you can optionally provide an electron density map instead, by choosing "map" from the "Partial model type" pulldown.

The input for ASU contents is identical to the Phaser-MR GUI. Heavy atoms may be either standard elements from the periodic table, or cluster compounds such as tantalum bromide or tungsten clusters.

Output

Depending on the choice of hand, Phaser will output one or two sets of files containing the heavy atom sites (PDB format) and phased reflections (MTZ), including map coefficients and Hendrickson-Lattman phase probability distributions (useful for refining with experimental phase restraints). This map is usually very noisy and difficult to interpret, and typically requires density modification to become useful.

Frequently Asked Questions

How do I tell Phaser to look for more anomalous scatterers? Click on the "Composition and sites" tab. A control at the bottom labeled "Enable substructure completion by log-likelihood gradient maps" turns on this feature (and is already on by default). However, Phaser also needs to be told what type of scatterers to look for. These can be entered into the "Scattering types" box either manually or by clicking the "Select scatterer type" button. You may enter as many of these as you think are present and ordered in your crystal - note however that at the wavelengths commonly used for macromolecular crystallography, some atom types may be difficult to distinguish (such as sulfur and chlorine). How do I phase a structure with tantalum bromide clusters in Phenix? This is possible, but not really optimal at present. Phaser has built-in support for Ta₆Br₁₂ clusters, which have the scatterer type "TX". These are represented as spherically-averaged scatterers, which are a better representation than conventional atoms but not as good as an oriented atomic model of the actual cluster. The output sites are represented as single TX pseudo-atoms.

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- How do I phase a structure with tantalum bromide clusters in Phenix? This is possible, but not really optimal at present. Phaser has built-in support for Ta₆Br₁₂ clusters, which have the scatterer type "TX". These are represented as spherically-averaged scatterers, which are a better representation than conventional atoms but not as good as an oriented atomic model of the actual cluster. The output sites are represented as single TX pseudo-atoms.

References

McCoy AJ, Grosse-Kunstleve RW, Adams PD, Winn Md, Storoni LC, Read RJ. Phaser crystallographic software. *J. Appl. Cryst.* (2007) 40:658-674. McCoy AJ, Storoni LC, Read RJ. Simple algorithm for a maximum-likelihood SAD function. *Acta Cryst.* (2004) D60:1220-8.

- McCoy AJ, Grosse-Kunstleve RW, Adams PD, Winn Md, Storoni LC, Read RJ. Phaser crystallographic software. *J. Appl. Cryst.* (2007) 40:658-674.
- McCoy AJ, Storoni LC, Read RJ. Simple algorithm for a maximum-likelihood SAD function. *Acta Cryst.* (2004) D60:1220-8.

Molecular replacement with the Phaser GUI

phaser_mr.html

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- Overview
- Video Tutorial
- Input files and mandatory parameters
- Additional parameters
- Re-using previous results
- How to read a Phaser molecular replacement log file (Phaser-MR)
- What to do next after Phaser-MR molecular replacement
- Frequently Asked Questions
- Reference

Overview

MRage presents a user-friendly and automated frontend to Phaser, but especially difficult cases may require finer control over parameters. The Phaser-MR GUI exposes all options available in the standalone command-line Phaser software, and allows use of additional modes besides MR_AUTO. It can, for instance, be used to search for individual components one at a time with modified parameters, or run each stage of the search (rotation, translation, packing, phasing+refinement) separately.

There is also a simplified GUI for structures with only one component in the crystal, allowing the relevant files to be dragged and dropped into the interface. This should only be used for such simple cases, and when no difficulties are anticipated.

This document only covers configuration of the Phaser GUI; for a detailed explanation of the various terms and search procedures referenced here as well as interpretation of results, consult the overview of molecular replacement in PHENIX (especially the discussion of "ensembles" and "ASU contents"), the Phaser documentation, or the Phaser WIKI. Some common user questions are discussed at the end of this section, and in the MR overview page. New users looking for a step-by-step introduction should read the Phaser-MR tutorial.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- Basic overview
- How to run Phaser MR using the tutorial data.

Input files and mandatory parameters

The input section is similar to that of MRage, with separate sections for reflection data, search ensembles, and scattering information. Unlike the command-line version of Phaser, the experimental data may be any file format, and both intensities and amplitudes (with or without anomalous data) are allowed. The default mode is "Automated molecular replacement" (MR_AUTO); some of the other modes require only a subset of the

parameters displayed.

All modes require a description of the search ensembles, either as PDB files (singly, or superimposed as an ensemble), or map coefficients in MTZ format (e.g. for phasing using a low-resolution EM map). Adding an ensemble does not automatically include it in the search; you may, for instance, supply several alternate ensembles for a chain, and instruct Phaser to try each in turn (see below).

You must also define the contents of the asymmetric unit (ASU); this is usually done by supplying information about the sequence (one chain at a time), mass, or type and number of residues. Alternatively, you may give the fractional ASU contents of each search ensemble, if known. It is also possible to additionally specify heavy atoms that contribute to scattering, but this is rarely necessary. Note that the ASU contents data does not necessarily have a 1:1 correspondence with the search ensembles (see FAQ list below for details). Even if you are only searching for a single ensemble out of several (e.g. the protein in a protein-DNA complex), you must still supply the expected ASU contents of the entire crystal, because Phaser needs to know what fraction of the whole crystal content each search model comprises.

Finally, the "Search procedure" tab defines the steps in the MR search, selected from each input ensemble. This is organized as a tree-like table with search components as main branches. Expanding a component branch displays the ensembles used for the searches. Expanding an ensemble branch displays the name of the ensemble's constituent coordinate files or the name of the electron density file. Specifying more than one ensemble per component will result in the ensemble yielding the best solution being used. You do not need to search for all ensembles at once, as described below.

Additional parameters

The "Output" button at the top of the window controls basic behavior such as the resulting file names and verbosity. All other parameters are set by clicking the "Other settings" button, which will open a large dialog window with assorted controls. (Like most such windows in the Phenix GUI, only a subset of basic controls may be displayed at first, but this can be changed by selecting options in the "User level" menu at the bottom of the window.) The available parameters will be slightly different depending on which mode you have selected.

Re-using previous results

Phaser can be instructed to start from a previous partial solution; this does not use a partial PDB file, but rather the saved results of a previous run. For example, a two-part MR solution may be obtained by running in MR_AUTO mode to place one ensemble, then modifying the parameters, selecting the previous run as the partial solution, and searching for the second ensemble. To do this, select the result which you want to use as input from the menu labeled "Use partial solution from previous job" (all previous Phaser jobs for the current project will be listed in the menu). It is best to do this by modifying the configuration of the GUI from the previous Phaser run instead of starting from scratch, because the stored solutions do not contain the definitions of the search ensembles. This means that the existing ensemble definitions must be left unchanged when defining any ensembles needed for further searches.

You may also run each mode separately, e.g. the fast rotation function followed by the fast translation function using the previous result, and so on. There are some restrictions on which modes may use which results, thus you may not follow a fast rotation function with packing analysis.

How to read a Phaser molecular replacement log file (Phaser-MR)

A summary of your Phaser molecular replacement results can be found in the very last section of your Phaser-MR log file. It may look something like this, where everything between "AUTOMATED MOLECULAR REPLACEMENT" and "Citations for Phaser:" is part of the summary:

```
*** Phaser Module: AUTOMATED MOLECULAR REPLACEMENT 2.8.3 ***
*****

** SINGLE solution

** Solution written to PDB file: GUI_crystals_phaser.1.pdb
** Solution written to MTZ file: GUI_crystals_phaser.1.mtz
Solution annotation (history):
SOLU SET RF*0 TF*0 LLG=8776 TFZ==67.6 PAK=1 LLG=8776 TFZ==67.6
SOLU SPAC I 4
SOLU 6DIM ENSE ense_1 EULER 0.0 0.0 0.0 FRAC -1.00 -0.00 -1.00 BFAC -1.03 #TFZ==67.6
SOLU ENSEMBLE ense_1 VRMS DELTA -0.1435 #RMSD 0.45 #VRMS 0.24

CPU Time: 0 days 0 hrs 0 mins 13.59 secs ( 13.59 secs)
Finished: Sun Sep 28 10:32:36 2025

-----
EXIT STATUS: SUCCESS
-----

CPU Time: 0 days 0 hrs 0 mins 13.59 secs ( 13.59 secs)
Finished: Sun Sep 28 10:32:36 2025

</pre>
Citations for Phaser:
```

Here are the elements of this Phaser-MR summary:

1. Summary. The first thing you should note about your Phaser-MR result is whether you have found a single solution or multiple solutions:

```
** SINGLE solution
```

The run in this example Phaser-MR run yielded a single solution. Typically a run with a single solution is a confident run. A run with more than one solution (indicated by a line such as "There were 21 solutions") is typically low-confidence.

- Output model and reflections file from a Phaser-MR run:

```
** Solution written to PDB file: GUI_crystals_phaser.1.pdb
** Solution written to MTZ file: GUI_crystals_phaser.1.mtz
```

The output mtz file from Phaser-MR molecular replacement will contain the input X-ray data as well as coefficients for a sigmaA-weighted map.

- Table of information about the Phaser-MR solution and its history

```
Solution annotation (history):
SOLU SET RF*0 TF*0 LLG=8776 TFZ==67.6 PAK=1 LLG=8779 TFZ==70.2
SOLU SPAC I 4
SOLU 6DIM ENSE ense_1 EULER 0.0 0.0 0.0 FRAC -1.00 -0.00 -1.00 BFAC -1.03 #TFZ==67.6
SOLU ENSEMBLE ense_1 VRMS DELTA -0.1 #RMSD 0.7 #VRMS 0.6
```

The first line of this history gives the initial LLG and TFZ of the solution (8776 and 67.6), followed by the values after refinement (8779 and 70.2). The space group of the solution was I4. The rotation and translations are represented by the EULER and FRAC values. The VRMS is the refined RMS variance, RMSD is the input RMSD value, and DELTA is the shift with respect to this RMSD.

After molecular replacement with Phaser-MR, you are looking for a high final LLG value (60 or more usually indicates a confident solution), and a high TFZ (a good value may be in the range of 7 or more). Make sure the space group is plausible. You should expect your VRMS value to be similar to the input RMSD (small DELTA), and typically the VRMS value will be in the range of 0.5-1 for AlphaFold models and 1-2 for homology models.

What to do next after Phaser-MR molecular replacement

After Phaser-MR molecular replacement if you have obtained a single solution with a high value of LLG (over 60) and a high TFZ (7 or more) then you will want to look at your solution in detail. Have a look at the sigmaA-weighted map that you can draw with the Phaser output mtz file, along with the output Phaser model in the pdb file, for example using Coot. The map should match the model reasonably well, though you should not expect an exact fit. Next you may want to refine the model with PhenixRefine, or you may want to automatically rebuild the model with AutoBuild, or you may want to rebuild by hand.

If you have obtained multiple solutions in Phaser-MR molecular replacement, or if your LLG value is low (under 40) or your TFZ is low (under 6), you may want to look carefully at your input data with Xtriage, check your data processing, and check your search models to make sure that you are working with the best possible data, the right space group, and the best available search models. You may want to try other sources of a search model, such as homology models or predicted models.

Frequently Asked Questions

These are now covered on a separate section.

Reference

Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. J Appl Crystallogr 40, 658-674 (2007).

- Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. J Appl Crystallogr 40, 658-674 (2007).

Model preparation with sculpt_ensemble

sculpt_ensemble.html

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- Purpose
- Usage Input files Output files
- Input files
- Output files
- Description Trimming
- Trimming
- Command line Command-line switches PHIL arguments
- Command-line switches
- PHIL arguments
- Additional information List of all available keywords
- List of all available keywords

Purpose

sculpt_ensemble is a unified model preparation program that internally uses Sculptor and Ensembler. When these two programs are run separately, the result is dependent on what order these programs are run, and no matter what order the programs are run, there are corner cases that it is suboptimal. In the unified program no information loss occurs and there are further benefits as well.

Usage

The program sculpt_ensemble is available from the command line and will also be available from the PHENIX GUI.

Input files

- Structures (compulsory): structures to be superposed and modified. For each structure, specific parts can be selected using PHENIX atom selection syntax, and assembly information can be associated with the using the assembly keyword, or the convenience keyword read_single_assembly_from_file (for more information please see the Ensembler documentation). Error files associated with the input PDB file can also be input using the errors scope. See the Sculptor documentation for more details (please note that there is no input for superposition errors, since those are calculated within the ensembling step and passed onto Sculptor). Accepted formats: PDB. Recognized extensions: .pdb, .ent.

Structures (compulsory): structures to be superposed and modified. For each structure, specific parts can be selected using PHENIX atom selection syntax, and assembly information can be associated with the using the assembly keyword, or the convenience keyword read_single_assembly_from_file (for more information please see the Ensembler documentation).

Error files associated with the input PDB file can also be input using the errors scope. See the Sculptor documentation for more details (please note that there is no input for superposition errors, since those are calculated within the ensembling step and passed onto Sculptor).

Accepted formats: PDB. Recognized extensions: .pdb, .ent.

- Sequence files: target sequences (for each assembly position). This is used to identify the target sequence for each alignment. In case no alignment is available, and the target sequence for the given assembly position is available, sculpt_ensemble prepares an alignment using the sequence of the model chain (if no target sequence available, and no suitable alignment is found, a dummy alignment will be used instead). Accepted extensions are .fasta, .faa or .fa for FASTA format, .pir for PIR-format and .seq or .dat for a relaxed PIR/FASTA-like format.

Sequence files: target sequences (for each assembly position). This is used to identify the target sequence for each alignment. In case no alignment is available, and the target sequence for the given assembly position is available, sculpt_ensemble prepares an alignment using the sequence of the model chain (if no target sequence available, and no suitable alignment is found, a dummy alignment will be used instead). Accepted extensions are .fasta, .faa or .fa for FASTA format, .pir for PIR-format and .seq or .dat for a relaxed PIR/FASTA-like format.

- Alignments: sequence alignment with the target sequence. The alignment contains information that can be exploited in model improvement (this is currently only implemented for protein chains). The chains are automatically associated with the corresponding alignment based on a sequence comparison. The target sequence is also automatically identified if it is provided through the target_sequence keyword (see above), otherwise the first sequence in the alignment is used as target. Alignment input is optional, in case it is not provided, an alignment will be made up using the chain sequence with itself. Accepted extensions (with the corresponding format) are .aln (CLUSTAL format), .pir (PIR-format) and .ali (relaxed PIR-like format).

Alignments: sequence alignment with the target sequence. The alignment contains information that can be exploited in model improvement (this is currently only implemented for protein chains). The chains are automatically associated with the corresponding alignment based on a sequence comparison. The target sequence is also automatically identified if it is provided through the target_sequence keyword (see above), otherwise the first sequence in the alignment is used as target. Alignment input is optional, in case it is not provided, an alignment will be made up using the chain sequence with itself. Accepted extensions (with the corresponding format) are .aln (CLUSTAL format), .pir (PIR-format) and .ali (relaxed PIR-like format).

- Homology search files: hits from a homology search. These work very similarly to a set of alignment files. It is assumed that the first sequence in each alignment is the target sequence (the sequence used for searching homologues). Accepted extensions (with the corresponding format) are .xml (BLAST-family XML output) and .hhr (HHPRED output).

Homology search files: hits from a homology search. These work very similarly to a set of alignment files. It is assumed that the first sequence in each alignment is the target sequence (the sequence used for searching homologues). Accepted extensions (with the corresponding format) are .xml (BLAST-family XML output) and .hhr (HHPRED output).

Output files

The fully processed structure is output. The file is named according to the following convention: root_merged.pdb, where root is a user-defined parameter (accessible from the output scope), and is similar to merged output style from Ensembler.

Description

The program reads in the input files, creates assembly, and runs ensembling first. The assemblies are then transformed, superposition error is associated with respective chains, and processed sequentially according to the corresponding Sculptor protocol (note that although the superposition errors are always present, these are not used unless requested explicitly by enabling a protocol). See the respective documentation for more details.

In case there is only a single assembly present, the superposition step is skipped, and the program works as the standalone Sculptor. It is also possible to disable the processing step (`sculpting.disable`), in which case the program works as the standalone Ensembler.

Trimming

`sculpt_ensemble` also supports chain trimming, but unlike the explicit option in Ensembler, this is available as the superposition option within the mainchain processing section in the sculpting machinery. In addition, this can also be used in combination with polishing options.

Command line

```
phenix.sculpt_ensemble \
[ command-line switches ] \
[ PHIL-format parameter files ] \
[ PHIL command-line assignments ] \
[ PDB-files ] \
[ alignment files ] \
[ sequence files ]
```

Command-line switches

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
-v, --verbosity set verbosity level (info,debug,verbose)
--version show program's version number and exit
--text-logfile FILE Verbatim copy of log stream
--html-logfile FILE Verbatim copy of log stream in HTML
```

PHIL arguments

Everything not starting with a dash ('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters, command-line assignment or a file whose type is automatically recognized (based on file extension). Note that sequence files are not accepted on the command line, since associated chains could not easily be guessed and require a fully specified parameter scope.

Additional information

List of all available keywords

inputInput filesalignment = None Input alignment filetarget_sequence = None Target sequence for assembly
positionstructureInput structuresremove_alternate_conformations = False Remove alternate
conformationssanitize_occupancies = False Sets occupancies > 1.0 to 1.0modelInput model filefile_name =
None Input file nameselection = None Selection stringassembly = None Assembly
informationread_single_assembly_from_file = False Consider contents of the file as a single
assemblyerrorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs
the error file corresponds tohomology_searchAlignment from homology search filefile_name = None

Homology search fileuse = None Which alignments to useoutputOutput file(s)root = sculpt_ensemble Output
 file rootgui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the
 command line.job_title = None Job title in PHENIX GUI, not used on command
 linechain_to_alignment_matchingChain-to-alignment matching optionsconsecutivity = *geometry numbering
 Consecutivity criterion to detect chain breaksmin_hss_length = 3 Minimum length of a sequence fragment to
 be included in chain alignmentmax_seed_hss_count = 12 Number of HSS to use in extensive
 searchmax_completion_hss_count = 6 Number of HSS to use in gap fillingmin_sequence_overlap = 10
 Minimum overlap between sequences to perform full alignmentmin_sequence_identity = 0.80 Minimum
 sequence identity of accepted chain alignmenterror_searchParameters for matching error files and/or
 estimating errorsmin_sequence_identity = 0.8 Minimum sequence identity to accept error
 filemin_sequence_overlap = 0.8 Minimum sequence overlap to accept error filecalculate_if_not_provided =
 False Use Rosetta and the ProQ2 server to get error estimatesoutput_prefix = None File name for results of
 intermediate stepshomology_modellingParameters for homology modellingmin_residue_margin = 2 Minimum
 number of residues to cut back on both sides of loopsresidue_distance = 2.5 Distance covered by a single
 residue (in Å)min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap
 margins)min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap
 margins)max_loop_length = 60 Maximum loop lengthrosetta_max_build_attempts = 1000 Maximum build
 attempts to close looprosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop
 closuresrosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closurerosetta_loop_refinement = *no
 default quick fast test Algorithm for Rosetta loop refinementensemblingOptions for ensemblingmapping =
 None Residue mapping methods. Valid choices: resid, ssm, ssm_multiple_alignment, muscle,
 multiple_alignmentalignment = None Alignment file (only necessary for 'alignments' modes)atoms = None
 Atom to include in superpositionclustering = None Cutoff distance for cluster
 analysissuperpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence =
 1.0E-4 Convergence criterion for superpositionweightingParameters for weighting schemescheme = unit
 *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight
 iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit
 recovery if cumulative damping factor is aboveRobust_resistantSetting for robust-resistance schemecritical = 3
 tolerancesculptingOptions for sculptingdisable = False No sculptingproteinOptions to process protein
 chainscompletion = cbeta cbeta_and_pro sidechain Method to build missing sidechain
 atomsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not
 be matched to an alignmentdeletionMainchain deletion algorithmssuse = *gap threshold_based_similarity
 completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to
 usegapDelete residue if aligned with gapthreshold_based_similarityDelete residue if sequen...

- inputInput filesalignment = None Input alignment filetarget_sequence = None Target sequence for
 assembly positionstructureInput structuresremove_alternate_conformations = False Remove alternate
 conformationssanitize_occupancies = False Sets occupancies > 1.0 to 1.0modelInput model
 filefile_name = None Input file nameselection = None Selection stringassembly = None Assembly
 informationread_single_assembly_from_file = False Consider contents of the file as a single
 assemblyerrorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain
 IDs the error file corresponds tohomology_searchAlignment from homology search filefile_name = None
 Homology search fileuse = None Which alignments to use
- alignment = None Input alignment file
- target_sequence = None Target sequence for assembly position
- structureInput structuresremove_alternate_conformations = False Remove alternate
 conformationssanitize_occupancies = False Sets occupancies > 1.0 to 1.0modelInput model
 filefile_name = None Input file nameselection = None Selection stringassembly = None Assembly

informationread_single_assembly_from_file = False Consider contents of the file as a single assemblyerrorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds to

- remove_alternate_conformations = False Remove alternate conformations
- sanitize_occupancies = False Sets occupancies > 1.0 to 1.0
- modelInput model filefile_name = None Input file nameselection = None Selection stringassembly = None Assembly informationread_single_assembly_from_file = False Consider contents of the file as a single assemblyerrorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds to
- file_name = None Input file name
- selection = None Selection string
- assembly = None Assembly information
- read_single_assembly_from_file = False Consider contents of the file as a single assembly
- errorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds to
- file_name = None Error file name
- chain_ids = None Which chain IDs the error file corresponds to
- homology_searchAlignment from homology search filefile_name = None Homology search fileuse = None Which alignments to use
- file_name = None Homology search file
- use = None Which alignments to use
- outputOutput file(s)root = sculpt_ensemble Output file rootgui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.job_title = None Job title in PHENIX GUI, not used on command line
- root = sculpt_ensemble Output file root
- gui_output_dir = None Sets base output directory for Phenix GUI - not used when run from the command line.
- job_title = None Job title in PHENIX GUI, not used on command line
- chain_to_alignment_matchingChain-to-alignment matching optionsconsecutivity = *geometry numbering Consecutivity criterion to detect chain breaksmin_hss_length = 3 Minimum length of a sequence fragment to be included in chain alignmentmax_seed_hss_count = 12 Number of HSS to use in extensive searchmax_completion_hss_count = 6 Number of HSS to use in gap fillingmin_sequence_overlap = 10 Minimum overlap between sequences to perform full alignmentmin_sequence_identity = 0.80 Minimum sequence identity of accepted chain alignment
- consecutivity = *geometry numbering Consecutivity criterion to detect chain breaks
- min_hss_length = 3 Minimum length of a sequence fragment to be included in chain alignment
- max_seed_hss_count = 12 Number of HSS to use in extensive search
- max_completion_hss_count = 6 Number of HSS to use in gap filling
- min_sequence_overlap = 10 Minimum overlap between sequences to perform full alignment
- min_sequence_identity = 0.80 Minimum sequence identity of accepted chain alignment

- `error_searchParameters` for matching error files and/or estimating errors
`min_sequence_identity = 0.8` Minimum sequence identity to accept error file
`min_sequence_overlap = 0.8` Minimum sequence overlap to accept error file
`calculate_if_not_provided = False` Use Rosetta and the ProQ2 server to get error estimates
`output_prefix = None` File name for results of intermediate steps
- `homology_modellingParameters` for homology modelling
`min_residue_margin = 2` Minimum number of residues to cut back on both sides of loops
`residue_distance = 2.5` Distance covered by a single residue (in Å)
`min_edge_segment_length = 5` Discard edge segments if shorter (after considering gap margins)
`min_internal_segment_length = 2` Discard internal segments if shorter (after considering gap margins)
`max_loop_length = 60` Maximum loop length
`rosetta_max_build_attempts = 1000` Maximum build attempts to close loop
`rosetta_bump_overlap_factor = 0.1` Allows some atomic overlap in initial loop closures
`rosetta_loop_closure = kic *ngk` Algorithm for Rosetta loop closure
`rosetta_loop_refinement = *no` default quick fast test Algorithm for Rosetta loop refinement
- `min_sequence_identity = 0.8` Minimum sequence identity to accept error file
- `min_sequence_overlap = 0.8` Minimum sequence overlap to accept error file
- `calculate_if_not_provided = False` Use Rosetta and the ProQ2 server to get error estimates
- `output_prefix = None` File name for results of intermediate steps
- `homology_modellingParameters` for homology modelling
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- `rosetta_bump_overlap_factor = 0.1` Allows some atomic overlap in initial loop closures
- `rosetta_loop_closure = kic *ngk` Algorithm for Rosetta loop closure
- `rosetta_loop_refinement = *no` default quick fast test Algorithm for Rosetta loop refinement
- `ensemblingOptions` for ensembling
`mapping = None` Residue mapping methods. Valid choices: `resid`, `ssm`, `ssm_multiple_alignment`, `muscle`, `multiple_alignment`
`alignment = None` Alignment file (only necessary for 'alignments' modes)
`atoms = None` Atom to include in superposition
`clustering = None` Cutoff distance for cluster analysis
`superposition` Superposition setup
`method = *gapless` gapped Superposition algorithm
`convergence = 1.0E-4` Convergence criterion for superposition
`weightingParameters` for weighting schemes
`scheme = unit *robust_resistant` Weighting scheme
`convergence = 1.0E-3` Convergence criterion for weight iteration
`incremental_damping_factor = 1.5` Damping factor in recovery cycle
`max_damping_factor = 3.34` Quit recovery if cumulative damping factor is above
`robust_resistant` Setting for robust-resistance scheme
`critical = 3` tolerance

- mapping = None Residue mapping methods. Valid choices: resid, ssm, ssm_multiple_alignment, muscle, multiple_alignment
- alignment = None Alignment file (only necessary for 'alignments' modes)
- atoms = None Atom to include in superposition
- clustering = None Cutoff distance for cluster analysis
- superpositionSuperposition setupmethod = *gapless gapped Superposition algorithmconvergence = 1.0E-4 Convergence criterion for superposition
- method = *gapless gapped Superposition algorithm
- convergence = 1.0E-4 Convergence criterion for superposition
- weightingParameters for weighting schemescheme = unit *robust_resistant Weighting schemeconvergence = 1.0E-3 Convergence criterion for weight iterationincremental_damping_factor = 1.5 Damping factor in recovery cyclemax_damping_factor = 3.34 Quit recovery if cumulative damping factor is above robust_resistantSetting for robust-resistance schemecritical = 3 tolerance
- scheme = unit *robust_resistant Weighting scheme
- convergence = 1.0E-3 Convergence criterion for weight iteration
- incremental_damping_factor = 1.5 Damping factor in recovery cycle
- max_damping_factor = 3.34 Quit recovery if cumulative damping factor is above
- robust_resistantSetting for robust-resistance schemecritical = 3 tolerance
- critical = 3 tolerance
- sculptingOptions for sculptingdisable = False No sculptingproteinOptions to process protein chainscompletion = cbeta cbeta_and_pro sidechain Method to build missing sidechain atomsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to usegapDelete residue if aligned with gapthreshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.20 Threshold to accept a residuesimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingcompleteness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingresidue_count_based_similarityDelete residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to

useLinearParameters for linear averaging window = 5 Averaging window width weighting = uniform
 *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment
 positions affected by gaps maximum_distance = 10 Spatial distance for averaging remove_long Delete gap
 segments if longer than a threshold minimum_length = 3 Minimum length for mainchain segment to
 remove residues Delete residues if estimated error is larger than a threshold threshold = 5.0 Maximum allowed
 error for a residue missing_value_substitutionPolicy to fill in missing values use = *maximum_value
 scaled_interpolated_value Algorithm to use maximum_value Substitute maximum from the
 sequence scaled_interpolated_value Substitute interpolated value scaled with the
 distance extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation interpolation_step_scale
 = 1.10 Stepwise scale factor for extrapolation superposition Delete residues if estimated error is larger
 than a threshold threshold = 2.0 Maximum allowed error for a residue missing_value_substitutionPolicy to
 fill in missing values use = *maximum_value scaled_interpolated_value Algorithm to
 use maximum_value Substitute maximum from the sequence scaled_interpolated_value Substitute
 interpolated value scaled with the distance extrapolation_step_scale = 1.20 Stepwise scale factor for
 extrapolation interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation polishing Mainchain
 polishing algorithm use = remove_short undo_short keep_regular Algorithm to use remove_short Delete
 short unconnected segments minimum_length = 3 Minimum length undo_short Delete short
 gaps maximum_length = 2 Maximum length keep_regular Keep residues in secondary s...

- disable = False No sculpting

- proteinOptions to process protein chains completion = cbeta cbeta_and_pro sidechain Method to build
 missing sidechain atoms mainchainOptions for main chain processing remove_unaligned = True Delete
 residues that could not be matched to an alignment deletion Mainchain deletion algorithm use = *gap
 threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long
 rms superposition Algorithm to use gap Delete residue if aligned with gap threshold_based_similarity Delete
 residue if sequence similarity is low threshold = -0.20 Threshold to accept a
 residue similarity_calculation Configure sequence similarity calculation matrix = identity dayhoff blosum50
 *blosum62 Similarity matrix smoothing Configure raw similarity smoothing use = *linear spatial Method to
 use linearParameters for linear averaging window = 5 Averaging window width weighting = uniform
 *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment
 positions affected by gaps maximum_distance = 10 Spatial distance for
 averaging completeness_based_similarity Delete residues based on sequence similarity to get same
 number of gaps as the Schwarzenbacher algorithm offset = 0.0 Completeness in fraction of model length
 (0.0 = completeness from Schwarzenbacher algorithm, useful range:
 +/-0.05) similarity_calculation Configure sequence similarity calculation matrix = identity dayhoff blosum50
 *blosum62 Similarity matrix smoothing Configure raw similarity smoothing use = *linear spatial Method to
 use linearParameters for linear averaging window = 5 Averaging window width weighting = uniform
 *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment
 positions affected by gaps maximum_distance = 10 Spatial distance for
 averaging residue_count_based_similarity Delete residues based on sequence similarity to delete the
 same number of residues as the Schwarzenbacher algorithm offset = 0.0 Completeness in fraction of
 model length (0.0 = completeness from Schwarzenbacher algorithm, useful range:
 +/-0.05) similarity_calculation Configure sequence similarity calculation matrix = identity dayhoff blosum50
 *blosum62 Similarity matrix smoothing Configure raw similarity smoothing use = *linear spatial Method to
 use linearParameters for linear averaging window = 5 Averaging window width weighting = uniform
 *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment
 positions affected by gaps maximum_distance = 10 Spatial distance for averaging remove_long Delete gap
 segments if longer than a threshold minimum_length = 3 Minimum length for mainchain segment to

remove_rms Delete residues if estimated error is larger than a threshold threshold = 5.0 Maximum allowed error for a residue missing_value_substitutionPolicy to fill in missing values use = *maximum_value scaled_interpolated_value Algorithm to use maximum_value Substitute maximum from the sequence scaled_interpolated_value Substitute interpolated value scaled with the distance extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation superposition Delete residues if estimated error is larger than a threshold threshold = 2.0 Maximum allowed error for a residue missing_value_substitutionPolicy to fill in missing values use = *maximum_value scaled_interpolated_value Algorithm to use maximum_value Substitute maximum from the sequence scaled_interpolated_value Substitute interpolated value scaled with the distance extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation polishing Main chain polishing algorithm use = remove_short undo_short keep_regular Algorithm to use remove_short Delete short unconnected segments minimum_length = 3 Minimum length undo_short Delete short gaps maximum_length = 2 Maximum length keep_regular Keep residues in secondary structure maximum_length = 1 Maximum length bfactor Configure ...

- completion = cbeta cbeta_and_pro sidechain Method to build missing sidechain atoms
- mainchainOptions for main chain processing remove_unaligned = True Delete residues that could not be matched to an alignment deletion Main chain deletion algorithm use = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to use gap Delete residue if aligned with gap threshold_based_similarity Delete residue if sequence similarity is low threshold = -0.20 Threshold to accept a residue similarity_calculation Configure sequence similarity calculation matrix = identity dayhoff blosum50 *blosum62 Similarity matrix smoothing Configure raw similarity smoothing use = *linear spatial Method to use linearParameters for linear averaging window = 5 Averaging window width weighting = uniform *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment positions affected by gaps maximum_distance = 10 Spatial distance for averaging completeness_based_similarity Delete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithm offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05) similarity_calculation Configure sequence similarity calculation matrix = identity dayhoff blosum50 *blosum62 Similarity matrix smoothing Configure raw similarity smoothing use = *linear spatial Method to use linearParameters for linear averaging window = 5 Averaging window width weighting = uniform *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment positions affected by gaps maximum_distance = 10 Spatial distance for averaging residue_count_based_similarity Delete residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithm offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05) similarity_calculation Configure sequence similarity calculation matrix = identity dayhoff blosum50 *blosum62 Similarity matrix smoothing Configure raw similarity smoothing use = *linear spatial Method to use linearParameters for linear averaging window = 5 Averaging window width weighting = uniform *triangular Weighting scheme spatialParameters for spatial averaging gap_bleed_length = 3 Alignment positions affected by gaps maximum_distance = 10 Spatial distance for averaging remove_long Delete gap segments if longer than a threshold minimum_length = 3 Minimum length for main chain segment to remove_rms Delete residues if estimated error is larger than a threshold threshold = 5.0 Maximum allowed error for a residue missing_value_substitutionPolicy to fill in missing values use = *maximum_value scaled_interpolated_value Algorithm to use maximum_value Substitute maximum from the sequence scaled_interpolated_value Substitute interpolated value scaled with the distance extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation interpolation_step_scale

= 1.10 Stepwise scale factor for extrapolationsuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationpolishingMainchain polishing algorithmsuse = remove_short undo_short keep_regular Algorithm to useremove_shortDelete short unconnected segmentsminimum_length = 3 Minimum lengthundo_shortDelete short gapsmaximum_length = 2 Maximum lengthkeep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length

- remove_unaligned = True Delete residues that could not be matched to an alignment

- deletionMainchain deletion algorithmsuse = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to usegapDelete residue if aligned with gapthreshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.20 Threshold to accept a residuesimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingcompleteness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingresidue_count_based_similarityDelete residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingremove_longDelete gap segments if longer than a thresholdminimum_length = 3 Minimum length for mainchain segment to removermsDelete residues if estimated error is larger than a thresholdthreshold = 5.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationsuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- use = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to use
- gapDelete residue if aligned with gap
- threshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.20 Threshold to accept a residue
 similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
 linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- threshold = -0.20 Threshold to accept a residue
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
 linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
 linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- completeness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
 similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
 linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
 linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment

positions affected by gapsmaximum_distance = 10 Spatial distance for averaging

- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to use linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- residue_count_based_similarityDelete residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to use linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging

- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- remove_longDelete gap segments if longer than a thresholdminimum_length = 3 Minimum length for mainchain segment to remove
- minimum_length = 3 Minimum length for mainchain segment to remove
- rmsDelete residues if estimated error is larger than a thresholdthreshold = 5.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- threshold = 5.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- superpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- threshold = 2.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- polishingMainchain polishing algorithmsuse = remove_short undo_short keep_regular Algorithm to use
remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
lengthundo_shortDelete short gapsmaximum_length = 2 Maximum lengthkeep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
- use = remove_short undo_short keep_regular Algorithm to use
- remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- minimum_length = 3 Minimum length
- undo_shortDelete short gapsmaximum_length = 2 Maximum length
- maximum_length = 2 Maximum length
- keep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
- maximum_length = 1 Maximum length
- bfactorConfigure B-factor modificationuse = *original asa similarity rms Algorithm to useminimum_b = 10.00 Minimum B-factororiginalUse original bfactors to predict new B-valuesfactor = 1.00 Scale factorasaUse accessible surface area to predict new B-valuesprecision = 960 Number of points per atomprobe_radius = 1.40 Radius for probing surface accessibilityfactor = 2.00 Scale factorsimilarityUse sequence similarity to predict new B-valuesfactor = -100.00 Scale factorsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to useParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingrmsUse external RMS error estimates to calculate B-valuesfactor = 1.00 Scale factormissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *original asa similarity rms Algorithm to use
- minimum_b = 10.00 Minimum B-factor
- originalUse original bfactors to predict new B-valuesfactor = 1.00 Scale factor
- factor = 1.00 Scale factor
- asaUse accessible surface area to predict new B-valuesprecision = 960 Number of points per atomprobe_radius = 1.40 Radius for probing surface accessibilityfactor = 2.00 Scale factor
- precision = 960 Number of points per atom
- probe_radius = 1.40 Radius for probing surface accessibility
- factor = 2.00 Scale factor
- similarityUse sequence similarity to predict new B-valuesfactor = -100.00 Scale factorsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to useParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- factor = -100.00 Scale factor

- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- rmsUse external RMS error estimates to calculate B-valuesfactor = 1.00 Scale factormissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- factor = 1.00 Scale factor
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- renumberResidue renumberinguse = *target model original Method to usestart = 1 Start residue number
- use = *target model original Method to use
- start = 1 Start residue number

- pruningOptions for sidechain pruninguse = null *schwarzenbacher similarity Algorithm to usepruning_level_unaligned = 2 Pruning level for residues that could not be matched to an alignmentnullDo not impose bond distance thresholdschwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncationsimilarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncationsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = null *schwarzenbacher similarity Algorithm to use
- pruning_level_unaligned = 2 Pruning level for residues that could not be matched to an alignment
- nullDo not impose bond distance threshold
- schwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncation
- pruning_level = 2 Level of truncation
- similarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncationsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- pruning_level = 2 Level of intermediate truncation
- full_length_limit = 0.2 Limit of no truncation
- full_truncation_limit = -0.2 Limit for full truncation
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 1 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging

- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- renameResidue renaminguse = original *target Method to usekeep_ptm = False Preserve post-translational modification of model residue if base residue types agreegapname = ALA Name residues corresponding to alignment gaps
- use = original *target Method to use
- keep_ptm = False Preserve post-translational modification of model residue if base residue types agree
- gapname = ALA Name residues corresponding to alignment gaps
- mappingOptions for sidechain mappinguse = *connectivity geometry Algorithm to usemap_if_identical = True Do mapping procedure for identical residues typesconnectivityMatch atoms by connectivitymatch_chemical_elements = True Take chemical element into accountgeometryMatch atoms by geometry considering all rotamersmatch_chemical_elements = False Take chemical element into accounttolerance = 0.1 Distance tolerancefine_sampling = False Use fine sampling for rotamers
- use = *connectivity geometry Algorithm to use
- map_if_identical = True Do mapping procedure for identical residues types
- connectivityMatch atoms by connectivitymatch_chemical_elements = True Take chemical element into account
- match_chemical_elements = True Take chemical element into account
- geometryMatch atoms by geometry considering all rotamersmatch_chemical_elements = False Take chemical element into accounttolerance = 0.1 Distance tolerancefine_sampling = False Use fine sampling for rotamers
- match_chemical_elements = False Take chemical element into account
- tolerance = 0.1 Distance tolerance
- fine_sampling = False Use fine sampling for rotamers
- dnaOptions to process DNA chainsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationpolishingMainchain polishing algorithmuse = remove_short Algorithm to useremove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- mainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationpolishingMainchain polishing algorithmuse = remove_short Algorithm to useremove_shortDelete short unconnected

segmentsminimum_length = 3 Minimum length

- remove_unaligned = True Delete residues that could not be matched to an alignment
- deletionMainchain deletion algorithmsuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *all superposition Algorithm to use
- allDelete all residues
- superpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- threshold = 2.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- polishingMainchain polishing algorithmsuse = remove_short Algorithm to useremove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- use = remove_short Algorithm to use
- remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- minimum_length = 3 Minimum length
- rnaOptions to process RNA chainsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale

= 1.10 Stepwise scale factor for extrapolation
polishingMainchain polishing algorithms
use = remove_short
Algorithm to useremove_shortDelete short unconnected segments
minimum_length = 3 Minimum length

- mainchainOptions for main chain processing
remove_unaligned = True Delete residues that could not be matched to an alignment
deletionMainchain deletion algorithms
use = *all superposition Algorithm to use
allDelete all residues
superpositionDelete residues if estimated error is larger than a threshold
threshold = 2.0 Maximum allowed error for a residue
missing_value_substitutionPolicy to fill in missing values
use = *maximum_value scaled_interpolated_value Algorithm to use
maximum_valueSubstitute maximum from the sequence
scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
polishingMainchain polishing algorithms
use = remove_short Algorithm to useremove_shortDelete short unconnected segments
minimum_length = 3 Minimum length

- remove_unaligned = True Delete residues that could not be matched to an alignment

- deletionMainchain deletion algorithms
use = *all superposition Algorithm to use
allDelete all residues
superpositionDelete residues if estimated error is larger than a threshold
threshold = 2.0 Maximum allowed error for a residue
missing_value_substitutionPolicy to fill in missing values
use = *maximum_value scaled_interpolated_value Algorithm to use
maximum_valueSubstitute maximum from the sequence
scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
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- use = *all superposition Algorithm to use

- allDelete all residues

- superpositionDelete residues if estimated error is larger than a threshold
threshold = 2.0 Maximum allowed error for a residue
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maximum_valueSubstitute maximum from the sequence
scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- threshold = 2.0 Maximum allowed error for a residue

- missing_value_substitutionPolicy to fill in missing values
use = *maximum_value scaled_interpolated_value Algorithm to use
maximum_valueSubstitute maximum from the sequence
scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- use = *maximum_value scaled_interpolated_value Algorithm to use

- maximum_valueSubstitute maximum from the sequence

- scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation

- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- polishingMainchain polishing algorithms
use = remove_short Algorithm to useremove_shortDelete short unconnected segments
minimum_length = 3 Minimum length

- use = remove_short Algorithm to use

- `remove_short` Delete short unconnected segments `minimum_length = 3` Minimum length
- `minimum_length = 3` Minimum length
- `hetero` Options to process hetero chains `keep = None` Keep named hetero residues
- `keep = None` Keep named hetero residues
- `monosaccharide` Options to process glycosyl chains `maximum_depth = 0` Keep chain up to maximum depth (None to keep all) `maximum_bond_length = 1.5` Maximum bond length for glycosidic bond
- `maximum_depth = 0` Keep chain up to maximum depth (None to keep all)
- `maximum_bond_length = 1.5` Maximum bond length for glycosidic bond

Model editing with Sculptor

sculptor.html

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Author(s)

- sculptor: Gabor Bunkoczi
- PHENIX GUI: Nathaniel Echols

Purpose

Sculptor can be used to improve a molecular replacement model using additional information available from an alignment and/or structure. It is based on an algorithm outlined in Schwarzenbacher et al. (2004).

See also

phenix.sculptor and phenix.ensembl have been superseded by phenix.sculpt_ensemble

Conventions

The following terms are used with the special meaning:

- target: the structure to be solved
- model: the structure used as a model for the target

Usage

Sculptor can be run from the PHENIX GUI and the command line, the only difference being the way commands are taken from the user.

The graphical user interface makes all settings accessible either as part of the main window (for frequently used options) or through a series of dialog boxes under Settings for::

- Main-chain removal
- Main-chain polishing
- Sidechain pruning
- B-factors
- Renumbering
- Renaming

The input PDB file is specified in the PDB file: input line, while alignments and target sequences can be added through the Sequence alignment files... and the Sequence files... dialog boxes, respectively.

Input files

- Structures (compulsory). The structure to be modified. Specific parts can be selected using CNS-style atom selection syntax. Chains are divided into protein and other chain categories, and processed according to instructions given for the appropriate chain type. Accepted formats: PDB. Recognized extensions: .pdb, .ent.
- Alignments Provides reliability information that can be used for calculating modifications for protein chains. The chains are automatically associated with the corresponding alignment based on a sequence comparison. No alignment information is used when processing other chains. If no calculation needs alignment information, alignment input is optional, otherwise compulsory. Multiple alignment files can be provided to cover all distinct protein chains in the structure. By convention, the first sequence of the alignment is the target sequence, but this can be overridden by using the target index parameter of the alignment scope. Accepted extensions (with the corresponding format) are .aln (CLUSTAL format), .pir (PIR-format) and .ali (relaxed PIR-like format).
- Sequence files In case no alignment is available, sculptor can prepare an alignment if the target sequence and the corresponding model chain selection is provided. Multiple sequence files can be provided with different target sequences for distinct protein chains. Accepted extensions are .fasta, .faa

or .fa for FASTA format, .pir for PIR-format and .seq or .dat for a relaxed PIR/FASTA-like format.

Output files

In flexible mode, the fully processed structure is output. The file is named according to the following convention: root_pdb.pdb, where root is a user-defined parameter (accessible from the output scope), and pdb is the basename of the input PDB file. In predefined mode, there is an output file produced for each requested protocol, and named according to root_pdb_N.pdb, where N is the number of the corresponding protocol.

Outline of the procedure

The workflow consists of several stages that can be independently configured. These are listed in order of execution. For a summary of all keywords with the corresponding defaults, see the Additional information section.

Preprocessing

- selection: selects a subset of the input PDB file, using CNS-style atom selection syntax. Default: all.
- remove_alterate_conformations: selects the first alternate conformation for disordered entities, and discards the rest. Also involves sanitize_occupancies.
- sanitize_occupancies: resets all occupancies to 1.0.

In addition, chains will be analysed, and solvent atoms will be separated from protein chains if they are not separated by TER cards.

Protein chains

Deletion

Discards residues from a model chain that are unlikely to improve signal in molecular replacement. This information is calculated from the alignment.

There are multiple algorithms available:

- gap: Deletes residues that are not present in the target (model residue is aligned with a gap). For this algorithm, the supplied alignment is used as a pairwise alignment.
- threshold_based_similarity: Deletes residues for which the sequence similarity is below a certain threshold. All sequences in a multiple alignment contribute to the score. Details of the sequence similarity calculation are given in the section Sequence similarity calculation.
- completeness_based_similarity: Deletes the same number of residues (modified by a fractional offset) as the gap algorithm would but residues that get removed are the ones with the lowest sequence similarity. This way the default values are valid over a much larger sequence similarity range than those in threshold_based_similarity. All sequences in a multiple alignment contribute to the score. Details of the sequence similarity calculation are given in the section Sequence similarity calculation.

These algorithms can also be used together in any combination. In this case, a residue will be deleted if assigned for deletion by any active algorithms.

Polishing

Makes small adjustments to the mainchain of a chain (taking results from deletion into account) to make it obey basic macromolecular features.

- **remove_short.** Deletes additional residue segments from the molecule so that no continuous segment is shorter than a preset limit (determined by the `minimum_length` parameter of the `remove_short` scope). Segment boundaries are determined from spatial connectivity of residues. This algorithm is primarily intended to remove "floating" residues that are the result of extensive loop truncation.
- **keep_regular.** Reinstates deleted residues if they are in regular secondary structure and the segment marked for deletion is shorter than the `maximum_length` parameter of the `keep_regular` scope.

These algorithms can also be used together in combination. In this case, the chain will be processed sequentially by both algorithms.

Pruning

This phase determines the level distance from the Calpha atom up to which a residue sidechain in the model is potentially similar to its counterpart in the target.

- **schwarzenbacher.** Implements the algorithm published by Schwarzenbacher et al. (2004), who propose that for optimal molecular replacement results a residue sidechain should be truncated if aligned with a non-identical residue, and not truncated otherwise. The level of truncation is controlled by the `pruning_level` parameter, and defaults to 3 (which corresponds to Cgamma) and can be controlled by the `pruning_level` parameter of the `schwarzenbacher` scope.
- **Similarity.** Uses sequence similarity values for deciding the level of truncation. Residues above `full_truncation_limit` are not truncated at all, those below the `full_truncation_limit` are truncated to Cbeta, and those in between are truncated according to the `pruning_level` parameter (all available from the similarity scope). Results tend to be similar to those given by the Schwarzenbacher algorithm; however, it is possible to get high similarity values (and full sidechain preservation) for certain substitutions (i.e. TYR to PHE), and low-sequence similarity zones can end up being truncated to Cbeta. Details of the sequence similarity calculation are given in the section [Sequence similarity calculation](#).

These algorithms can also be used together in any combination, in which case the sidechain will be truncated to the shortest value suggested.

B-factor prediction

B-factor prediction tries to increase B-factors for atoms that are likely to be more flexible or more in error. The calculation takes simple physical properties into account, and these are linearly transformed to B-factors (controlled by the `factor` parameter of the corresponding scope). If this value is lower than the minimum (from the `bfactorscope`) parameter, a constant is added to all B-factors so that the lowest of those equals to minimum (this is primarily intended to avoid negative B-factors).

- **original.** This uses the original B-factor of atoms. This is primarily intended as a contributor to a combination, but can also be used to manipulate current B-factors, e.g. set them to a constant value.
- **asa.** This calculates accessible surface area for an isolated chain and transforms the raw values to B-factors. A high ASA-value indicates a potential for flexibility. The calculation can be configured by the `precision` and `probe_radius` parameters of the `asa` scope.
- **similarity.** Low sequence similarity regions tend to be more dissimilar. Details of the sequence similarity calculation are given in the section [Sequence similarity calculation](#).

Algorithms can be used in combination, in which case the sum of the predicted B-factors is used. This mode can also be used to map sequence similarity or accessible surface area to residues/atoms for display purposes.

Renumber

Renumbers residues according to the target or model sequence. It is also possible to turn renumbering off (option original).

Rename

Renames residues according their counterpart in the target sequence. It also "morphs" the sidechain, i.e. renames atoms and deletes atoms that are not present. It can also generate missing atoms, if their positions are determined unambiguously by present atoms (available via the completion parameter of the macromolecule scope).

- cbeta. Adds Cbeta atom if the residue is not glycine, and C, N and Calpha atoms are all present.

Non-macromolecular chains

Residues in these chains are normally deleted, unless an exception is made by specifying the residue codes that are to be retained. This is primarily intended to keep a known ligands of protein classes (e.g. HEM).

Sequence similarity calculation

Sequence similarity is calculated from the full alignment supplied (taking all present sequences into account), using a scoring matrix (currently blosum50, blosum62, dayhoff and identity are available). Raw scores are then averaged over a window of residues (defaults to 5 residues in both directions) that is weighted using either uniform or triangular weights. The resulting scores are "normalized" so that 1.0 would indicate a perfect alignment, 0.0 would be a random match, and -1.0 a (locally) fully gapped alignment (on average). Note, it is possible to obtain values outside this range. This helps to ensure that defaults are sensible irrespective of the choice for the scoring matrix.

Sequence similarity calculation is configured individually for the steps that are using it.

Command line

```
phenix.sculptor \  
[ command-line switches ] \  
[ PHIL-format parameter files ] \  
[ PHIL command-line assignments ] \  
[ PDB-files ] \  
[ alignment files ]
```

Command-line switches:

```
-h, --help show this help message and exit  
--show-defaults print PHIL and exit  
-i, --stdin read PHIL from stdin as well  
-v, --verbosity set verbosity level (DEBUG, INFO, WARNING, VERBOSE)  
--mode set mode (flexible, predefined)
```

PHIL arguments:

Everything not starting with a dash ('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters, command-line assignment or a file whose type is automatically recognized (based on

file extension). Note that sequence files are not accepted on the command line, since associated chains could not easily be guessed and require a fully specified parameter scope.

Specific limitations and possible problems

Processing features

- Very short residue segments (shorter than 3 consecutive residues) cannot be reliably aligned to the sequence, and these will be discarded from the model.
- The similarity algorithm from the deletion scope may result in residues that are aligned with a gap being included in the model. Although this possibly indicates an error in the alignment and is potentially beneficial for molecular replacement, this causes a problem at the rename stage, as there is no 3-letter residue name for a "-"; these residues are tentatively named GAP.
- Residue numbers for GAP residues are built up using the residue number of the previous non-GAP residue and an insertion code (A-Z, depending on the number of GAP residues after the previous non-GAP residue).

Error messages

- No pdb files specified: there are no PDB files to process.
- No atoms left after atom selection: the atom selection provided results in an empty structure.
- No alignment: no alignments have been provided and the calculation requires alignment information.
- No sufficiently similar alignment sequences have been found: the longest exact overlap between the chain sequence and any alignment sequences is lower than the min_hssp_length parameter (typically 6 residues), therefore no alignment corresponds to this chain.
- Unable to align: matching fraction < min_matching_fraction: the best matching alignment does not match the chain sufficiently (specified by the min_matching_fraction parameter, typically 40%), and it is likely that the alignment is incorrect.

Warning messages

- There are N sequences with longest overlap: N sequences give identical matching statistics with the current chain. Lacking other criteria, the first is selected. Ordering is affected by the sequence of alignment files passed to the program and the order of sequences in the alignment file.
- Unaligned residues: certain residues could not be aligned reliably with the sequence because they appear in very short segments and sequence matching can be arbitrary. The minimum length of an accepted segments is controlled by the min_hssp_length parameter.
- Sequence mismatches: there are mismatches between the alignment and the chain sequence. These are typically caused by unknown residues codes assigned to uncommon or modified residues. If the sequence identity falls below a preset threshold (controlled by the min_matching_fraction parameter), and error is raised.
- File contains multiple sequences, only the first will be used: the sequence file used to provide the target sequence for given chains contains multiple sequences. The first sequence will be accepted as correct.

References

Citation

Improvement of molecular-replacement models with Sculptor. G. Bunkoczi and R. J. Read Acta Cryst. D67, 303-312 (2011)

Additional information

List of all available keywords

hetero = None Keep named hetero residues
min_hssp_length = 6 Length of residue segment that indicates a reliable match
min_matching_fraction = 0.4 Minimum matching fraction in residue-to-alignment
matching_input Input file
alignment_from_homology_search = None Input homology search to use hits as alignments
model Input pdb file
file_name = None PDB file name
selection = all Selection
string_remove_alternate_conformations = False Remove alternate conformations
sanitize_occupancies = False Sets occupancies > 1.0 to 1.0
keep_crystal_symmetry = False Keeps the crystal symmetry of the model
alignment Input alignment file
file_name = None target_index = 1 Index of target sequence in alignment
sequence Input sequence file
file_name = None Sequence file
chain_ids = None output Output options
job_title = None Job title in PHENIX GUI, not used on command line
folder = . Output file folder
root = sculpt Output file root
format = *pdb coot Output file format
macromolecule Workflow step
configuration completion = *cbeta Sidechain completion algorithms (* = active)
rename = True True: enable; False: disable
keep_ptm_if_base_residues_agree = False Keep post-translational modification if residues agree
deletion Configure mainchain deletion
use = *gap threshold_based_similarity completeness_based_similarity remove_long Available algorithms (* = active)
gap Delete residue if aligned with gap
threshold_based_similarity Delete residue if sequence similarity is low
threshold = -0.2 Threshold to accept a residue
calculation Configure sequence similarity calculation
matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
window = 5 Averaging window width
weighting = uniform *triangular Weighting scheme
completeness_based_similarity Delete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithm
offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
calculation Configure sequence similarity calculation
matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
window = 5 Averaging window width
weighting = uniform *triangular Weighting scheme
remove_long Delete residue if aligned with gap
min_length = 3 Minimum length for mainchain segment to remove
polishing Configure mainchain polishing
use = remove_short keep_regular Available algorithms (* = active)
remove_short Delete short unconnected segments
minimum_length = 3 Minimum length
keep_regular Keep residues in secondary structure
maximum_length = 1 Maximum length
pruning Configure sidechain pruning
use = *schwarzenbacher similarity Available algorithms (* = active)
schwarzenbacher Truncate atoms if target residue != source residue
pruning_level = 2 Level of truncations
similarity Truncate atoms based on sequence similarity
pruning_level = 2 Level of intermediate truncation
full_length_limit = 0.2 Limit of no truncation
full_truncation_limit = -0.2 Limit for full truncation
calculation Configure sequence similarity calculation
matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
window = 5 Averaging window width
weighting = uniform *triangular Weighting scheme
bfactor Configure bfactor prediction
use = *original similarity asa Available algorithms (* = active)
minimum = 10 Minimum allowed value (a constant is added if any B-factors would fall below this value)
original Use original b-factors to predict new B-values
bfactor = 1 Transform values by multiplying with a factor
similarity Use sequence similarity to predict new B-values
bfactor = -100 Transform values by multiplying with a factor
calculation Configure sequence similarity calculation
matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
window = 5 Averaging window width
weighting = uniform *triangular Weighting scheme
asa Use accessible surface area to predict new B-values
bfactor = 2 Transform values by multiplying with a factor
precision = 960 Number of points per atom
probe_radius = 1.4 Radius for

probing surface accessibility
renumberuse = *target model original Mainchain numbering; (* = selec...

- hetero = None Keep named hetero residues
- min_hssp_length = 6 Length of residue segment that indicates a reliable match
- min_matching_fraction = 0.4 Minimum matching fraction in residue-to-alignment matching
- inputInput filesalignment_from_homology_search = None Input homology search to use hits as alignmentsmodelInput pdb filefile_name = None PDB file nameselection = all Selection stringremove_alternate_conformations = False Remove alternate conformationssanitize_occupancies = False Sets occupancies > 1.0 to 1.0keep_crystal_symmetry = False Keeps the crystal symmetry of the modelalignmentInput alignment filefile_name = None target_index = 1 Index of target sequence in alignmentsequenceInput sequence filefile_name = None Sequence filechain_ids = None
- alignment_from_homology_search = None Input homology search to use hits as alignments
- modelInput pdb filefile_name = None PDB file nameselection = all Selection stringremove_alternate_conformations = False Remove alternate conformationssanitize_occupancies = False Sets occupancies > 1.0 to 1.0keep_crystal_symmetry = False Keeps the crystal symmetry of the model
- file_name = None PDB file name
- selection = all Selection string
- remove_alternate_conformations = False Remove alternate conformations
- sanitize_occupancies = False Sets occupancies > 1.0 to 1.0
- keep_crystal_symmetry = False Keeps the crystal symmetry of the model
- alignmentInput alignment filefile_name = None target_index = 1 Index of target sequence in alignment
- file_name = None
- target_index = 1 Index of target sequence in alignment
- sequenceInput sequence filefile_name = None Sequence filechain_ids = None
- file_name = None Sequence file
- chain_ids = None
- outputOutput optionsjob_title = None Job title in PHENIX GUI, not used on command linefolder = . Output file folderroot = sculpt Output file rootformat = *pdb coot Output file format
- job_title = None Job title in PHENIX GUI, not used on command line
- folder = . Output file folder
- root = sculpt Output file root
- format = *pdb coot Output file format
- macromoleculeWorkflow step configurationcompletion = *cbeta Sidechain completion algorithms (* = active)rename = True True: enable; False: disablekeep_ptm_if_base_residues_agree = False Keep post-translational modification if residues agreedeletionConfigure mainchain deletionuse = *gap threshold_based_similarity completeness_based_similarity remove_long Available algorithms (* = active)gapDelete residue if aligned with gapthreshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.2 Threshold to accept a residuecalculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemecompleteness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0

Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
 calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
 remove_longDelete residue if aligned with gapmin_length = 3 Minimum length for mainchain segment to remove
 polishingConfigure mainchain polishinguse = remove_short keep_regular Available algorithms (* = active)
 remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum lengthkeep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
 pruningConfigure sidechain pruninguse = *schwarzenbacher similarity Available algorithms (* = active)
 schwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncationsimilarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncation
 calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
 bfactorConfigure bfactor predictionuse = *original similarity asa Available algorithms (* = active)minimum = 10 Minimum allowed value (a constant is added if any B-factors would fall below this value)
 originalUse original b-factors to predict new B-valuesfactor = 1 Transform values by multiplying with a factor
 similarityUse sequence similarity to predict new B-valuesfactor = -100 Transform values by multiplying with a factor
 calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
 asaUse accessible surface area to predict new B-valuesfactor = 2 Transform values by multiplying with a factor
 precision = 960 Number of points per atomprobe_radius = 1.4 Radius for probing surface accessibility
 renumberuse = *target model original Mainchain numbering; (* = selected; None: disable)start = 1 Number for first residue

- completion = *cbeta Sidechain completion algorithms (* = active)
- rename = True True: enable; False: disable
- keep_ptm_if_base_residues_agree = False Keep post-translational modification if residues agree
- deletionConfigure mainchain deletionuse = *gap threshold_based_similarity completeness_based_similarity remove_long Available algorithms (* = active)
 gapDelete residue if aligned with gapthreshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.2 Threshold to accept a residue
 calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
 completeness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
 calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
 remove_longDelete residue if aligned with gapmin_length = 3 Minimum length for mainchain segment to remove
- use = *gap threshold_based_similarity completeness_based_similarity remove_long Available algorithms (* = active)
- gapDelete residue if aligned with gap
- threshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.2 Threshold to accept a residue
 calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- threshold = -0.2 Threshold to accept a residue

- calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- completeness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
- calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- remove_longDelete residue if aligned with gapmin_length = 3 Minimum length for mainchain segment to remove
- min_length = 3 Minimum length for mainchain segment to remove
- polishingConfigure mainchain polishinguse = remove_short keep_regular Available algorithms (* = active)remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum lengthkeep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
- use = remove_short keep_regular Available algorithms (* = active)
- remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- minimum_length = 3 Minimum length
- keep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
- maximum_length = 1 Maximum length
- pruningConfigure sidechain pruninguse = *schwarzenbacher similarity Available algorithms (* = active)schwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncationsimilarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncationcalculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- use = *schwarzenbacher similarity Available algorithms (* = active)
- schwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncation
- pruning_level = 2 Level of truncation
- similarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncationcalculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting

scheme

- pruning_level = 2 Level of intermediate truncation
- full_length_limit = 0.2 Limit of no truncation
- full_truncation_limit = -0.2 Limit for full truncation
- calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- bfactorConfigure bfactor predictionuse = *original similarity asa Available algorithms (* = active)minimum = 10 Minimum allowed value (a constant is added if any B-factors would fall below this value)originalUse original bfactors to predict new B-valuesfactor = 1 Transform values by multiplying with a factorsimilarityUse sequence similarity to predict new B-valuesfactor = -100 Transform values by multiplying with a factorcalculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemeasaUse accessible surface area to predict new B-valuesfactor = 2 Transform values by multiplying with a factorprecision = 960 Number of points per atomprobe_radius = 1.4 Radius for probing surface accessibility
- use = *original similarity asa Available algorithms (* = active)
- minimum = 10 Minimum allowed value (a constant is added if any B-factors would fall below this value)
- originalUse original bfactors to predict new B-valuesfactor = 1 Transform values by multiplying with a factor
- factor = 1 Transform values by multiplying with a factor
- similarityUse sequence similarity to predict new B-valuesfactor = -100 Transform values by multiplying with a factorcalculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- factor = -100 Transform values by multiplying with a factor
- calculationConfigure sequence similarity calculationmatrix = *identity dayhoff blosum50 blosum62 Similarity matrixwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- matrix = *identity dayhoff blosum50 blosum62 Similarity matrix
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- asaUse accessible surface area to predict new B-valuesfactor = 2 Transform values by multiplying with a factorprecision = 960 Number of points per atomprobe_radius = 1.4 Radius for probing surface accessibility
- factor = 2 Transform values by multiplying with a factor
- precision = 960 Number of points per atom
- probe_radius = 1.4 Radius for probing surface accessibility
- renumberuse = *target model original Mainchain numbering; (* = selected; None: disable)start = 1 Number for first residue

- use = *target model original Mainchain numbering; (* = selected; None: disable)
- start = 1 Number for first residue

Model editing with Sculptor

sculptor_new.html

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Purpose

Sculptor can be used to improve a molecular replacement model using additional information available from an alignment and/or structure. It is based on an algorithm outlined in Schwarzenbacher et al. (2004).

Conventions

The following terms are used with the special meaning:

- target: the structure to be solved
- model: the structure used as a model for the target

Usage

There is a standalone `sculptor_new` program available from the command line. The unified model preparation program `sculpt_ensemble` is available from the command line and also from the PHENIX GUI.

Input files

- Structure (compulsory): the structure to be modified. Specific parts can be selected using CNS-style atom selection syntax. Chains are divided into protein, DNA, RNA, hetero, monosaccharide and other chain categories, and processed according to instructions given for the appropriate chain type. Accepted formats: PDB. Recognized extensions: `.pdb`, `.ent`.
- Alignments: sequence alignment with the target sequence. The alignment contains information that can be exploited in model improvement (this is currently only implemented for protein chains). The chains are automatically associated with the corresponding alignment based on a sequence comparison. The target sequence is also automatically identified if it is provided through the `target_sequence` keyword, otherwise the first sequence in the alignment is used as target. Alignment input is optional, in case it is not provided, an alignment will be made up using the chain sequence with itself. Accepted extensions (with the corresponding format) are `.aln` (CLUSTAL format), `.pir` (PIR-format) and `.ali` (relaxed PIR-like format).

- Homology search files: hits from a homology search. These work very similarly to a set of alignment files. It is assumed that the first sequence in each alignment is the target sequence (the sequence used for searching homologues). Accepted extensions (with the corresponding format) are .xml (BLAST-family XML output) and .hhr (HHPRED output).
- Sequence files: target sequence. In case no alignment is available, Sculptor can be instructed to prepare an alignment using the sequences of model chains assigned on the chain_ids parameter (please note that in this case this alignment will be used for the specified chains, and any user-supplied alignments will be ignored). Multiple sequence files can be provided when target sequences for distinct protein chains differ. Accepted extensions are .fasta, .faa or .fa for FASTA format, .pir for PIR-format and .seq or .dat for a relaxed PIR/FASTA-like format.
- Error files and Superposition error files: error estimates coming either from a model quality assessment program (MQAP) or from a superposition. Errors from MQAPs can be used in mainchain processing and also in B-factor weighting, while those from a superposition can only be used for mainchain processing. The inputs are kept separate, and hence it is possible to use two different error estimates for each macromolecule chain.

Output files

The fully processed structure is output. The file is named according to the following convention: root_pdb.pdb, where root is a user-defined parameter (accessible from the output scope), and pdb is the basename of the input PDB file.

Outline of the procedure

The workflow consists of several stages that can be independently configured. These are listed in order of execution. For a summary of all keywords with the corresponding defaults, see the Additional information section.

Preprocessing

- selection: selects a subset of the input PDB file, using CNS-style atom selection syntax. Default: all.

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CNS-style atom selection syntax. Default: all.

- remove_alternate_conformations: selects the first alternate conformation for disordered entities, and discards the rest. Also involves sanitize_occupancies.

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conformation for disordered entities, and discards the rest. Also involves sanitize_occupancies.

- sanitize_occupancies: resets all occupancies to 1.0.
- keep_crystal_symmetry: retain the CRYST record of the model structure.

In addition, chains will be analysed, hetero, sugar and solvent atoms will be separated from protein/DNA/RNA chains if they are not separated by TER cards.

Chain-to-alignment matching

Parameters in the chain_to_alignment_matching scope control how a sequence from an alignment is matched to the sequence of a macromolecule chain, and what constitutes an acceptable match. The sequence from

the alignment is considered strictly consecutive, while gaps are allowed in the sequence derived from the protein chain (this is governed by the consecutivity parameter; geometry means that a chain segment is strictly consecutive if there is a bond to the neighbouring residue, and numbering means that residue numbering should be used to decide whether a residue is connected to neighbouring ones). The `min_sequence_identity` parameter is used as a threshold to accept a possible match between the two sequences.

Error estimates

These parameters control how error estimates input are matched to macromolecule chains. In addition, if no external errors are available, the modelling program * Rosetta * is installed and configured to be used with PHENIX, and there is network connectivity, Sculptor can be instructed to calculate an error estimate (`calculate_if_not_provided` parameter and `homology_modelling` scope) by first making a homology model using * Rosetta * and then submitting this to the ProQ2 server. Raw results * obtained will be written out if the `output_prefix` parameter is set. Please note that this only works for protein chains. For more information, see the `simple_homology_model` documentation.

Please note that the input/obtained error estimates are only used if a suitable processing method is selected. Such methods are available in the Deletion and B-factor prediction sections.

Chain processing

Protein chains

Deletion

Discards residues from a model chain that are unlikely to improve signal in molecular replacement. This information is calculated either from the alignment or from estimated errors.

There are multiple algorithms available:

- `gap`: Deletes residues that are not present in the target (model residue is aligned with a gap). For this algorithm, the supplied alignment is used as a pairwise alignment.
- `threshold_based_similarity`: Deletes residues for which the sequence similarity is below a certain threshold. All sequences in a multiple alignment contribute to the score. Details of the sequence similarity calculation are given in the section Sequence similarity calculation.
- `completeness_based_similarity`: Deletes the same number of residues (modified by a fractional offset) as the gap algorithm would but residues that get removed are the ones with the lowest sequence similarity. This way the default values are valid over a much larger sequence similarity range than those in `threshold_based_similarity`. All sequences in a multiple alignment contribute to the score. Details of the sequence similarity calculation are given in the section Sequence similarity calculation.
- `remove_long`: Deletes residues from the model if these are aligned with gaps and at least `minimum_length` long.
- `rms`: Deletes residues whose estimated error is over a certain threshold. Missing values are filled in using parameters in the `missing_value_substitution` scope (see Missing value substitution).
- `superposition`: Deletes residues whose superposition error is over a certain threshold. Missing values are filled in using parameters in the `missing_value_substitution` scope (see Missing value substitution).

These algorithms can also be used together in any combination. In this case, a residue will be deleted if assigned for deletion by any active algorithms.

Polishing

Makes small adjustments to the mainchain of a chain (taking results from deletion into account) to make it obey basic macromolecular features.

- `remove_short`. Deletes additional residue segments from the molecule so that no continuous segment is shorter than a preset limit (determined by the `minimum_length` parameter of the `remove_short` scope). Segment boundaries are determined from spatial connectivity of residues. This algorithm is primarily intended to remove "floating" residues that are the result of extensive loop truncation.
- `undo_short`. Reinstate short segment that are assigned for deletion. The maximum length is controlled by the `maximum_length` parameter.
- `keep_regular`. Reinstate deleted residues if they are in regular secondary structure and the segment marked for deletion is shorter than the `maximum_length` parameter of the `keep_regular` scope.

These algorithms can also be used together in combination. In this case, the chain will be processed sequentially by both algorithms.

Atom mapping

This governs how the sidechain of an amino acid residue in the model is morphed into the target type.

- `connectivity`. This uses atom-connectivity-based similarity to match one sidechain to another. The matching can take into account whether the chemical element of the model and target type agree (`match_chemical_elements` keyword).
- `geometry`. This uses the actual observed geometry of the sidechain in the model structure and tries to map it onto all known rotamers of the target amino acid, and selects the rotamer that results in the highest number of atoms mapped. Geometric matching is done via matching distances, and the tolerance is controlled by the `tolerance` keyword. The algorithm can also take into account the chemical elements explicitly, but this is probably not as useful, since implicitly these are always taken into account through bond lengths and angles. Rotamer information is obtained through the monomer library, and hence this algorithm only works well if the monomer library contains this information. With this algorithm, matching a particular sidechain with the same type can result in a partial match if the observed sidechain conformation is not rotameric. The keyword `match_if_identical` controls whether matching should be performed in this case or just an identity atom map should be used.

Pruning

This phase determines the level distance from the C α atom up to which a residue sidechain in the model is potentially similar to its counterpart in the target.

- `null`. No pruning is applied (unmatched atoms will still be discarded).
- `schwarzenbacher`. Implements the algorithm published by Schwarzenbacher et al. (2004), who propose that for optimal molecular replacement results a residue sidechain should be truncated if aligned with a non-identical residue, and not truncated otherwise. The level of truncation is controlled by the `pruning_level` parameter, and defaults to 3 (which corresponds to C γ) and can be controlled by the `pruning_level` parameter of the `schwarzenbacher` scope.

- **Similarity.** Uses sequence similarity values for deciding the level of truncation. Residues above `full_truncation_limit` are not truncated at all, those below the `full_truncation_limit` are truncated to Cbeta, and those in between are truncated according to the `pruning_level` parameter (all available from the similarity scope). Results tend to be similar to those given by the Schwarzenbacher algorithm; however, it is possible to get high similarity values (and full sidechain preservation) for certain substitutions (i.e. TYR to PHE), and low-sequence similarity zones can end up being truncated to Cbeta. Details of the sequence similarity calculation are given in the section Sequence similarity calculation.

These algorithms can also be used together in any combination, in which case the sidechain will be truncated to the shortest value suggested.

B-factor prediction

B-factor prediction tries to increase B-factors for atoms that are likely to be more flexible or more in error. The calculation takes simple physical properties into account, and these are linearly transformed to B-factors (controlled by the `factor` parameter of the corresponding scope). If this value is lower than the minimum (from the `bfactorscope`) parameter, a constant is added to all B-factors so that the lowest of those equals to minimum (this is primarily intended to avoid negative B-factors).

- **original.** This uses the original B-factor of atoms. This is primarily intended as a contributor to a combination, but can also be used to manipulate current B-factors, e.g. set them to a constant value.
- **asa.** This calculates accessible surface area for an isolated chain and transforms the raw values to B-factors. A high ASA-value indicates a potential for flexibility. The calculation can be configured by the `precision` and `probe_radius` parameters of the `asa` scope.
- **similarity.** Low sequence similarity regions tend to be more dissimilar. Details of the sequence similarity calculation are given in the section Sequence similarity calculation.
- **rms.** Use error estimates (either input or those calculated within the program) to weight atoms. The theoretical scale factor of $\frac{8}{3} * \pi^2$ is already built into the calculation, but this can be adjusted if necessary. Missing values are filled in using parameters in the `missing_value_substitution` scope (see Missing value substitution).

Algorithms can be used in combination, in which case the sum of the predicted B-factors is used. This mode can also be used to map sequence similarity or accessible surface area to residues/atoms for display purposes.

Renumber

Renumbers residues according to the target or model sequence. It is also possible to turn renumbering off (option `original`).

Rename

Renames residues according their counterpart in the target sequence. Please note that this is only a name change. Sidechain atoms are always mapped onto their target counterparts, and deleted if not present in the target. On the other hand, the addition of atoms that are present in the target and not in the model does not take place if renaming is not performed.

Completion

Controls the addition of missing atoms.

- **cbeta.** Adds Cbeta atom if the residue is not glycine, and C, N and Calpha atoms are all present. A common Cbeta position is used for all non-proline residues, and a slightly differing ones for prolines.
- **cbeta_and_pro.** Fills in Cbeta where possible for non-proline residues, and the whole sidechain for prolines.
- **sidechain.** Fills in missing atoms for all sidechains. The algorithm tries to get as close as possible to the atoms present in the structure, but in the general case, all sidechain atoms within the residue may move slightly.

DNA/RNA chains

Discards residues from a model chain that are unlikely to improve signal in molecular replacement.

There are two algorithms available:

- **all:** Deletes all residues.
- **superposition:** Deletes residues whose superposition error is over a certain threshold. Missing values are filled in using parameters in the `missing_value_substitution` scope (see Missing value substitution).

These algorithms can also be used together in any combination. In this case, a residue will be deleted if assigned for deletion by any active algorithms.

Makes small adjustments to the mainchain of a chain (taking results from deletion into account) to make it obey basic macromolecular features.

- **remove_short.** Deletes additional residue segments from the molecule so that no continuous segment is shorter than a preset limit (determined by the `minimum_length` parameter of the `remove_short` scope). Segment boundaries are determined from spatial connectivity of residues. This algorithm is primarily intended to remove "floating" residues that are the result of extensive loop truncation.

These algorithms can also be used together in combination. In this case, the chain will be processed sequentially by both algorithms.

Monosaccharide chains

This can be used to trim existing glucosyl chains based on the distance from the residue to which they are attached. Connectivity of glycosyl chains is worked out from distance tests, and the `maximum_bond_length` parameter can be used to adjust this slightly. Branched glycosyl chains are also handled.

Hetero chains

Residues in these chains are normally deleted, unless an exception is made by specifying the residue codes that are to be retained. This is primarily intended to keep a known ligands of protein classes (e.g. HEM).

Other chains

These are removed from the model.

Sequence similarity calculation

Sequence similarity is calculated from the full alignment supplied (taking all present sequences into account), using a scoring matrix (currently `blosum50`, `blosum62`, `dayhoff` and `identity` are available). Raw scores are then smoothed using one of two algorithms:

- **linear:** averaging is done over a window of residues (this is measured from the residue in question, i.e. $2 \times \text{window} + 1$ residues scores are used), and is weighted using either uniform or triangular weights. The resulting scores are "normalized" so that 1.0 would indicate a perfect alignment, 0.0 would be a random match, and -1.0 a (locally) fully gapped alignment (on average). Note, it is possible to obtain values outside this range. This helps to ensure that defaults are sensible irrespective of the choice for the scoring matrix.
- **spatial:** averaging is done over a sphere with radius `maximum_distance` using all residue scores that are within this limit. Residues that are not present in the model and hence no spatial coordinate can be associated with them affect the raw scores of neighbouring residues (using a convolution with a triangular weight function).

Sequence similarity calculation is configured individually for the steps that are using it.

Missing value substitution

Governs how missing error values are substituted.

- **maximum_value:** uses the maximum value encountered for the particular chain.
- **scaled_interpolated_value:** interpolates within a chain or extrapolates at the termini. Extrapolation is done by increasing the B-factor of the closest residue with a factor that is calculated as the n th power of a constant value (`extrapolation_step_scale` parameter), where n is the distance from the closest residue. Interpolation is done in two steps. First linear interpolation is performed between the two bridge residues, and then these are multiplied with a factor that is calculated as the n th power of a constant value (`interpolation_step_scale`), where n is the distance from the nearest bridge residue.

Missing value substitution is configured individually for the steps that are using it.

Command line

```
phenix.sculptor \
[ command-line switches ] \
[ PHIL-format parameter files ] \
[ PHIL command-line assignments ] \
[ PDB-files ] \
[ alignment files ]
```

Command-line switches

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
-v, --verbosity set verbosity level (info,debug,verbose)
--version show program's version number and exit
--text-logfile FILE Verbatim copy of log stream
--html-logfile FILE Verbatim copy of log stream in HTML
```

PHIL arguments

Everything not starting with a dash ('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters, command-line assignment or a file whose type is automatically recognized (based on file extension). Note that sequence files are not accepted on the command line, since associated chains could not easily be guessed and require a fully specified parameter scope.

Specific limitations and possible problems

Processing features

- Very short residue segments (shorter in min_hss_length consecutive residues than min_hss_length parameter in the chain_to_alignment_matching scope) cannot be reliably aligned to the sequence, and these may be discarded from the model.
- The similarity algorithm from the deletion scope may result in residues that are aligned with a gap being included in the model. Although this possibly indicates an error in the alignment and is potentially beneficial for molecular replacement, this causes a problem at the rename stage, as there is no 3-letter residue name for a "-"; these residues are named according to the gapname parameter (default: ALA).
- Residue numbers for gap residues are built up using the residue number of the previous non-gap residue and an insertion code (A-Z, depending on the number of gap residues after the previous non-gap residue).

Error messages

- No pdb files specified: there are no PDB files to process.
- Applicable chain_ids not specified for ...: the input file requires the specification chainIDs (sequence file, error files and superposition error files).
- No atoms left after atom selection: the atom selection provided results in an empty structure.
- Error while reading alignment: ...: some error occurred while trying to read an alignment file: ...: input alignment is empty: no alignment sequences were found in alignment file: no alignment sequences match: the target sequence specified for alignment file .. does not match any alignment sequences.
- ...: input alignment is empty: no alignment sequences were found in alignment file ..
- ...: no alignment sequences match: the target sequence specified for alignment file .. does not match any alignment sequences.
- Cannot open file: a file cannot be opened for reading (possibly does not exist).
- No hit with index .. in ...: the requested homology search hit from homology search file .. does not exist.

Warning messages

Alignment

- Aligner: no aligned candidates: no alignment sequence match the observed chain sequence. In this case a dummy alignment will be used.
- No sequences read from file: the sequence file given as the target sequence does not contain any sequences.
- Multiple sequences found, using first: the sequence file given as the target sequence contains multiple sequences.

Sequence files

- File contains multiple sequences, only the first will be used: the sequence file given as the target sequence contains multiple sequences.

Error files

- Duplicate resid: '..': resid .. occurs multiple times.
- Cannot convert '..' to floating-point number: the error value is non-numeric.
- Error file specified for chain fail criteria: the error values that are specified for a particular chain are rejected, because they fail the set criteria.

Error estimation

- Homology modelling failed: no homology model can be created.
- ProQ2 error estimation failed: no error values could be obtained from the ProQ2 server.
- ProQ2 structure contains multiple chains: although a single chain was submitted to ProQ2, the results returned contain multiple chains.
- Chain '..' is not '..': chain .. is not recognised as a macromolecular chain.
- PDB could not be matched to sequence: sequence-to-structure matching fail for the returned results.

References

Citation

Improvement of molecular-replacement models with Sculptor. G. Bunkoczi and R. J. Read Acta Cryst. D67, 303-312 (2011)

Additional information

List of all available keywords

inputInput filesmodelInput pdb filefile_name = None PDB file nameselection = all Selection
stringremove_alternate_conformations = False Remove alternate conformationsssanitize_occupancies = False
Sets occupancies > 1.0 to 1.0keep_crystal_symmetry = False Keeps the crystal symmetry of the
modelalignmentInput alignment filefile_name = None Alignment file nametarget_sequence = None Target
sequence file name (to locate target sequence in alignment)homology_searchAlignment from homology
search filefile_name = None Homology search fileuse = None Which alignments to usesequenceInput
sequence filefile_name = None Sequence filechain_ids = None Which chain IDs the target sequence
applieserrorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs
the error file corresponds tosuperposition_errorsSuperposition errors for chainfile_name = None Error file
namechain_ids = None Which chain IDs the error file corresponds toooutputOutput optionsjob_title = None Job
title in PHENIX GUI, not used on command linefolder = . Output file folderroot = sculpt Output file rootformat =
*pdb Output file formatchain_to_alignment_matchingChain-to-alignment matching optionsconsecutivity =
*geometry numbering Consecutivity criterion to detect chain breaksmin_hss_length = 3 Minimum length of a
sequence fragment to be included in chain alignmentmax_seed_hss_count = 12 Number of HSS to use in
extensive searchmax_completion_hss_count = 6 Number of HSS to use in gap fillingmin_sequence_overlap =
10 Minimum overlap between sequences to perform full alignmentmin_sequence_identity = 0.80 Minimum
sequence identity of accepted chain alignmenterror_searchParameters for matching error files and/or
estimating errorsmin_sequence_identity = 0.8 Minimum sequence identity to accept error
filemin_sequence_overlap = 0.8 Minimum sequence overlap to accept error filecalculate_if_not_provided =
False Use Rosetta and the ProQ2 server to get error estimatesoutput_prefix = None File name for results of
intermediate stepshomology_modellingParameters for homology modellingmin_residue_margin = 2 Minimum
number of residues to cut back on both sides of loopsresidue_distance = 2.5 Distance covered by a single

residue (in A)min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap margins)min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)max_loop_length = 60 Maximum loop lengthrosetta_max_build_attempts = 1000 Maximum build attempts to close looprosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closuresrosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closurerosetta_loop_refinement = *no default quick fast test Algorithm for Rosetta loop refinementsculptParameters for sculptingproteinOptions to process protein chainscompletion = cbeta cbeta_and_pro sidechain Method to build missing sidechain atomsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to usegapDelete residue if aligned with gapthreshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.20 Threshold to accept a residuesimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingcompleteness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset ...

- inputInput filesmodellInput pdb filefile_name = None PDB file nameselection = all Selection stringremove_alternate_conformations = False Remove alternate conformationsssanitize_occupancies = False Sets occupancies > 1.0 to 1.0keep_crystal_symmetry = False Keeps the crystal symmetry of the modelalignmentInput alignment filefile_name = None Alignment file nametarget_sequence = None Target sequence file name (to locate target sequence in alignment)homology_searchAlignment from homology search filefile_name = None Homology search fileuse = None Which alignments to usesequenceInput sequence filefile_name = None Sequence filechain_ids = None Which chain IDs the target sequence applieserrorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds tosuperposition_errorsSuperposition errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds to
- modelInput pdb filefile_name = None PDB file nameselection = all Selection stringremove_alternate_conformations = False Remove alternate conformationsssanitize_occupancies = False Sets occupancies > 1.0 to 1.0keep_crystal_symmetry = False Keeps the crystal symmetry of the model
- file_name = None PDB file name
- selection = all Selection string
- remove_alternate_conformations = False Remove alternate conformations
- sanitize_occupancies = False Sets occupancies > 1.0 to 1.0
- keep_crystal_symmetry = False Keeps the crystal symmetry of the model
- alignmentInput alignment filefile_name = None Alignment file nametarget_sequence = None Target sequence file name (to locate target sequence in alignment)
- file_name = None Alignment file name
- target_sequence = None Target sequence file name (to locate target sequence in alignment)
- homology_searchAlignment from homology search filefile_name = None Homology search fileuse = None Which alignments to use
- file_name = None Homology search file

- use = None Which alignments to use
- sequenceInput sequence filefile_name = None Sequence filechain_ids = None Which chain IDs the target sequence applies
- file_name = None Sequence file
- chain_ids = None Which chain IDs the target sequence applies
- errorsEstimated errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds to
- file_name = None Error file name
- chain_ids = None Which chain IDs the error file corresponds to
- superposition_errorsSuperposition errors for chainfile_name = None Error file namechain_ids = None Which chain IDs the error file corresponds to
- file_name = None Error file name
- chain_ids = None Which chain IDs the error file corresponds to
- outputOutput optionsjob_title = None Job title in PHENIX GUI, not used on command linefolder = . Output file folderroot = sculpt Output file rootformat = *pdb Output file format
- job_title = None Job title in PHENIX GUI, not used on command line
- folder = . Output file folder
- root = sculpt Output file root
- format = *pdb Output file format
- chain_to_alignment_matchingChain-to-alignment matching optionsconsecutivity = *geometry numbering Consecutivity criterion to detect chain breaksmin_hss_length = 3 Minimum length of a sequence fragment to be included in chain alignmentmax_seed_hss_count = 12 Number of HSS to use in extensive searchmax_completion_hss_count = 6 Number of HSS to use in gap fillingmin_sequence_overlap = 10 Minimum overlap between sequences to perform full alignmentmin_sequence_identity = 0.80 Minimum sequence identity of accepted chain alignment
- consecutivity = *geometry numbering Consecutivity criterion to detect chain breaks
- min_hss_length = 3 Minimum length of a sequence fragment to be included in chain alignment
- max_seed_hss_count = 12 Number of HSS to use in extensive search
- max_completion_hss_count = 6 Number of HSS to use in gap filling
- min_sequence_overlap = 10 Minimum overlap between sequences to perform full alignment
- min_sequence_identity = 0.80 Minimum sequence identity of accepted chain alignment
- error_searchParameters for matching error files and/or estimating errorsmin_sequence_identity = 0.8 Minimum sequence identity to accept error filemin_sequence_overlap = 0.8 Minimum sequence overlap to accept error filecalculate_if_not_provided = False Use Rosetta and the ProQ2 server to get error estimatesoutput_prefix = None File name for results of intermediate stepshomology_modellingParameters for homology modellingmin_residue_margin = 2 Minimum number of residues to cut back on both sides of loopsresidue_distance = 2.5 Distance covered by a single residue (in Å)min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap margins)min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)max_loop_length = 60 Maximum loop lengthrosetta_max_build_attempts = 1000 Maximum build attempts to close looprosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closuresrosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closurerosetta_loop_refinement = *no

default quick fast test Algorithm for Rosetta loop refinement

- min_sequence_identity = 0.8 Minimum sequence identity to accept error file
- min_sequence_overlap = 0.8 Minimum sequence overlap to accept error file
- calculate_if_not_provided = False Use Rosetta and the ProQ2 server to get error estimates
- output_prefix = None File name for results of intermediate steps
- homology_modellingParameters for homology modellingmin_residue_margin = 2 Minimum number of residues to cut back on both sides of loopsresidue_distance = 2.5 Distance covered by a single residue (in Å)min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap margins)min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)max_loop_length = 60 Maximum loop lengthrosetta_max_build_attempts = 1000 Maximum build attempts to close looprosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closuresrosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closure
rosetta_loop_refinement = *no default quick fast test Algorithm for Rosetta loop refinement
- min_residue_margin = 2 Minimum number of residues to cut back on both sides of loops
- residue_distance = 2.5 Distance covered by a single residue (in Å)
- min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap margins)
- min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)
- max_loop_length = 60 Maximum loop length
- rosetta_max_build_attempts = 1000 Maximum build attempts to close loop
- rosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closures
- rosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closure
- rosetta_loop_refinement = *no default quick fast test Algorithm for Rosetta loop refinement
- sculptParameters for sculptingproteinOptions to process protein chainscompletion = cbeta
cbeta_and_pro sidechain Method to build missing sidechain atomsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsmuse = *gap threshold_based_similarity
completeness_based_similarity residue_count_based_similarity remove_long rms superposition
Algorithm to usegapDelete residue if aligned with gapthreshold_based_similarityDelete residue if
sequence similarity is lowthreshold = -0.20 Threshold to accept a residuesimilarity_calculationConfigure
sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity
matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters
for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting
schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by
gapsmaximum_distance = 10 Spatial distance for averagingcompleteness_based_similarityDelete
residues based on sequence similarity to get same number of gaps as the Schwarzenbacher
algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from
Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity
calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw
similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5
Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial
averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial
distance for averagingresidue_count_based_similarityDelete residues based on sequence similarity to
delete the same number of residues as the Schwarzenbacher algorithmoffset = 0.0 Completeness in
fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range:

+/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50
 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to
 uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform
 *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment
 positions affected by gapsmaximum_distance = 10 Spatial distance for averagingremove_longDelete gap
 segments if longer than a thresholdminimum_length = 3 Minimum length for mainchain segment to
 removermsDelete residues if estimated error is larger than a thresholdthreshold = 5.0 Maximum allowed
 error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value
 scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the
 sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the
 distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale
 = 1.10 Stepwise scale factor for extrapolationsuperpositionDelete residues if estimated error is larger
 than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to
 fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to
 usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute
 interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for
 extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationpolishingMainchain
 polishing algorithmsuse = remove_short undo_short keep_regular Algorithm to useremove_shortDelete
 short unconnected segmentsminimum_length = 3 Minimum lengthundo_shortDelete short
 gapsmaximum_length = 2 Maximum lengthkeep_regularKeep residues in secondary
 structuremaximum_length = 1 M...

- proteinOptions to process protein chainscompletion = cbeta cbeta_and_pro sidechain Method to build
 missing sidechain atomsmainchainOptions for main chain processingremove_unaligned = True Delete
 residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *gap
 threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long
 rms superposition Algorithm to usegapDelete residue if aligned with gapthreshold_based_similarityDelete
 residue if sequence similarity is lowthreshold = -0.20 Threshold to accept a

residuesimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50
 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to
 uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform
 *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment
 positions affected by gapsmaximum_distance = 10 Spatial distance for
 averagingcompleteness_based_similarityDelete residues based on sequence similarity to get same
 number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length
 (0.0 = completeness from Schwarzenbacher algorithm, useful range:

+/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50
 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to
 uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform
 *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment
 positions affected by gapsmaximum_distance = 10 Spatial distance for
 averagingresidue_count_based_similarityDelete residues based on sequence similarity to delete the
 same number of residues as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of
 model length (0.0 = completeness from Schwarzenbacher algorithm, useful range:

+/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50
 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to
 uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform
 *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment
 positions affected by gapsmaximum_distance = 10 Spatial distance for averagingremove_longDelete gap

segments if longer than a threshold `minimum_length = 3` Minimum length for mainchain segment to remove `remove_rms` Delete residues if estimated error is larger than a threshold `threshold = 5.0` Maximum allowed error for a residue `missing_value_substitutionPolicy` to fill in missing values `use = *maximum_value scaled_interpolated_value` Algorithm to use `maximum_valueSubstitute` maximum from the sequence `scaled_interpolated_valueSubstitute` interpolated value scaled with the distance `extrapolation_step_scale = 1.20` Stepwise scale factor for extrapolation `interpolation_step_scale = 1.10` Stepwise scale factor for extrapolation `superposition` Delete residues if estimated error is larger than a threshold `threshold = 2.0` Maximum allowed error for a residue `missing_value_substitutionPolicy` to fill in missing values `use = *maximum_value scaled_interpolated_value` Algorithm to use `maximum_valueSubstitute` maximum from the sequence `scaled_interpolated_valueSubstitute` interpolated value scaled with the distance `extrapolation_step_scale = 1.20` Stepwise scale factor for extrapolation `interpolation_step_scale = 1.10` Stepwise scale factor for extrapolation `polishingMainchain` polishing algorithm `suse = remove_short undo_short keep_regular` Algorithm to use `remove_shortDelete` short unconnected segments `minimum_length = 3` Minimum length `undo_shortDelete` short gaps `maximum_length = 2` Maximum length `keep_regularKeep` residues in secondary structure `maximum_length = 1` Maximum length `bfactorConfigure ...`

- `completion = cbeta cbeta_and_pro` sidechain Method to build missing sidechain atoms

- `mainchainOptions` for main chain processing `remove_unaligned = True` Delete residues that could not be matched to an alignment `deletionMainchain` deletion algorithm `suse = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition` Algorithm to use `gapDelete` residue if aligned with `gapthreshold_based_similarityDelete` residue if sequence similarity is low `threshold = -0.20` Threshold to accept a residue `similarity_calculationConfigure` sequence similarity calculation `matrix = identity dayhoff blosum50 *blosum62` Similarity `matrixsmoothingConfigure` raw similarity smoothing `use = *linear spatial` Method to use `linearParameters` for linear averaging `window = 5` Averaging window width `weighting = uniform *triangular` Weighting scheme `spatialParameters` for spatial averaging `gap_bleed_length = 3` Alignment positions affected by gaps `maximum_distance = 10` Spatial distance for averaging `completeness_based_similarityDelete` residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithm `offset = 0.0` Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05) `similarity_calculationConfigure` sequence similarity calculation `matrix = identity dayhoff blosum50 *blosum62` Similarity `matrixsmoothingConfigure` raw similarity smoothing `use = *linear spatial` Method to use `linearParameters` for linear averaging `window = 5` Averaging window width `weighting = uniform *triangular` Weighting scheme `spatialParameters` for spatial averaging `gap_bleed_length = 3` Alignment positions affected by gaps `maximum_distance = 10` Spatial distance for averaging `residue_count_based_similarityDelete` residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithm `offset = 0.0` Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05) `similarity_calculationConfigure` sequence similarity calculation `matrix = identity dayhoff blosum50 *blosum62` Similarity `matrixsmoothingConfigure` raw similarity smoothing `use = *linear spatial` Method to use `linearParameters` for linear averaging `window = 5` Averaging window width `weighting = uniform *triangular` Weighting scheme `spatialParameters` for spatial averaging `gap_bleed_length = 3` Alignment positions affected by gaps `maximum_distance = 10` Spatial distance for averaging `remove_longDelete` gap segments if longer than a threshold `minimum_length = 3` Minimum length for mainchain segment to remove `remove_rms` Delete residues if estimated error is larger than a threshold `threshold = 5.0` Maximum allowed error for a residue `missing_value_substitutionPolicy` to fill in missing values `use = *maximum_value scaled_interpolated_value` Algorithm to use `maximum_valueSubstitute` maximum from the sequence `scaled_interpolated_valueSubstitute` interpolated value scaled with the

distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
 interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
 superpositionDelete residues if estimated error is larger than a threshold
 threshold = 2.0 Maximum allowed error for a residue
 missing_value_substitutionPolicy to fill in missing values
 use = *maximum_value scaled_interpolated_value Algorithm to use
 maximum_valueSubstitute maximum from the sequence
 scaled_interpolated_valueSubstitute interpolated value scaled with the distance
 extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
 interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
 polishingMainchain polishing algorithms
 use = remove_short undo_short keep_regular Algorithm to use
 remove_shortDelete short unconnected segments
 minimum_length = 3 Minimum length
 undo_shortDelete short gaps
 maximum_length = 2 Maximum length
 keep_regularKeep residues in secondary structure
 maximum_length = 1 Maximum length

- remove_unaligned = True Delete residues that could not be matched to an alignment

- deletionMainchain deletion algorithms
 use = *gap threshold_based_similarity completeness_based_similarity
 residue_count_based_similarity remove_long rms superposition
 Algorithm to use
 gapDelete residue if aligned with gap
 threshold_based_similarityDelete residue if sequence similarity is low
 threshold = -0.20 Threshold to accept a residue
 similarity_calculationConfigure sequence similarity calculation
 matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
 smoothingConfigure raw similarity smoothing
 use = *linear spatial Method to use
 linearParameters for linear averaging
 window = 5 Averaging window width
 weighting = uniform *triangular Weighting scheme
 spatialParameters for spatial averaging
 gap_bleed_length = 3 Alignment positions affected by gaps
 maximum_distance = 10 Spatial distance for averaging
 completeness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithm
 offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
 similarity_calculationConfigure sequence similarity calculation
 matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
 smoothingConfigure raw similarity smoothing
 use = *linear spatial Method to use
 linearParameters for linear averaging
 window = 5 Averaging window width
 weighting = uniform *triangular Weighting scheme
 spatialParameters for spatial averaging
 gap_bleed_length = 3 Alignment positions affected by gaps
 maximum_distance = 10 Spatial distance for averaging
 residue_count_based_similarityDelete residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithm
 offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
 similarity_calculationConfigure sequence similarity calculation
 matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
 smoothingConfigure raw similarity smoothing
 use = *linear spatial Method to use
 linearParameters for linear averaging
 window = 5 Averaging window width
 weighting = uniform *triangular Weighting scheme
 spatialParameters for spatial averaging
 gap_bleed_length = 3 Alignment positions affected by gaps
 maximum_distance = 10 Spatial distance for averaging
 remove_longDelete gap segments if longer than a threshold
 minimum_length = 3 Minimum length for mainchain segment to remove
 rmsDelete residues if estimated error is larger than a threshold
 threshold = 5.0 Maximum allowed error for a residue
 missing_value_substitutionPolicy to fill in missing values
 use = *maximum_value scaled_interpolated_value Algorithm to use
 maximum_valueSubstitute maximum from the sequence
 scaled_interpolated_valueSubstitute interpolated value scaled with the distance
 extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
 interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
 superpositionDelete residues if estimated error is larger than a threshold
 threshold = 2.0 Maximum allowed error for a residue
 missing_value_substitutionPolicy to fill in missing values
 use = *maximum_value scaled_interpolated_value Algorithm to use
 maximum_valueSubstitute maximum from the sequence
 scaled_interpolated_valueSubstitute interpolated value scaled with the distance
 extrapolation_step_scale = 1.20 Stepwise scale factor for

extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- use = *gap threshold_based_similarity completeness_based_similarity residue_count_based_similarity remove_long rms superposition Algorithm to use
- gapDelete residue if aligned with gap
- threshold_based_similarityDelete residue if sequence similarity is lowthreshold = -0.20 Threshold to accept a residue
similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- threshold = -0.20 Threshold to accept a residue
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- completeness_based_similarityDelete residues based on sequence similarity to get same number of gaps as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to use
linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform

*triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging

- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- residue_count_based_similarityDelete residues based on sequence similarity to delete the same number of residues as the Schwarzenbacher algorithmoffset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- offset = 0.0 Completeness in fraction of model length (0.0 = completeness from Schwarzenbacher algorithm, useful range: +/-0.05)
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging

- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- remove_longDelete gap segments if longer than a thresholdminimum_length = 3 Minimum length for mainchain segment to remove
- minimum_length = 3 Minimum length for mainchain segment to remove
- rmsDelete residues if estimated error is larger than a thresholdthreshold = 5.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- threshold = 5.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- superpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- threshold = 2.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- polishingMainchain polishing algorithmsuse = remove_short undo_short keep_regular Algorithm to use
remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
lengthundo_shortDelete short gapsmaximum_length = 2 Maximum lengthkeep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
- use = remove_short undo_short keep_regular Algorithm to use
- remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- minimum_length = 3 Minimum length
- undo_shortDelete short gapsmaximum_length = 2 Maximum length
- maximum_length = 2 Maximum length
- keep_regularKeep residues in secondary structuremaximum_length = 1 Maximum length
- maximum_length = 1 Maximum length
- bfactorConfigure B-factor modificationuse = *original asa similarity rms Algorithm to useminimum_b = 10.00 Minimum B-factororiginalUse original bfactors to predict new B-valuesfactor = 1.00 Scale factorasaUse accessible surface area to predict new B-valuesprecision = 960 Number of points per atomprobe_radius = 1.40 Radius for probing surface accessibilityfactor = 2.00 Scale factorsimilarityUse sequence similarity to predict new B-valuesfactor = -100.00 Scale factorsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to useParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averagingrmsUse external RMS error estimates to calculate B-valuesfactor = 1.00 Scale factormissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *original asa similarity rms Algorithm to use
- minimum_b = 10.00 Minimum B-factor
- originalUse original bfactors to predict new B-valuesfactor = 1.00 Scale factor
- factor = 1.00 Scale factor
- asaUse accessible surface area to predict new B-valuesprecision = 960 Number of points per atomprobe_radius = 1.40 Radius for probing surface accessibilityfactor = 2.00 Scale factor
- precision = 960 Number of points per atom
- probe_radius = 1.40 Radius for probing surface accessibility
- factor = 2.00 Scale factor
- similarityUse sequence similarity to predict new B-valuesfactor = -100.00 Scale factorsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to useParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- factor = -100.00 Scale factor

- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 5 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 5 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- rmsUse external RMS error estimates to calculate B-valuesfactor = 1.00 Scale factormissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- factor = 1.00 Scale factor
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- renumberResidue renumberinguse = *target model original Method to usestart = 1 Start residue number
- use = *target model original Method to use
- start = 1 Start residue number

- pruningOptions for sidechain pruninguse = null *schwarzenbacher similarity Algorithm to usepruning_level_unaligned = 2 Pruning level for residues that could not be matched to an alignmentnullDo not impose bond distance thresholdschwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncationsimilarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncationsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = null *schwarzenbacher similarity Algorithm to use
- pruning_level_unaligned = 2 Pruning level for residues that could not be matched to an alignment
- nullDo not impose bond distance threshold
- schwarzenbacherTruncate atoms if target residue != source residuepruning_level = 2 Level of truncation
- pruning_level = 2 Level of truncation
- similarityTruncate atoms based on sequence similaritypruning_level = 2 Level of intermediate truncationfull_length_limit = 0.2 Limit of no truncationfull_truncation_limit = -0.2 Limit for full truncationsimilarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- pruning_level = 2 Level of intermediate truncation
- full_length_limit = 0.2 Limit of no truncation
- full_truncation_limit = -0.2 Limit for full truncation
- similarity_calculationConfigure sequence similarity calculationmatrix = identity dayhoff blosum50 *blosum62 Similarity matrixsmoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- matrix = identity dayhoff blosum50 *blosum62 Similarity matrix
- smoothingConfigure raw similarity smoothinguse = *linear spatial Method to uselinearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting schemespatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging
- use = *linear spatial Method to use
- linearParameters for linear averagingwindow = 1 Averaging window widthweighting = uniform *triangular Weighting scheme
- window = 1 Averaging window width
- weighting = uniform *triangular Weighting scheme
- spatialParameters for spatial averaginggap_bleed_length = 3 Alignment positions affected by gapsmaximum_distance = 10 Spatial distance for averaging

- gap_bleed_length = 3 Alignment positions affected by gaps
- maximum_distance = 10 Spatial distance for averaging
- renameResidue renaminguse = original *target Method to usekeep_ptm = False Preserve post-translational modification of model residue if base residue types agreegapname = ALA Name residues corresponding to alignment gaps
- use = original *target Method to use
- keep_ptm = False Preserve post-translational modification of model residue if base residue types agree
- gapname = ALA Name residues corresponding to alignment gaps
- mappingOptions for sidechain mappinguse = *connectivity geometry Algorithm to usemap_if_identical = True Do mapping procedure for identical residues typesconnectivityMatch atoms by connectivitymatch_chemical_elements = True Take chemical element into accountgeometryMatch atoms by geometry considering all rotamersmatch_chemical_elements = False Take chemical element into accounttolerance = 0.1 Distance tolerancefine_sampling = False Use fine sampling for rotamers
- use = *connectivity geometry Algorithm to use
- map_if_identical = True Do mapping procedure for identical residues types
- connectivityMatch atoms by connectivitymatch_chemical_elements = True Take chemical element into account
- match_chemical_elements = True Take chemical element into account
- geometryMatch atoms by geometry considering all rotamersmatch_chemical_elements = False Take chemical element into accounttolerance = 0.1 Distance tolerancefine_sampling = False Use fine sampling for rotamers
- match_chemical_elements = False Take chemical element into account
- tolerance = 0.1 Distance tolerance
- fine_sampling = False Use fine sampling for rotamers
- dnaOptions to process DNA chainsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationpolishingMainchain polishing algorithmsuse = remove_short Algorithm to useremove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- mainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencesscaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolationpolishingMainchain polishing algorithmsuse = remove_short Algorithm to useremove_shortDelete short unconnected

segmentsminimum_length = 3 Minimum length

- remove_unaligned = True Delete residues that could not be matched to an alignment
- deletionMainchain deletion algorithmsuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *all superposition Algorithm to use
- allDelete all residues
- superpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- threshold = 2.0 Maximum allowed error for a residue
- missing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- use = *maximum_value scaled_interpolated_value Algorithm to use
- maximum_valueSubstitute maximum from the sequence
- scaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
- polishingMainchain polishing algorithmsuse = remove_short Algorithm to useremove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
- use = remove_short Algorithm to use
- remove_shortDelete short unconnected segmentsminimum_length = 3 Minimum length
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- rnaOptions to process RNA chainsmainchainOptions for main chain processingremove_unaligned = True Delete residues that could not be matched to an alignmentdeletionMainchain deletion algorithmsuse = *all superposition Algorithm to useallDelete all residuesuperpositionDelete residues if estimated error is larger than a thresholdthreshold = 2.0 Maximum allowed error for a residuemissing_value_substitutionPolicy to fill in missing valuesuse = *maximum_value scaled_interpolated_value Algorithm to usemaximum_valueSubstitute maximum from the sequencescaled_interpolated_valueSubstitute interpolated value scaled with the distanceextrapolation_step_scale = 1.20 Stepwise scale factor for extrapolationinterpolation_step_scale

= 1.10 Stepwise scale factor for extrapolation
polishingMainchain polishing algorithms
use = remove_short
Algorithm to useremove_shortDelete short unconnected segments
minimum_length = 3 Minimum length

- mainchainOptions for main chain processing
remove_unaligned = True Delete residues that could not be matched to an alignment
deletionMainchain deletion algorithms
use = *all superposition Algorithm to use
allDelete all residues
superpositionDelete residues if estimated error is larger than a threshold
threshold = 2.0 Maximum allowed error for a residue
missing_value_substitutionPolicy to fill in missing values
use = *maximum_value scaled_interpolated_value Algorithm to use
maximum_valueSubstitute maximum from the sequence
scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation
polishingMainchain polishing algorithms
use = remove_short Algorithm to useremove_shortDelete short unconnected segments
minimum_length = 3 Minimum length

- remove_unaligned = True Delete residues that could not be matched to an alignment

- deletionMainchain deletion algorithms
use = *all superposition Algorithm to use
allDelete all residues
superpositionDelete residues if estimated error is larger than a threshold
threshold = 2.0 Maximum allowed error for a residue
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- use = *all superposition Algorithm to use

- allDelete all residues

- superpositionDelete residues if estimated error is larger than a threshold
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- missing_value_substitutionPolicy to fill in missing values
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maximum_valueSubstitute maximum from the sequence
scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- use = *maximum_value scaled_interpolated_value Algorithm to use

- maximum_valueSubstitute maximum from the sequence

- scaled_interpolated_valueSubstitute interpolated value scaled with the distance
extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation
interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- extrapolation_step_scale = 1.20 Stepwise scale factor for extrapolation

- interpolation_step_scale = 1.10 Stepwise scale factor for extrapolation

- polishingMainchain polishing algorithms
use = remove_short Algorithm to useremove_shortDelete short unconnected segments
minimum_length = 3 Minimum length

- use = remove_short Algorithm to use

- `remove_short` Delete short unconnected segments `minimum_length = 3` Minimum length
- `minimum_length = 3` Minimum length
- `hetero` Options to process hetero chains `keep = None` Keep named hetero residues
- `keep = None` Keep named hetero residues
- `monosaccharide` Options to process glycosyl chains `maximum_depth = 0` Keep chain up to maximum depth (None to keep all) `maximum_bond_length = 1.5` Maximum bond length for glycosidic bond
- `maximum_depth = 0` Keep chain up to maximum depth (None to keep all)
- `maximum_bond_length = 1.5` Maximum bond length for glycosidic bond